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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The following text is from the 1913 story “The King’s Coin” by Emily Pauline Johnson, a Kanienkahagen (Mohawk) writer also known as Tekahionwake.

Fox-Foot, a young Ojibwe man, is guiding a group of fur traders who are traveling by canoe and suspects that they are being followed.

At supper time, Fox-Foot would allow no fire to be built, no landing to be made, no trace of their passing to be left. They ate canned meat and marmalade, drank again of the stream and pushed on, until just at dusk they reached the edge of a long, still lake, with shores of granite and dense fir forest.

As used in the text, what does the word “trace” most nearly mean?

- A) Evidence
- B) Blemish
- C) Amount
- D) Sketch

2

Many ancient sculptures of people’s heads are missing their noses. This is because the nose is the most _____ part of a sculpture of a person’s head. It is delicate and sticks out from the rest of the sculpture, making it especially easy to break.

Which choice completes the text with the most logical and precise word or phrase?

- A) recognizable
- B) fragile
- C) common
- D) sophisticated

3

K.D. Leka and colleagues found that the Sun's corona provides an advance indication of solar flares—intense eruptions of electromagnetic radiation that emanate from active regions in the Sun's photosphere and can interfere with telecommunications on Earth. Preceding a flare, the corona temporarily exhibits increased brightness above the region where the flare is _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) antecedent
- B) impending
- C) innocuous
- D) perpetual

4

To demonstrate that the integrity of underground metal pipes can be assessed without unearthing the pipes, engineer Aroba Saleem and colleagues _____ the tendency of some metals' internal magnetic fields to alter under stress: the team showed that such alterations can be measured from a distance and can reveal concentrations of stress in the pipes.

Which choice completes the text with the most logical and precise word or phrase?

- A) hypothesized
- B) discounted
- C) redefined
- D) exploited

5

Despite the generalizations about human behavior they have produced, many studies of behavioral psychology have used highly unrepresentative subject pools: students at the colleges and universities where the researchers are employed. To _____ this situation, it is necessary to actively recruit subjects from diverse backgrounds and locations.

Which choice completes the text with the most logical and precise word or phrase?

- A) sanction
- B) ameliorate
- C) rationalize
- D) postulate

6

The following text is adapted from Jean Webster's 1912 novel *Daddy-Long-Legs*. The narrator is a young college student writing letters detailing her weekly experiences.

[The college is] organizing the Freshman basket-ball team and there's just a chance that I shall make it. I'm little of course, but terribly quick and wiry and tough. While the others are hopping about in the air, I can dodge under their feet and grab the ball.

Which choice best states the main purpose of the text?

- A) To compare basketball with other sports
- B) To provide details of how to play basketball
- C) To state how players will be chosen for the basketball team
- D) To explain why the narrator thinks she might make the basketball team

7

In the late 1800s, Spanish-language newspapers flourished in cities across Texas. San Antonio alone produced eleven newspapers in Spanish between 1890 and 1900. But El Paso surpassed all other cities in the state. This city produced twenty-two newspapers in Spanish during that period. El Paso is located on the border with Mexico and has always had a large population of Spanish speakers. Thus, it is unsurprising that this city became such a rich site for Spanish-language journalism.

Which choice best states the main purpose of the text?

- A) To compare Spanish-language newspapers published in Texas today with ones published there during the late 1800s
- B) To explain that Spanish-language newspapers thrived in Texas and especially in El Paso during the late 1800s
- C) To argue that Spanish-language newspapers published in El Paso influenced the ones published in San Antonio during the late 1800s
- D) To explain why Spanish-language newspapers published in Texas were so popular in Mexico during the late 1800s

8

Chile's Atacama Desert is one of the driest places on Earth. Mary Beth Wilhelm and other astrobiologists search for life, or its remains, in this harsh place because the desert closely mirrors the extreme environment on Mars. The algae and bacteria found in Atacama's driest regions may offer clues about Martian life. By studying how these and other microorganisms survive such extreme conditions on Earth, Wilhelm's team hopes to determine whether similar life might have existed on Mars and to develop the best tools to look for evidence of it.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) To contrast the conditions in the Atacama Desert with those on Mars
- B) To explain why many life-forms cannot survive in the Atacama Desert
- C) To indicate why astrobiologists choose to conduct research in the Atacama Desert
- D) To describe certain limitations to conducting scientific study in the Atacama Desert

9

More than 60% of journeys in Mexico City occur via public transit, but simply reproducing a feature of the city's transit system—e.g., its low fares—is unlikely to induce a significant increase in another city's transit ridership. As Erick Guerra et al. have shown, transportation mode choice in urban areas of Mexico is the product of a complex mix of factors, including population density, the spatial distribution of jobs, and demographic characteristics of individuals. System features do affect ridership, of course, but there is an irreducibly contextual dimension of transportation mode choice.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It presents an objection to the argument of Guerra et al. about transportation mode choice in urban areas of Mexico.
- B) It explains why it is challenging to influence transit ridership solely by altering characteristics of a transit system.
- C) It illustrates the claim that a characteristic associated with high transit ridership in Mexico City is not associated with high transit ridership elsewhere.
- D) It substantiates the assertion that population density, the spatial distribution of jobs, and demographic characteristics are important factors in transportation mode choice.

10

Changes to vegetation cover and other human activities influence carbon and nitrogen levels in soil, though how deep these effects extend is unclear. Hypothesizing that differences in land use lead to differences in carbon and nitrogen levels that are not restricted to the topsoil layer (0–30 cm deep), Chukwuebuka Okolo and colleagues sampled soils across multiple land-use types (e.g., grazing land, cropland, forest) within each of several Ethiopian locations. They found, though, that across land-use types, carbon and nitrogen decreased to comparably low levels beyond depths of 30 cm.

Which choice best describes the overall structure of the text?

- A) It describes a phenomenon that scientists do not fully understand, explains a research team's hypothesis about that phenomenon, and then describes a finding that led the team to refine the hypothesis.
- B) It introduces an unresolved scientific question, presents a research team's hypothesis pertaining to that question, and then describes an observation made by the team that conflicts with that hypothesis.
- C) It discusses a process that scientists are somewhat unclear about, introduces competing hypotheses about that process, and then explains how a research team concluded that one of those hypotheses is likely correct.
- D) It explains a hypothesis that has been the subject of scientific debate, discusses how a research team tested that hypothesis, and then presents data the team collected that validate the hypothesis.

11

Recently, scientists looked at data collected by NASA’s InSight lander to learn more about seismic activity on Mars, known as marsquakes. The data show that the marsquakes all started from the same location on the planet. This discovery was surprising to scientists, as they expected that the marsquakes would originate from all over the planet because of the cooling of the planet’s surface. Now, scientists believe that there could be areas of active magma flows deep beneath the planet’s surface that trigger the marsquakes.

According to the text, what was surprising to scientists studying the seismic activity data from NASA’s InSight lander?

- A) The surface temperature of Mars has been rising.
- B) There were different types of seismic waves causing marsquakes.
- C) NASA’s InSight lander collected less data than scientists had expected.
- D) All the marsquakes started from the same location on the planet.

12

The ancient writing system used in the Maya kingdoms of southern Mexico and Central America had a symbol for the number zero. The earliest known example of the symbol dates to more than 2,000 years ago. At that time, almost none of the writing systems elsewhere in the world possessed a zero symbol. And the use of zero in Mexico and Central America may be even more ancient. Some historians suggest that Maya mathematicians inherited it from the Olmec civilization, which flourished in the region 2,400–3,600 years ago.

According to the text, what do some historians suggest about Maya civilization?

- A) Maya civilization acquired the use of zero from the Olmec civilization.
- B) Maya civilization respected its historians more than it respected its mathematicians.
- C) Maya civilization was highly secretive about its intellectual achievements.
- D) Maya civilization tried to introduce its writing system to other civilizations.

13

“The Bet” is an 1889 short story by Anton Chekhov. In the story, a banker is described as being very upset about something: _____

Which quotation from “The Bet” most effectively illustrates the claim?

- A) “Then the banker cautiously broke the seals off the door and put the key in the keyhole.”
- B) “It struck three o’clock, the banker listened; everyone was asleep in the house and nothing could be heard outside but the rustling of the chilled trees.”
- C) “The banker, spoilt and frivolous, with millions beyond his reckoning, was delighted at the bet.”
- D) “When [the banker] got home he lay on his bed, but his tears and emotion kept him for hours from sleeping.”

14

Partial List of Candidate Species for De-extinction

Common name	Scientific name	Became extinct
Huia	<i>Heteralocha acutirostris</i>	1907
Caribbean monk seal	<i>Monachus tropicalis</i>	1952
Passenger pigeon	<i>Ectopistes migratorius</i>	1914
Saber-toothed cat	<i>Smilodon</i>	11,000 years before present
Woolly mammoth	<i>Mammuthus primigenius</i>	6,400 years before present

The passage of time is among the many obstacles faced by scientists who are pursuing de-extinction efforts—that is, efforts to use breeding or a mixture of cloning and genetic engineering to bring back extinct species. Specifically, researchers are concerned that the longer a species has been extinct, the less likely it is that a suitable habitat still exists for that species. Among candidate species for de-extinction, this problem would be especially concerning for the _____

Which choice most effectively uses data from the table to complete the statement?

- A) passenger pigeon (*Ectopistes migratorius*), which became extinct only a few years after the huia (*Heteralocha acutirostris*).
- B) saber-toothed cat (*Smilodon*), which became extinct 11,000 years ago.
- C) woolly mammoth (*Mammuthus primigenius*), which became extinct several thousand years before the saber-toothed cat (*Smilodon*).
- D) Caribbean monk seal (*Monachus tropicalis*), which became extinct in 1952.

15

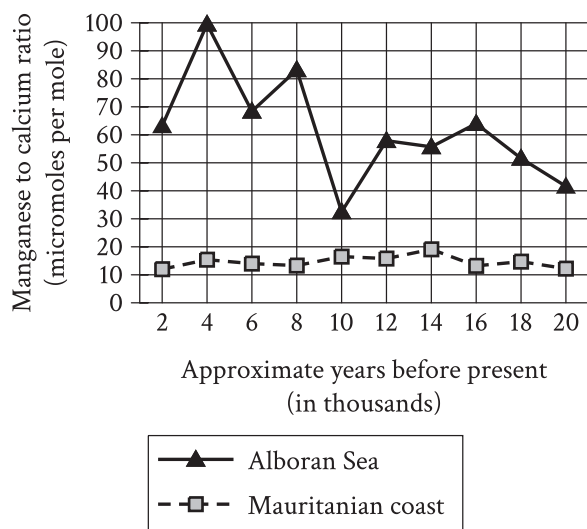
“The Yellow Wallpaper” is an 1892 short story by Charlotte Perkins Gilman. In the story, the narrator expresses mixed feelings about her surroundings:

Which quotation from “The Yellow Wallpaper” most effectively illustrates the claim?

- A) “This wallpaper has a kind of sub-pattern in a different shade, a particularly irritating one, for you can only see it in certain lights, and not clearly then.”
- B) “By moonlight—the moon shines in all night when there is a moon—I wouldn’t know it was the same paper.”
- C) “I’m really getting quite fond of the big room, all but that horrid [wall]paper.”
- D) “The color is repellant, almost revolting; a smouldering, unclean yellow, strangely faded by the slow-turning sunlight.”

16

Ratio of Manganese to Calcium in Samples
from Alboran Sea and Mauritanian Coast



The population of the coral *Lophelia pertusa* declined significantly around 9,000 years ago in the Alboran Sea and around 11,000 years ago near the Mauritanian coast. Using the ratio of manganese to calcium, which inversely correlates with ocean oxygenation levels, marine scientist Rodrigo da Costa Portilho-Ramos and colleagues evaluated whether oxygenation played a role in the declines of *L. pertusa*. The researchers concluded that oxygenation may have been important in the Alboran Sea but not the Mauritanian coast, since _____

Which choice most effectively uses data from the graph to complete the statement?

- A) a substantial increase in oxygenation in the Alboran Sea corresponded with the local decline in *L. pertusa*, but the opposite relationship between oxygenation and *L. pertusa* was found near the Mauritanian coast.
- B) *L. pertusa* declined in the Alboran Sea during a period of substantial local decline in oxygenation, but *L. pertusa* declined near the Mauritanian coast during a period of little local change in oxygenation.
- C) oxygenation in the Alboran Sea was higher before the decline in *L. pertusa* than after the decline, whereas oxygenation near the Mauritanian coast was relatively low both before and after the decline in *L. pertusa*.
- D) oxygenation in the Alboran Sea tended to be substantially higher than oxygenation near the Mauritanian coast during the period studied.

17

The Younger Dryas was a period of extreme cooling from 11,700 to 12,900 years ago in the Northern Hemisphere. Some scientists argue that a comet fragment hitting Earth brought about the cooling. Others disagree, partly because there is no known crater from such an impact that dates to the beginning of the period. In 2015, a team led by Kurt Kjær detected a 19-mile-wide crater beneath a glacier in Greenland. The scientists who believe an impact caused the Younger Dryas claim that this discovery supports their view. However, Kjær's team hasn't yet been able to determine the age of the crater. Therefore, the team suggests that _____

Which choice most logically completes the text?

- A) it can't be concluded that the impact that made the crater was connected to the beginning of the Younger Dryas.
- B) it can't be determined whether a comet fragment could make a crater as large as 19 miles wide.
- C) scientists have ignored the possibility that something other than a comet fragment could have made the crater.
- D) the scientists who believe an impact caused the Younger Dryas have made incorrect assumptions about when the period began.

18

In 2016 biological anthropologist Heather F. Smith and her team investigated the evolution of the appendix, an intestinal organ that is present in some mammals, including humans, but is generally thought to have no function. Studying 533 mammal species, the team found that the appendix has emerged independently across multiple lineages in separate instances and, significantly, hasn't disappeared after emerging in specific lineages. Moreover, the team determined that species with the organ tend to have higher concentrations of lymphoid tissue, which supports immune responses, in the cecum, the organ the appendix is attached to. Therefore, the team hypothesized that the appendix likely _____

Which choice most logically completes the text?

- A) was once present in many nonmammal species but has since disappeared from those lineages.
- B) has been preserved in certain mammal species because it benefits their immune systems.
- C) will emerge in a greater number of mammal species because it may serve a necessary function in the immune system.
- D) produced higher concentrations of lymphoid tissue in mammals in the past than it does currently.

19

Some ethicists hold that the moral goodness of an individual's actions depends solely on whether the actions themselves are good, irrespective of the context in which they are carried out. Philosopher L. Sebastian Purcell has shown that surviving works of Aztec (Nahua) philosophy express a very different view. Purcell reveals that these works posit an ethical system in which an individual's actions are judged in light of how well they accord with the individual's role in society and how well they contribute to the community. To the extent that these works are representative of Aztec thought, Purcell's analysis suggests that _____

Which choice most logically completes the text?

- A) the Aztecs would have disputed the idea that the morality of an individual's actions can be assessed by appealing to standards of behavior that are independent of the individual's social circumstances.
- B) the Aztecs would not have accepted the notion that the morality of an individual's actions can be fairly evaluated by people who do not live in the same society as that individual.
- C) actions by members of Aztec society who contributed a great deal to their community could be judged as morally good even if those actions were inconsistent with behaviors the Aztecs regarded as good in all contexts.
- D) similar actions performed by people in different social roles in Aztec society would have been regarded as morally equivalent unless those actions led to different outcomes for the community.

20

Lê Lươ ng Minh became the thirteenth secretary-general of the Association of Southeast Asian Nations (ASEAN) in January 2013, making _____ the first time the organization would appoint a Vietnamese leader.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) these
- B) those
- C) this
- D) some

21

In 1929, Edwin Herbert Land invented a polarizing filter that was featured in a number of products, from sunglasses to 3D movies. A decade later, Land _____ his technology to invent the world's first instant camera, the Polaroid Land camera.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) used
- B) to have used
- C) to use
- D) using

22

An online content creator who uses copyrighted songs without permission risks being demonetized (prohibited from including paid advertisements in content). The best way to avoid demonetization is to choose music from the public domain. Using one of these noncopyrighted songs _____ a creator won't lose advertising revenue.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are ensuring
- B) have ensured
- C) ensure
- D) ensures

23

What makes the theremin a unique musical instrument? You play it without touching it. When you place your _____ the pitch will shift as your hands move through the air.

Which choice completes the text with the most logical transition?

- A) hand's between the two antenna's,
- B) hands between the two antennas,
- C) hands' between the two antennae's,
- D) hands' between the two antennas,

24

In the music video for the song "We Didn't Start the Fire" by Billy Joel, the singer lists 118 political and cultural references. Such iconic references, cited in rapid and frenetic procession by the musician, who is seated impassively at a dinner table, _____ key moments and personalities of the twentieth century.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) represents
- B) has represented
- C) was representing
- D) represent

25

Long attributed to Jacques-Louis David, the preeminent Neoclassical painter of his day, the 1801 painting *Marie Joséphine Charlotte du Val d'Ognes* gained fresh attention in the 1990s when art historians discovered that the painting—which depicts a solitary young woman sketching—was actually the work of little-known French portrait _____ Marie-Denise Villers (1774–1821).

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) artist—
- B) artist
- C) artist:
- D) artist,

26

In 1986, conceptual artist Sophie Calle asked twenty-three people, all of whom had been born without sight, to describe “their image of beauty” in rich detail. Calle paired excerpts of these conversations with photographs—both of interviewees and the items they _____ to powerful effect in her exhibition *The Blind*.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) described, from hair to grass to sculptures
- B) described, from hair to grass to sculptures—
- C) described—from hair to grass to sculptures,
- D) described: from hair to grass to sculptures

27

Celebrated Tewa potter Maria Martinez (1887–1980) made her signature all-black ceramic vessels using a heating technique called reduction firing. This technique involves smothering the flame surrounding the clay vessel. _____ the vessel takes on a shiny, black hue.

Which choice completes the text with the most logical transition?

- A) On the contrary,
- B) For example,
- C) Previously,
- D) As a result,

28

Historians agree that the jazz pianist Jelly Roll Morton was exaggerating when he claimed to have invented jazz music. No one can deny, _____ that Morton’s innovative compositions and remarkable improvisational skills helped shape jazz as a genre during its early years.

Which choice completes the text with the most logical transition?

- A) therefore,
- B) in the second place,
- C) in other words,
- D) though,

29

According to Duverger’s law, countries with single-ballot majoritarian elections for single-member districts tend to polarize into two-party systems, wherein dueling political parties consistently dominate the political system. _____ countries with proportional-representation electoral systems tend to support multi-partyism, under which power gets distributed among many political parties.

Which choice completes the text with the most logical transition?

- A) Subsequently,
- B) Conversely,
- C) For instance,
- D) In other words,

30

A turtle shell appears external to the animal, protecting its body like armor. _____ the shell is often incorrectly assumed to be an exoskeleton, a rigid outer casing like that of a crustacean or an insect, when in fact it is an endoskeleton, a part of the turtle's internal bone structure, more akin to a spine or a pair of ribs.

Which choice completes the text with the most logical transition?

- A) That being said,
- B) However,
- C) For instance,
- D) Hence,

31

While researching a topic, a student has taken the following notes:

- In 1859, the novel *Adam Bede* was published in England.
- According to the novel's title page, the author's name was George Eliot.
- George Eliot was widely assumed to be a pseudonym.
- A pseudonym is a fake name used to conceal an author's identity.
- A woman named Mary Ann Evans later revealed herself as the novel's real author.

The student wants to identify the real author of *Adam Bede*. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The real author of *Adam Bede* was Mary Ann Evans, who published the novel using the pseudonym George Eliot.
- B) George Eliot, which *Adam Bede*'s title page indicated was the name of the novel's author, was widely assumed to be a pseudonym.
- C) The title page of the novel *Adam Bede* indicated that the author's name was George Eliot.
- D) A woman who had used a pseudonym to conceal her identity later revealed herself as the real author of *Adam Bede*.

32

While researching a topic, a student has taken the following notes:

- Scientists have developed a “freeze-thaw” battery that can retain 92% of its charge after twelve weeks.
- The battery contains molten salt (a type of salt that liquifies when heated and solidifies at room temperature).
- When the salt is in a liquid state, energy flows through the battery.
- When the salt is in a solid state, energy stops flowing and is stored in the battery.
- The stored (frozen) energy can be used by reheating (thawing) the battery.

The student wants to specify how the salt enables energy storage. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Scientists have developed a freeze-thaw battery that contains molten salt, which liquifies when heated and solidifies at room temperature.
- B) The stored energy in a freeze-thaw battery, which contains molten salt, can be used by reheating the battery.
- C) When the molten salt in a freeze-thaw battery solidifies at room temperature, energy stops flowing and can be stored in the battery.
- D) Molten salt allows a freeze-thaw battery to retain 92% of its charge after twelve weeks.

33

While researching a topic, a student has taken the following notes:

- The US Fish and Wildlife Service (FWS) keeps a list of all at-risk species.
- Species on the list are classified as either endangered or threatened.
- A species that is in danger of extinction throughout most or all of its range is classified as endangered.
- A species that is likely to soon become endangered is classified as threatened.
- The California red-legged frog (*Rana draytonii*) is likely to soon become endangered, according to the FWS.

The student wants to indicate the California red-legged frog’s FWS classification category. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Species on the FWS list, which includes the California red-legged frog (*Rana draytonii*), are classified as either endangered or threatened.
- B) The California red-legged frog (*Rana draytonii*) appears on the FWS list of at-risk species.
- C) According to the FWS, the California red-legged frog is in the endangered category, in danger of extinction throughout most or all of its range.
- D) Likely to soon become endangered, the California red-legged frog is classified as threatened by the FWS.

STOP

If you finish before time is called, you may check your work on this module only.

Do not turn to any other module in the test.

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The following text is adapted from Elizabeth von Arnim's 1922 novel *The Enchanted April*. Mrs. Wilkins and her friend Rose are traveling in Italy.

"I'm going to have one of these gorgeous oranges," said Mrs. Wilkins, staying where she was and reaching across to a black bowl piled with them. "Rose, how can you resist them. Look—have this one. Do have this beauty—" And she held out a big one.

As used in the text, what does the phrase "reaching across to" most nearly mean?

- A) Joining with
- B) Gaining on
- C) Stretching toward
- D) Arriving at

2

A team of paleontologists has found a rich fossil deposit near Gulgong, Australia. The fossils are so well preserved that the team has been able to _____ detailed information about the life forms that left them behind, such as color patterns and how they interacted with other species.

Which choice completes the text with the most logical and precise word or phrase?

- A) occupy
- B) hoard
- C) reserve
- D) obtain

3

According to a team of neuroeconomists from the University of Zurich, ease of decision making may be linked to communication between two brain regions, the prefrontal cortex and the parietal cortex. Individuals tend to be more decisive if the information flow between the regions is intensified, whereas they make choices more slowly when information flow is _____.

Which choice completes the text with the most logical and precise word or phrase?

- A) reduced
- B) evaluated
- C) determined
- D) acquired

4

The War of 1812 has _____ place in historical memory in Britain, partly because it is overshadowed by the much larger concurrent conflict against Napoleonic France and partly because it essentially maintained the geopolitical status quo for Britain: the country neither gained nor lost significant territory or position as a result of its participation in the war.

Which choice completes the text with the most logical and precise word or phrase?

- A) a tenuous
- B) an enduring
- C) a contentious
- D) a conspicuous

5

Claims about the original significance of Minoan bull-leaping rituals—depicted in paintings and sculptures from the second millennium BCE—are difficult to successfully _____. We know so little about the people archaeologists call the Minoans that assertions about what bull-leaping meant to them will almost inevitably rely on significant speculation and guesswork.

Which choice completes the text with the most logical and precise word or phrase?

- A) imagine
- B) summarize
- C) defend
- D) adjust

6

The following text is adapted from Jerome K. Jerome's 1889 novel *Three Men in a Boat (To Say Nothing of the Dog)*. The narrator is traveling by boat with Harris and another friend.

[Harris] told us anecdotes of how he had gone across the [English] Channel when it was so rough that the passengers had to be tied into their [beds], and he and the captain were the only two living souls on board who were not ill. Sometimes it was he and the second mate who were not ill; but it was generally he and one other man. If not he and another man, then it was he by himself.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It indicates the reason for Harris's eagerness to resume traveling.
- B) It hints at Harris's feeling that during an earlier boat trip, others didn't include him in activities.
- C) It emphasizes that Harris always boasts about his own constitution when speaking of a previous boat trip.
- D) It reveals that although Harris claims to prefer solitary activities when traveling, he actually enjoys having company.

7

Several studies have found negligible electoral consequences for governments that impose fiscal austerity measures, yet some European governments recently suffered electorally due to their austerity programs. Evelyn Huebscher and colleagues attribute this incongruity to governments' tendency—not followed in the recent European cases—to implement austerity programs strategically to avoid electoral costs (e.g., setting spending cuts to take effect only after the next election), which has obscured the inherent political risks of austerity measures in the election data scholars have examined.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It explains a discrepancy between what has been observed in study settings and what has been observed in real-world settings that the text goes on to assert is attributable to the studies not using real-world data.
- B) It identifies a conflict between research findings and recent events that the text goes on to suggest is a consequence of a complicating factor in the data used to generate those findings.
- C) It presents a long-standing divergence in research findings that the text goes on to say is due to different groups of researchers using data that derive from different electoral circumstances.
- D) It describes a recent exception to a general pattern in research findings that the text goes on to explain is a result of researchers underestimating the significance of inconsistencies in the data they've analyzed.

8

Text 1

Graphic novels are increasingly popular in bookstores and libraries, but they shouldn't be classified as literature. By definition, literature tells a story or conveys meaning through language only; graphic novels tell stories through illustrations and use language only sparingly, in captions and dialogue. Graphic novels are experienced as series of images and not as language, making them more similar to film than to literature.

Text 2

Graphic novels present their stories through both language and images. Without captions and dialogue, readers would be unable to understand what is depicted in the illustrations: the story results from the interaction of text and image. Moreover, Alison Bechdel's *Fun Home* and many other graphic novels feature text that is as beautifully written as the prose found in many standard novels. Therefore, graphic novels qualify as literary texts.

Based on the texts, how would the author of Text 2 most likely respond to the overall argument presented in Text 1?

- A) By asserting that language plays a more important role in graphic novels than the author of Text 1 recognizes
- B) By acknowledging that the author of Text 1 has identified a flaw that is common to all graphic novels
- C) By suggesting that the story lines of certain graphic novels are more difficult to understand than the author of Text 1 claims
- D) By agreeing with the author of Text 1 that most graphic novels aren't as well crafted as most literary works are

9

The following text is adapted from Ann Petry's 1946 novel *The Street*. Lutie lives in an apartment in Harlem, New York.

The glow from the sunset was making the street radiant. The street is nice in this light, [Lutie] thought. It was swarming with children who were playing ball and darting back and forth across the sidewalk in complicated games of tag. Girls were skipping double dutch rope, going tirelessly through the exact center of a pair of ropes, jumping first on one foot and then the other.

©1946 by Ann Petry

Which choice best describes what is happening in the text?

- A) Lutie is observing the appearance of the street at a particular time of day and the events occurring on it.
- B) Lutie is annoyed by the noise of children playing games on her street.
- C) Lutie is puzzled by the rules of certain children's games.
- D) Lutie is spending time alone in her apartment because she doesn't want to interact with her neighbors.

10

Since its completion in 2014, Bosco Verticale (Vertical Forest)—a pair of residential towers in Milan, Italy, covered by vegetation—has become a striking symbol of environmental sustainability in architecture. Stefano Boeri intended his design, which features balconies that are home to hundreds of trees, to serve as a model for promoting urban biodiversity. However, the concept has faced skepticism: critics note that although the trees used in Bosco Verticale were specifically cultivated for the project, it's too early to tell if they can thrive in this unusual setting.

According to the text, why are some critics skeptical of the concept behind Bosco Verticale?

- A) Some essential aspects of Bosco Verticale's design are difficult to adapt to locations other than Milan.
- B) The plant life on Bosco Verticale ended up being less varied than Boeri had envisioned it would be.
- C) The construction of Bosco Verticale was no less environmentally damaging than the construction of more conventional buildings is.
- D) It is unclear whether Bosco Verticale can support the plant life included in its design.

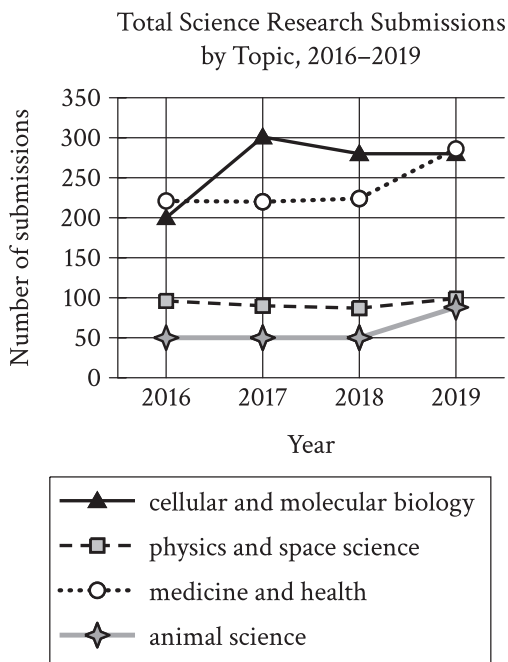
11

Many literary theorists distinguish between *fabula*, a narrative's content, and *syuzhet*, a narrative's arrangement and presentation of events. In the film *The Godfather Part II*, the *fabula* is the story of the Corleone family, and the *syuzhet* is the presentation of the story as it alternates between two timelines in 1901 and 1958. But literary theorist Mikhail Bakhtin maintained that *fabula* and *syuzhet* are insufficient to completely describe a narrative—he held that systematic categorizations of artistic phenomena discount the subtle way in which meaning is created by interactions between the artist, the work, and the audience.

Which choice best states the main idea of the text?

- A) Literary theorist Mikhail Bakhtin argued that there are important characteristics of narratives that are not fully encompassed by two concepts that other theorists have used to analyze narratives.
- B) Literary theorist Mikhail Bakhtin claimed that meaning is not inherent in a narrative but is created when an audience encounters a narrative so that narratives are interpreted differently by different people.
- C) The storytelling methods used in *The Godfather Part II* may seem unusually complicated, but they can be easily understood when two concepts from literary theory are utilized.
- D) Narratives that are told out of chronological order are more difficult for audiences to understand than are narratives presented chronologically.

12



A student is researching the trends in the topics submitted to a national science fair for high school students. The graph shows the number of submissions by topic that were made each year. Based on the data in the graph, the student claims that there were more medicine and health research topics submitted in 2019 than in any other year.

Which choice most effectively uses data from the graph to support the underlined claim?

- A) In 2016, the number of cellular and molecular biology topic submissions was the same as the number of animal science topic submissions.
- B) In 2019, there were more physics and space science topic submissions than there were medicine and health topic submissions.
- C) The lowest number of animal science topic submissions in a year was approximately 95 in 2016.
- D) The highest number of medicine and health topic submissions during the period shown is approximately 285 in 2019.

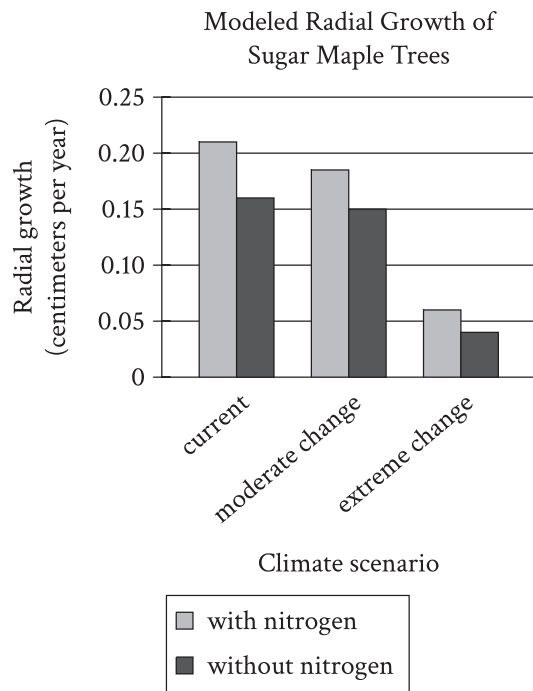
13

Fish whose DNA has been modified to include genetic material from other species are known as transgenic. Some transgenic fish have genes from jellyfish that result in fluorescence (that is, they glow in the dark). Although these fish were initially engineered for research purposes in the 1990s, they were sold as pets in the 2000s and can now be found in the wild in creeks in Brazil. A student in a biology seminar who is writing a paper on these fish asserts that their escape from Brazilian fish farms into the wild may have significant negative long-term ecological effects.

Which quotation from a researcher would best support the student's assertion?

- A) "In one site in the wild where transgenic fish were observed, females outnumbered males, while in another the numbers of females and males were equivalent."
- B) "Though some presence of transgenic fish in the wild has been recorded, there are insufficient studies of the impact of those fish on the ecosystems into which they are introduced."
- C) "The ecosystems into which transgenic fish are known to have been introduced may represent a subset of the ecosystems into which the fish have actually been introduced."
- D) "Through interbreeding, transgenic fish might introduce the trait of fluorescence into wild fish populations, making those populations more vulnerable to predators."

14



Inés Ibáñez and colleagues studied a forest site in which some sugar maple trees receive periodic fertilization with nitrogen to mimic the broader trend of increasing anthropogenic nitrogen deposition in soil. Ibáñez and colleagues modeled the radial growth of the trees with and without nitrogen fertilization under three different climate scenarios (the current climate, moderate change, and extreme change). Although they found that climate change would negatively affect growth, they concluded that anthropogenic nitrogen deposition could more than offset that effect provided that change is moderate rather than extreme.

Which choice best describes data from the graph that support Ibáñez and colleagues' conclusion?

- A) Growth with nitrogen under the current climate exceeded growth with nitrogen under moderate change, but the latter exceeded growth without nitrogen under extreme change.
- B) Growth without nitrogen under the current climate exceeded growth without nitrogen under moderate change, but the latter exceeded growth with nitrogen under extreme change.
- C) Growth with nitrogen under moderate change exceeded growth without nitrogen under moderate change, but the latter exceeded growth without nitrogen under extreme change.
- D) Growth with nitrogen under moderate change exceeded growth without nitrogen under the current climate, but the latter exceeded growth with nitrogen under extreme change.

15

“Poetry” is a 1919 poem by Marianne Moore. The poem highlights an ambivalence toward poetry as the speaker acknowledges its merits while also expressing a sense of displeasure, writing _____

Which quotation from “Poetry” most effectively illustrates the claim?

- A) “nor is it valid / to discriminate against ‘business documents and / school-books’; all these phenomena are important.”
- B) “One must make a distinction / however: when dragged into prominence by half poets, the result is not / poetry”
- C) “when [poems] become so derivative as to become unintelligible, the / same thing may be said for all of us—that we / do not admire what / we cannot understand.”
- D) “Reading [poetry], however, with a perfect contempt for it, one discovers that there is in / it after all, a place for the genuine.”

Simulated Change in Annual Aquifer Input and Irrigation Output if
Precipitation Concentration Increases as Climate Models Predict

Baseline concentration of annual precipitation	% change in water entering aquifers	% change in surface water used for irrigation	% change in groundwater used for irrigation
Precipitation is currently somewhat concentrated	4.9	0.4	0.9
Precipitation is currently evenly distributed	11.0	9.0	7.9

Some climate models for the western United States predict that while total annual precipitation may remain unchanged from the present level, precipitation will become concentrated into fewer but more intense rain and snow events. University of Texas climate scientist Geeta Persad and her colleagues simulated how the amount of water entering aquifers and the amount being used for irrigation purposes would change if this were to occur. Persad and her colleagues concluded that concentration of precipitation into fewer events would result in a higher number of dry days, triggering more irrigation, but that this change in irrigation output is highly sensitive to the baseline concentration of precipitation that currently exists in an area.

Which choice best describes data from the table that support Persad and her colleagues' conclusion?

- A) If baseline precipitation is somewhat concentrated, the amount of water being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater, whereas the amount of water entering aquifers will increase 11.0% if baseline precipitation is evenly distributed.
- B) If baseline precipitation is somewhat concentrated, water use for irrigation will increase only slightly, whereas it will increase 9.0% for surface water and 7.9% for groundwater if baseline precipitation is evenly distributed.
- C) If baseline precipitation is somewhat concentrated, the amount of water entering aquifers will increase 4.9%, while the amount being used for irrigation will increase 0.4% for surface water and 0.9% for groundwater.
- D) If baseline precipitation is somewhat concentrated, water use for irrigation will decline by a small amount, whereas it will increase 11.0% for surface water and 9.0% for groundwater if baseline precipitation is evenly distributed.

17

In dialects of English spoken in Scotland, the “r” sound is strongly emphasized when it appears at the end of syllables (as in “car”) or before other consonant sounds (as in “bird”). English dialects of the Upland South, a region stretching from Oklahoma to western Virginia, place similar emphasis on “r” at the ends of syllables and before other consonant sounds. Historical records show that the Upland South was colonized largely by people whose ancestors came from Scotland. Thus, linguists have concluded that _____

Which choice most logically completes the text?

- A) the English dialects spoken in the Upland South acquired their emphasis on the “r” sound from dialects spoken in Scotland.
- B) emphasis on the “r” sound will eventually spread from English dialects spoken in the Upland South to dialects spoken elsewhere.
- C) the English dialects spoken in Scotland were influenced by dialects spoken in the Upland South.
- D) people from Scotland abandoned their emphasis on the “r” sound after relocating to the Upland South.

18

How did whales, once no bigger than seals, evolve to become the largest animals on Earth? Brazilian biologist Mariana Nery believes the answer might be found in whales’ DNA. In January 2023, Nery and her colleagues _____ a study showing changes over time in four whale genes associated with body size.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) published
- B) publishing
- C) having published
- D) to publish

19

A harpsichord may look just like a piano, but the difference between the two instruments is easy to hear. When a harpsichord’s keys are pressed, the strings inside the _____ are plucked, not struck.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) instrument:
- B) instrument
- C) instrument—
- D) instrument,

20

When they were first introduced to western Europe from Byzantium in the eleventh century, table forks were met with much resistance. The Bishop of Ostia, St. Peter Damian, condemned the eating utensils because he considered _____ dangerous and unnecessary luxury items.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) them
- B) this
- C) that
- D) it

21

Julia Alvarez’s 1994 novel *In the Time of the Butterflies*, a fictionalized account of the lives of the Mirabal _____ can serve as a starting point for those wanting to explore how the rule of dictator Rafael Trujillo has been represented in Dominican American literature.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) sisters, and
- B) sisters and
- C) sisters,
- D) sisters

22

On March 23, 2021, a gust of wind wreaked havoc on global trade. *Ever Given*, an international shipping container vessel, became lodged in Egypt’s Suez Canal, a major shipping route between Europe and Asia. The vessel took six days to _____ it’s as heavy as two thousand blue whales when fully loaded.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) dislodge in part due to its sheer size,
- B) dislodge, in part due to its sheer size:
- C) dislodge, in part due to its sheer size,
- D) dislodge, in part, due to its sheer size

23

Fans of the film *Moana* (2016) may not know that the deep and humorous voice behind the _____ belongs to comedian, actor, and musician Jemaine Clement. The versatile performer has appeared in everything from television commercials to action movies, but voice acting, specifically, has become a notable part of his career.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) character Tamatoa the crab
- B) character Tamatoa the crab,
- C) character: Tamatoa the crab,
- D) character, Tamatoa the crab

24

In her 1983 book *The Managed Heart: Commercialization of Human Feeling*, sociologist Arlie Russell Hochschild first explored at length her conception of a “sociology of emotions”—the idea that the various cultural and ideological frameworks a person has internalized (class, gender, political affiliation, etc.) _____ each emotional reaction that person has within a situation.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) underlies
- B) is underlying
- C) underlie
- D) has been underlying

25

American abstract artist Richard _____ his installations to make passersby keenly aware of how one’s movements are affected by the physical features of one’s environment, assembles large-scale steel plates into sculptures that dominate the outdoor spaces they occupy.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Serra is intending
- B) Serra, intends
- C) Serra, intending
- D) Serra intends

26

In 1949, Frank Zamboni developed an ice rink resurfacing machine. As Zamboni's machine moved along the rink's surface, it first scraped off the top layer of ice. _____ it sprayed water into the deep grooves left behind by customers' skates. Lastly, it smoothed over the newly formed ice.

Which choice completes the text with the most logical transition?

- A) For example,
- B) Next,
- C) Similarly,
- D) In contrast,

27

In 2014, Nestor Gomez won his first-ever storytelling competition, relating a tale about his life as a Guatemalan immigrant living in Chicago. _____ in 2017, Gomez created the show *80 Minutes Around the World* as a platform for others to share stories about their immigration experiences.

Which choice completes the text with the most logical transition?

- A) Instead,
- B) For example,
- C) Later,
- D) In other words,

28

To guarantee the validity of experimental results, scientists rely on precise, unchanging standards of measurement. _____ metrologists (scientists who study measurement) developed the SI, or International System of Units. The SI's units of measurement are based on unchanging values in nature, such as the mass of an electron or the speed of light.

Which choice completes the text with the most logical transition?

- A) In contrast,
- B) Regardless,
- C) In addition,
- D) For this reason,

29

In retrospect, one of the lessons of the 2003 Human Genome Project is that a gene is affected by many factors, not the least of which is its interactions with the protein products of other genes. _____ rather than just focusing on the human genome, efforts to better understand gene mutations related to disease have begun to consider the human proteome, the complete set of proteins expressed by human genes.

Which choice completes the text with the most logical transition?

- A) In other words,
- B) That said,
- C) For example,
- D) Accordingly,

30

While researching a topic, a student has taken the following notes:

- Iranian scholar Abu Rayhan al-Biruni studied Earth's physical features.
- He theorized that a large landmass existed west of Europe and east of Asia.
- Al-Biruni published his landmass theory in 1037 CE.

The student wants to specify when al-Biruni published his landmass theory. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) In 1037 CE, al-Biruni published his theory that a large landmass existed west of Europe and east of Asia.
- B) Al-Biruni, who studied Earth's physical features, published a theory about a large landmass.
- C) Al-Biruni was an Iranian scholar who studied Earth's physical features.
- D) An Iranian scholar who studied Earth's physical features, al-Biruni theorized that a large landmass existed west of Europe and east of Asia.

31

While researching a topic, a student has taken the following notes:

- In astronomy, the mass of stars can be described in units called solar masses.
- One solar mass is roughly equal to the mass of the Sun.
- The mass of the star Proxima Centauri is 0.122 solar masses.
- The mass of the star Sirius A is 2.063 solar masses.

The student wants to emphasize the mass of Sirius A. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The mass of stars, like Proxima Centauri, can be described in units called solar masses.
- B) In astronomy, the mass of stars can be described in units called solar masses, and one solar mass is roughly equal to the mass of the Sun.
- C) The Sun is more massive than Proxima Centauri, which has a mass of 0.122 solar masses.
- D) With a mass of 2.063 solar masses, Sirius A is more massive than the Sun.

32

While researching a topic, a student has taken the following notes:

- A lever is a simple machine consisting of a rigid beam and a fulcrum.
- The fulcrum is the point about which the beam pivots.
- The input force (effort) is the force applied to the lever.
- The output force (load) is the force that the lever exerts on another object.
- In first-class levers, the fulcrum is located between the effort and the load.
- In second-class levers, the load is located between the effort and the fulcrum.

The student wants to contrast first-class levers and second-class levers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) In levers, the effort is the force applied to the lever; the load, in contrast, is the force that the lever exerts on another object.
- B) In first-class and second-class levers, the fulcrum and the load are in different locations.
- C) First-class levers are simple machines consisting of a rigid beam and a fulcrum, but then again, the same is true of second-class levers.
- D) In first-class levers, the fulcrum is located between the effort and the load, but in second-class levers, the load is located between the effort and the fulcrum.

33

While researching a topic, a student has taken the following notes:

- Earthquakes start at a point called a “focus” and spread out from there as seismic waves.
- The two types of seismic waves that travel beneath Earth’s surface are primary waves (P waves) and secondary waves (S waves).
- P waves travel more quickly beneath Earth’s surface than do S waves.
- P waves compress and expand the ground, causing it to move backward and forward.
- S waves cause the ground to move from side to side.

The student wants to emphasize a similarity between P waves and S waves. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) P waves and S waves both travel beneath Earth’s surface, causing the ground to move.
- B) P waves travel away from an earthquake’s starting point at a higher rate of speed than do S waves.
- C) Spreading out from the focus of an earthquake, P waves move the ground backward and forward.
- D) Although P waves and S waves start at the same point, they behave very differently.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**



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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Though not closely related, the hedgehog tenrecs of Madagascar share basic _____ true hedgehogs, including protective spines, pointed snouts, and small body size—traits the two groups of mammals independently developed in response to equivalent roles in their respective habitats.

Which choice completes the text with the most logical and precise word or phrase?

- A) examples of
- B) concerns about
- C) indications of
- D) similarities with

2

In editor Lisa Yaszek's introduction to her anthology *The Future Is Female! More Classic Science Fiction Stories by Women*, Yaszek identifies an increasing sense of _____ feminist mode of writing in the 1970s, in contrast to many woman-authored science fiction stories of the 1920s to 1960s whose politics were less deliberately signaled.

Which choice completes the text with the most logical and precise word or phrase?

- A) a prudently
- B) an overtly
- C) a cordially
- D) an inadvertently

3

_____ the long-standing trend of overemphasizing teenagers and young adults in research on social media use, scholars have recently begun to expand their focus to include the fastest-growing cohort of social media users: senior citizens.

Which choice completes the text with the most logical and precise word or phrase?

- A) Exacerbating
- B) Redressing
- C) Epitomizing
- D) Precluding

4

The following text is adapted from James Baldwin's 1956 novel *Giovanni's Room*. The narrator is riding in a taxi down a street lined with food vendors and shoppers in Paris, France.

The multitude of Paris seems to be dressed in blue every day but Sunday, when, for the most part, they put on an unbelievably festive black. Here they were now, in blue, disputing, every inch, our passage, with their wagons, handtrucks, their bursting baskets carried at an angle steeply self-confident on the back.

©1956 by James Baldwin

As used in the text, what does the word “disputing” most nearly mean?

- A) Arguing about
- B) Disapproving of
- C) Asserting possession of
- D) Providing resistance to

5

While recent scholarship has undermined claims that the works of twelfth-century Islamic philosopher Ibn Rushd were _____ other Muslim philosophers of his time, it is indisputable that his location in the Muslim-ruled area of what is now Spain meant that his works were primarily available thousands of miles west of the era's center of Islamic thought.

Which choice completes the text with the most logical and precise word or phrase?

- A) controversial among
- B) antagonistic toward
- C) imitated by
- D) inconsequential to

6

On painter William H. Johnson's return to the United States in 1938 after a decade in Europe, his style underwent an abrupt transformation. Turning away from landscapes painted in an expressionist style—a style that often involves using fluid, distorted shapes and thick, textured brushstrokes to express the artist's subjective experience of reality—Johnson began painting portraits of Black Americans in a bold new way. Evocative of African sculpture and American and Scandinavian folk art, these portraits feature flat, deliberately oversimplified figures in a vibrant but limited color palette.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It elaborates on the previous sentence's statement about a transitional moment in Johnson's artistic career.
- B) It provides information about Johnson's travels in support of a claim about his artistic influences, which is advanced in the following sentence.
- C) It recounts a moment in Johnson's personal life that enabled the success of his subsequent career, which is summarized in the following sentence.
- D) It presents evidence that calls into question the previous sentence's characterization of Johnson's artistic development.

7

When classical pianist Martha Argerich performs, it appears as if the music is coming to her spontaneously. She's highly skilled technically, but because of how freely she plays and her willingness to take risks, she seems relaxed and natural. Her apparent ease, however, is due to a tremendous amount of preparation. Despite Argerich's experience and virtuosity, she never takes for granted that she knows a piece of music. Instead, she approaches the music as if encountering it for the first time and tries to understand it anew.

Which choice best states the main purpose of the text?

- A) To provide details about how Argerich identifies which pieces of music she will perform
- B) To assert that Argerich's performances look effortless because of how she prepares for them
- C) To discuss the kinds of music Argerich feels most comfortable encountering for the first time
- D) To describe the unique way that Argerich approaches music she hasn't performed before

8

The following text is adapted from Herman Melville's 1855 novel *Israel Potter*. Israel is a young man wandering through New England during the late eighteenth century.

He hired himself out for three months; at the end of that time to receive for his wages two hundred acres of land lying in New Hampshire. [...] His employer proving false to the contract in the matter of the land, and there being no law in the country to force him to fulfil it, Israel—who, however brave-hearted, and even much of a dare-devil upon a pinch, seems nevertheless to have evinced, throughout many parts of his career, a singular patience and mildness—was obliged to look round for other means of livelihood than clearing out a farm for himself in the wilderness.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It implies that Israel treasures a particular characteristic of his personality when that characteristic should usually be regarded as a flaw.
- B) It suggests that if not for a certain aspect of his character, Israel might not have been as easily thwarted in his ambition to establish a farm.
- C) It shows why Israel would not have been able to undertake the enormous amount of labor necessary to run a farm even if he had owned the necessary property.
- D) It explains why, when the situation requires it, Israel is able to undertake courageous acts that others would generally avoid.

9

Text 1

In 1954 George Balanchine choreographed a production of *The Nutcracker*, a ballet by Pyotr Ilyich Tchaikovsky. It has since become a tradition for hundreds of dance companies in North America to stage *The Nutcracker* each year. But the show is stuck in the past, with an old-fashioned story and references, so it should no longer be produced. Ballet needs to create new traditions if it wants to stay relevant to contemporary audiences.

Text 2

The Nutcracker is outdated, but it should be kept because it's a holiday favorite and provides substantial income for some dance companies. Although it can be behind the times, there are creative ways to update the show. For example, Debbie Allen successfully modernized the story. Her show *Hot Chocolate Nutcracker* combines ballet, tap, hip-hop, and other styles, and it has been gaining in popularity since it opened in 2009.

Based on the texts, how would the author of Text 2 most likely respond to the underlined claim in Text 1?

- A) By questioning the idea that the story of *The Nutcracker* is stuck in the past and by rejecting the suggestion that contemporary audiences would enjoy an updated version
- B) By agreeing that contemporary audiences have largely stopped going to see performances of *The Nutcracker* because it's so old-fashioned
- C) By pointing out that most dance companies could increase their incomes by offering modernized versions of *The Nutcracker*
- D) By suggesting that dance companies should consider offering revised versions of *The Nutcracker* instead of completely rejecting the show

10

To understand how Paleolithic artists navigated dark caves, archaeologist M^a Ángeles Medina-Alcaide and her team tested different lighting methods in a cave in Spain using replicas of artifacts found in European caves with art. They used three different Paleolithic light sources—torches, animal-fat lamps, and fireplaces—determining that each likely had a specific purpose. For instance, the team learned that the animal-fat lamps were less useful than torches while walking because the lamps didn't illuminate the cave floor.

Which choice best states the main idea of the text?

- A) Medina-Alcaide and her team's study demonstrated that fireplaces were essential to the creators of Paleolithic cave art.
- B) Medina-Alcaide and her team discovered that Paleolithic cave artists in Spain used animal-fat lamps more often than they used torches.
- C) Medina-Alcaide and her team were reluctant to draw many conclusions from their study because of the difficulty they had replicating light sources based on known artifacts.
- D) Medina-Alcaide and her team tested Paleolithic light sources and learned some details about how Paleolithic artists traveled within dark caves.

11

Annual Car Production in the United States, 1910–1925

Year	Number of cars produced	Number of companies producing cars
1910	123,990	320
1915	548,139	224
1920	1,651,625	197
1925	3,185,881	80

A student is using the table as part of a social studies class presentation on the US auto industry in the early twentieth century. The student notes that, according to the table, from 1910 to 1925 _____

Which choice most effectively uses data from the table to complete the statement?

- A) the number of cars produced increased but the number of companies producing cars decreased.
- B) both the number of cars produced and the number of companies producing cars remained unchanged.
- C) the number of cars produced decreased but the number of companies producing cars remained unchanged.
- D) both the number of cars produced and the number of companies producing cars increased.

12

External shopping cues are a type of marketing that uses obvious messaging—a display featuring a new product, for example, or a “buy one, get one free” offer—to entice consumers to make spontaneous purchases. In a study, data scientist Sam K. Hui and colleagues found that this effect can also be achieved with a less obvious cue: rearranging a store’s layout. The researchers explain that trying to find items in new locations causes shoppers to move through more of the store, exposing them to more products and increasing the likelihood that they’ll buy an item they hadn’t planned on purchasing.

Which response from a survey given to shoppers who made a purchase at a retail store best supports the researchers’ explanation?

- A) “I needed to buy some cleaning supplies, but they weren’t in their regular place. While I was looking for them, I saw this interesting notebook and decided to buy it, too.”
- B) “I didn’t buy everything on my shopping list today. I couldn’t find a couple of the items in the store, even though I looked all over for them.”
- C) “The store sent me a coupon for a new brand of soup, so I came here to find out what kinds of soup that brand offers. I decided to buy a few cans because I had the coupon.”
- D) “This store is larger than one that’s closer to where I live, and it carries more products. I came here to buy some things that the other store doesn’t always have.”

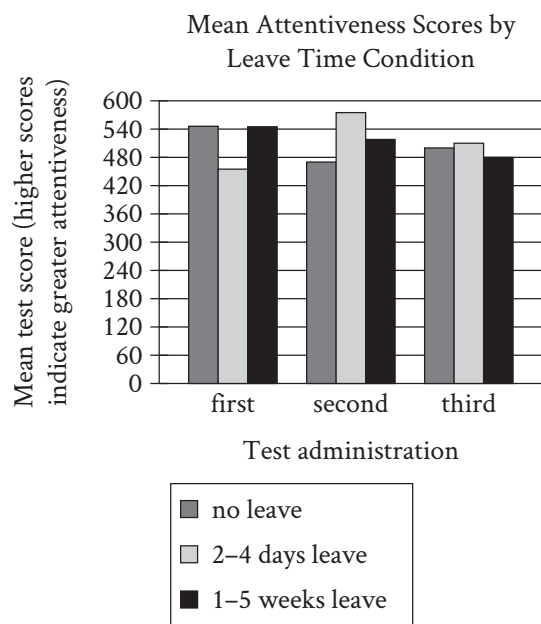
13

The 2021 exhibition *This Is the Day* at Arkansas’s Crystal Bridges Museum of American Art featured works dealing with expressions of faith and spirituality in the Black community. The museum’s 2022 exhibition *The Dirty South*, meanwhile, focused on Black culture in the American South from 1920 to 2020, with a particular focus on the intersections between visual arts and music. Together, these exhibitions don’t merely highlight the diversity of the Black experience in the US; they also showcase the diverse media through which artists have depicted and engaged with that experience.

Which statement about the exhibitions, if true, would most directly support the underlined claim?

- A) Between them, *This Is the Day* and *The Dirty South* included drawings, paintings, photographs, sculptures, textiles, videos, costumes, and music.
- B) *This Is the Day* included works by fewer than two dozen artists, whereas *The Dirty South* included works by more than 80 artists.
- C) *This Is the Day* exclusively included works in the permanent collection of the museum, whereas *The Dirty South* included works from multiple sources outside the museum.
- D) Between them, *This Is the Day* and *The Dirty South* included works depicting more than 300 years of Black experience in the United States.

14

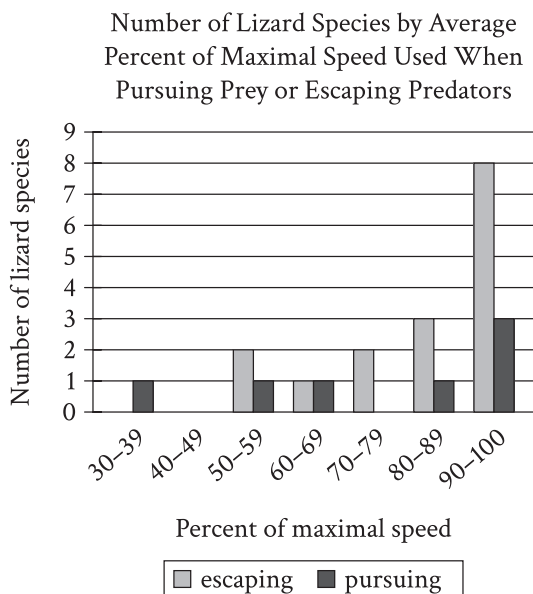


To investigate potential cognitive benefits of taking leave from work, psychologist Jan Packer and colleagues conducted a six-month study of Australian university staff members who took no leave from work during the study, took 2–4 days of leave, or took 1–5 weeks of leave. Tests of attentiveness were administered to participants three times during the study: at random for the no-leave staff, and for the rest, one week before their leave, one week following their return to work, and one week after the second test administration. After analyzing the results, the researchers concluded that longer leave times might not confer a greater cognitive benefit than shorter leave times do.

Which choice best describes data from the graph that support the researchers' conclusion?

- A) In the second test administration, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave, but in the third test administration, those who took no leave had higher average scores than those who took 1–5 weeks of leave.
- B) In the first test administration, participants who took 2–4 days of leave had lower average attentiveness scores than did those who took 1–5 weeks of leave and those who took no leave.
- C) In both the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did participants who took 1–5 weeks of leave.
- D) In the second and third test administrations, participants who took 2–4 days of leave had higher average attentiveness scores than did those who took no leave.

15



It may seem that the optimal strategy for an animal pursuing prey or escaping predators is to move at maximal speed, but the energy expense of exploiting full speed capacity can disfavor such a strategy even in escape contexts, as evidenced by the fact that

Which choice most effectively uses data from the graph to complete the text?

- A) most lizard species use about the same percentage of their maximal speed when escaping predation as they do when pursuing prey.
- B) multiple lizard species move at an average of less than 90% of their maximal speed while escaping predation.
- C) more lizard species use, on average, 90%–100% of their maximal speed while escaping predation than use any other percentage of their maximal speed.
- D) at least 4 lizard species use, on average, less than 100% of their maximal speed while pursuing prey.

16

Under normal atmospheric pressure at Earth's surface, water molecules form a tetrahedral network stabilized by hydrogen bonds between adjacent molecules. Extreme high pressure, such as can be found in deep ocean waters, destabilizes these bonds and compresses water's structure, allowing water molecules within organisms to permeate proteins and impede crucial biological functions; yet deep-sea organisms known as piezophiles have adapted to extreme pressure. Studies have found a positive correlation between the depths that various piezophiles inhabit and concentrations of a compound called trimethylamine N-oxide (TMAO) in their muscle tissues, which has led a team of researchers to hypothesize that TMAO reduces water's compressibility.

Which finding, if true, would most directly support the researchers' hypothesis?

- A) Water molecules are found to be impervious to TMAO even when the water molecules' tetrahedral configuration has been distorted by high pressure.
- B) Examination of TMAO's molecular structure shows that TMAO molecules retain their shape even as pressure increases.
- C) A positive correlation is found between concentrations of TMAO and the rate at which water's molecular structure compresses as pressure increases.
- D) Analysis of water's molecular structure under high pressure reveals that hydrogen bonds are more stable when TMAO is present than when it is not.

17

The Cretaceous pterosaur *Tupandactylus navigans* is known for having an anomalously oversized head crest. Until an almost complete fossil skeleton was found in Brazil, paleontologists had been able to study only skull specimens from *T. navigans*, though it was presumed that, like other pterosaurs, the species's primary form of locomotion was powered flight. Examining the fuller skeleton in 2016, Victor Beccari and his team determined that *T. navigans* had long hind legs, short wings, and an unusually long neck—characteristics that, combined with the creature's large-crested head, would have made sustained flight difficult and walking upright relatively comfortable. Based on these findings the team suggests that *T. navigans* likely _____

Which choice most logically completes the text?

- A) flew for longer distances than did other pterosaur species that had oversized head crests.
- B) had longer wings than other pterosaur species considered to have been comfortable walking.
- C) had a smaller head than researchers expected based on the earlier *T. navigans* skull specimens.
- D) flew for shorter distances and spent more time walking than researchers previously thought.

18

Consumer psychologists have theorized that the likelihood that people who identify as ethical consumers—meaning that they strive to purchase goods and services with positive or neutral social and ecological effects—will purchase a given product positively correlates with their perception of that product's effects. In a recent study of the attitudes of self-identified ethical consumers toward purchasing a specific mobile phone coming to market, researchers found that, on average, study participants in their twenties rated the phone's social and ecological effects much less positively than did participants in other age groups. All other things being equal, if consumer psychologists' theory is correct, this finding suggests that _____

Which choice most logically completes the text?

- A) the phone is less appealing to ethical consumers in their twenties than other similar phones on the market are.
- B) ethical consumers in their twenties are less likely to purchase the phone than ethical consumers in other age groups are.
- C) there is not a meaningful difference in the likelihood of purchasing the phone among ethical consumers in different age groups.
- D) ethical consumers in their twenties are more likely than ethical consumers in other age groups to consider a phone's social and ecological effects when deciding whether to purchase that phone.

19

The Boston Saloon was one of the most popular African American–owned establishments in nineteenth-century Nevada. _____ by businessman William A.G. Brown, the saloon was known to offer elegant accommodations and an inclusive environment.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Created
- B) Creates
- C) Creating
- D) Create

20

Louise Bennett (1919–2006), also known as “Miss Lou,” was an influential Jamaican poet and folklorist. Her innovative poems _____ the use of Jamaican Creole (a spoken language) in literature.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) popularized;
- B) popularized,
- C) popularized
- D) popularized:

21

“He was just the man for such a place, and it was just the place for such a man.” This line is from Frederick Douglass’s autobiography *Narrative of the Life of Frederick Douglass* (1845). It’s an example of antimetabole, a writing technique that _____ emphasis by repeating a statement in a reversed order.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) create
- B) are creating
- C) have created
- D) creates

22

Researchers Amit Kumar and Nicholas Epley investigated how _____. In a series of experiments conducted in 2022, they found that people performing small acts of kindness underestimated the positive effect their actions had on others.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) do people perceive acts of kindness.
- B) do people perceive acts of kindness?
- C) people perceive acts of kindness?
- D) people perceive acts of kindness.

23

In a painting titled “The Milkmaid” by Johannes Vermeer, the artist prominently features a bread basket, milk pitcher, and bowl. Such quotidian objects, depicted in exquisite detail by Vermeer, a painter celebrated for his naturalism, _____ the daily minutiae of a seventeenth-century Dutch household.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) was revealing
- B) has revealed
- C) reveals
- D) reveal

24

Jamaican British artist Willard Wigan is known for his remarkable _____ so small that they are best viewed through a microscope, Wigan’s sculptures are made from tiny natural materials, such as spiderweb strands.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) microsculptures creations
- B) microsculptures, creations
- C) microsculptures. Creations
- D) microsculptures and creations

25

Consider the mechanics of the pinhole camera: light passes through a small hole, resulting in a focused projected image. A ray diagram reveals how this _____ the hole’s small size restricts light to a single ray, all light passing through the hole can only arrive at a single destination, eliminating diffraction and ensuring a clear image.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) works because
- B) works. Because
- C) works, it’s because
- D) works: it’s because

26

In the search for extraterrestrial life, astrobiologists Stuart Bartlett and Michael L. Wong propose that scientists avoid using the term “life.” _____ researchers should use another word: “lyfe.” This new term, they argue, could be used to draw distinctions between the known characteristics of life on Earth and the potentially differing characteristics of lyfe on other planets.

Which choice completes the text with the most logical transition?

- A) Previously,
- B) Regardless,
- C) There,
- D) Instead,

27

Before it unveiled a massive new gallery in 2009, the Art Institute of Chicago was only able to display about 5% of its art collection. _____ the museum is able to display close to 30% of its collection.

Which choice completes the text with the most logical transition?

- A) Additionally,
- B) For example,
- C) Nevertheless,
- D) Today,

28

Working together with the Navajo Nation Department of Water Resources, Dr. Lani Tsinnajinnie analyzed data about snowpack levels in the Chuska Mountains. She found that the snowpack (the amount of snow on the ground) was deepest in early March at lower elevations. At higher elevations, _____ the snowpack was deepest in mid-March.

Which choice completes the text with the most logical transition?

- A) in other words,
- B) for instance,
- C) on the other hand,
- D) in summary,

29

The Inca of South America used intricately knotted string devices called quipus to record countable information, like population data and payments. _____ they may have used quipus to record more complex information, like stories and myths, according to researchers.

Which choice completes the text with the most logical transition?

- A) As a result,
- B) In other words,
- C) In addition,
- D) For example,

30

In hindsight, given the ideas about the natural world circulating among British scientists in the 1800s, the theory of natural selection was an obvious next step. It may not have been a coincidence, _____ that Charles Darwin and Alfred Wallace arrived at the concept independently. Indeed, contrary to the popular myth of the lone genius, theirs is not the first paradigm-shifting theory to have emerged from multiple scholars working in parallel.

Which choice completes the text with the most logical transition?

- A) however,
- B) then,
- C) moreover,
- D) for example,

31

While researching a topic, a student has taken the following notes:

- In the 1930s, the Imperial Sugar Cane Institute in India sought to limit the country's dependence on imported sugarcane.
- The institute enlisted botanist Janaki Ammal to breed a local variety of sugarcane.
- She crossbred the imported sugarcane species *Saccharum officinarum* with grasses native to India.
- She succeeded in creating sugarcane hybrids well suited to India's climate.

The student wants to emphasize Janaki Ammal's achievement. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) By crossbreeding the imported sugarcane species *Saccharum officinarum* with grasses native to India, Ammal succeeded in creating sugarcane hybrids well suited to India's climate.
- B) In the 1930s, the Imperial Sugar Cane Institute, which enlisted Ammal, sought to limit dependence on imported sugarcane.
- C) Ammal was enlisted by the Imperial Sugar Cane Institute at a time when a local variety of sugarcane needed to be produced.
- D) As part of efforts to breed a local variety of sugarcane, an imported sugarcane species called *Saccharum officinarum* was crossbred with grasses native to India.

32

While researching a topic, a student has taken the following notes:

- Elizabeth Catlett's sculpture *Recognition* (1970) shows two African American figures with rounded, indistinct features.
- The figures reach out to each other in a pose that symbolizes a close, supportive relationship.
- Her sculpture *Students Aspire* (1978) shows two African American figures with sharply defined features.
- The figures hold an equal sign above their heads with one hand and embrace each other with the other hand.
- This pose symbolizes their support for each other in the pursuit of equality.

The student wants to emphasize a similarity between the two sculptures. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Catlett's *Students Aspire* depicts two figures supporting each other in the pursuit of equality.
- B) *Recognition* and *Students Aspire* both show African American figures in poses that symbolize supportive relationships.
- C) Catlett completed *Recognition* in 1970 and *Students Aspire* in 1978.
- D) The figures in *Recognition* have features that are rounded and indistinct, while the figures in *Students Aspire* have sharply defined features.

33

While researching a topic, a student has taken the following notes:

- The ancient Arab dhow was a sailing vessel distinguishable by its triangular sails and stitched hull construction.
- Dhows were used primarily for trade along the coasts of Arab, South Asian, and East African countries.
- Contemporary shipbuilders in Oman use a mix of modern and traditional materials to build replicas of ancient dhows.
- Most of the materials used are traditional.
- Replica hulls are stitched together using the same traditional coconut palm fiber rope used on the hulls of ancient dhows.

The student wants to make a generalization about the materials used in dhow replicas. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A traditional material that was used to stitch together the hulls of ancient dhows, coconut palm fiber rope is still used by shipbuilders.
- B) The ancient Arab dhow was a sailing vessel used primarily for trade and distinguishable by its triangular sails.
- C) Although most materials used in dhow replicas are traditional, some modern materials are used.
- D) Contemporary shipbuilders in Oman build replicas of the dhow, which was an ancient sailing vessel with a stitched hull construction.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The works of Chicana artist Ester Hernandez are now _____ in museums both in the United States and abroad, but the murals she contributed to as a member of Las Mujeres Muralistas early in her artistic career were displayed in outdoor public spaces across San Francisco.

Which choice completes the text with the most logical and precise word or phrase?

- A) invented
- B) adjusted
- C) featured
- D) recommended

2

Whether Carmen Lomas Garza is creating small paintings and illustrations or large public artworks—such as *Baile*, a copper cutout of traditional Mexican dance in the San Francisco International Airport—she is _____ direct experience, drawing from memories of her childhood in Texas or details of her current surroundings in California.

Which choice completes the text with the most logical and precise word or phrase?

- A) complimented by
- B) uncertain about
- C) unbothered by
- D) inspired by

3

Animal researcher Amalia P.M. Bastos led a 2021 study about a wild kea parrot that used small stones as tools to preen its feathers. Skeptical colleagues had initially suggested to Bastos that the kea's interactions with the stones might simply be _____, but Bastos and her team showed that the kea was using the stones deliberately.

Which choice completes the text with the most logical and precise word or phrase?

- A) intriguing
- B) obvious
- C) accidental
- D) observable

4

In 1891, design artist William Morris cofounded the Kelmscott Press, which printed editions of books using preindustrial methods. Historians argue that Morris's repudiation of industrialization is _____ the Kelmscott editions' use of handmade materials and intricate ornamentation reminiscent of medieval manuscripts: these meticulously handcrafted elements exemplify the artistry involved.

Which choice completes the text with the most logical and precise word or phrase?

- A) insensible to
- B) manifest in
- C) scrutinized by
- D) complicated by

5

Mary Engle Pennington, a chemist who helped advance home refrigeration, undoubtedly made a substantial impact on society, but her place in our historical memory is perhaps more _____ than that of Stephanie Kwolek, who invented the incredibly strong material known as Kevlar, an accomplishment for which she will long be remembered.

Which choice completes the text with the most logical and precise word or phrase?

- A) permanent
- B) tentative
- C) warranted
- D) prominent

6

The following text is from Betty Smith's 1943 novel *A Tree Grows in Brooklyn*. Francie, a young girl, visits the library often.

Francie thought that all the books in the world were in that library and she had a plan about reading all the books in the world. She was reading a book a day in alphabetical order and not skipping the dry ones. She remembered that the first author had been Abbott. She had been reading a book a day for a long time now and she was still in the B's. Already she had read about bees and buffaloes, Bermuda vacations and Byzantine architecture. For all her enthusiasm, she had to admit that some of the B's had been hard going. But Francie was a reader.

©1947 by Betty Smith

Which choice best states the main purpose of the text?

- A) To illustrate Francie's enjoyment of an unusual topic
- B) To explain why Francie prefers reading over other activities
- C) To portray Francie's determination to meet a goal
- D) To describe a book that Francie greatly admires

7

Researchers have long hypothesized that woolly mammoths were hunted to extinction in North America by humans using spears with grooved tips known as Clovis points. One anthropologist set out to test this hypothesis. Using a mechanical spear-thrower, he launched spears with Clovis points into mounds of clay—substitutes for the animals' large bodies. The projectiles generally penetrated only a few inches into the clay, an amount insufficient to have harmed most woolly mammoths. This led the anthropologist to conclude that hunters using spears with Clovis points likely weren't the principal drivers of the extinction.

Which choice best states the main purpose of the text?

- A) To argue for the significance of new findings amid an ongoing debate among researchers
- B) To discuss the advantages and disadvantages of the method used in an experiment
- C) To summarize two competing hypotheses and a major finding associated with each one
- D) To describe an experiment whose results cast doubt on an established hypothesis

8

The people of medieval Europe have traditionally been seen as uninterested in cleanliness and hygiene, but modern research has shown that this is largely a myth. According to historian Eleanor Janega, most medieval towns in Europe had at least one public bathhouse, which often offered both full-immersion baths and—more affordably—steam baths. While such amenities were available mainly to town dwellers, regular bathing in rivers and streams or daily sponge baths at home were common practices throughout medieval Europe.

Which choice best describes the function of the underlined portion?

- A) It asserts that in medieval Europe steam baths were more popular in rural areas than in urban ones.
- B) It describes a limitation of earlier historians' studies of medieval European bathing habits.
- C) It concedes that not all people in medieval Europe had access to public bathhouses.
- D) It explains why Janega decided to study the popularity of public bathhouses in medieval Europe.

9

Scholarly accounts of the Chicano movement—a movement that advocated for the social, political, and cultural empowerment of Mexican Americans and reached its zenith in the 1960s and 1970s—tend to focus on the most militant, outspoken figures in the movement, making it seem uniformly radical. Geographer Juan Herrera has shown, however, that if we shift our focus toward the way the movement manifested in comparatively low-profile neighborhood institutions and projects, we see participants espousing an array of political orientations and approaches to community activism.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It presents a trend in scholarship on the Chicano movement that the text claims has been reevaluated by researchers in light of Herrera’s work on the movement’s participants.
- B) It identifies an aspect of the Chicano movement that the text implies was overemphasized by scholars due to their own political orientations.
- C) It describes a common approach to studying the Chicano movement that, according to the text, obscures the ideological diversity of the movement’s participants.
- D) It summarizes the conventional method for analyzing the Chicano movement, which the text suggests creates a misleading impression of the effectiveness of neighborhood institutions and projects.

10

Elizabeth Asiedu has identified a negative correlation between the share of developing countries’ economies derived from natural-resource extraction and those countries’ receipts of foreign investment. This may appear counterintuitive—resource extraction requires initial investments (in extractive technology, for instance) at scales best met by multinational corporations—but Asiedu notes that natural-resource industries’ boom-bust cycle can destabilize local currencies and increase developing countries’ vulnerability to external shocks, creating levels of uncertainty to which foreign investors are typically averse.

Which choice best states the main idea of the text?

- A) Although it may seem surprising that foreign investment declines in developing countries as natural-resource extraction makes up a larger share of those countries’ economies, that decline happens because resource extraction requires initial investments too large for foreign investors to supply.
- B) Although developing countries tend to become less dependent on foreign investment as natural-resource industries make up a larger share of their economies, this change may not occur if the boom-bust cycle of those industries destabilizes local currencies or increases countries’ vulnerability to external shocks.
- C) Although one might expect that foreign investment would increase as natural-resource extraction makes up a larger share of developing countries’ economies, the opposite happens because heavy reliance on natural resources can lead to unattractive conditions for investors.
- D) Although foreign investors tend to avoid initial investments in natural-resource industries in developing countries, foreign investment may increase significantly as those industries stabilize and the risks associated with them decline.

11

To understand how temperature change affects microorganism-mediated cycling of soil nutrients in alpine ecosystems, Eva Kaštovská et al. collected plant-soil cores in the Tatra Mountains at elevations around 2,100 meters and transplanted them to elevations of 1,700–1,800 meters, where the mean air temperature was warmer by 2°C. Microorganism-mediated nutrient cycling was accelerated in the transplanted cores; crucially, microorganism community composition was unchanged, allowing Kaštovská et al. to attribute the acceleration to temperature-induced increases in microorganism activity.

It can most reasonably be inferred from the text that the finding about the microorganism community composition was important for which reason?

- A) It provided preliminary evidence that microorganism-mediated nutrient cycling was accelerated in the transplanted cores.
- B) It suggested that temperature-induced changes in microorganism activity may be occurring at increasingly high elevations.
- C) It ruled out a potential alternative explanation for the acceleration in microorganism-mediated nutrient cycling.
- D) It clarified that microorganism activity levels in the plant-soil cores varied depending on which microorganisms comprised the community.

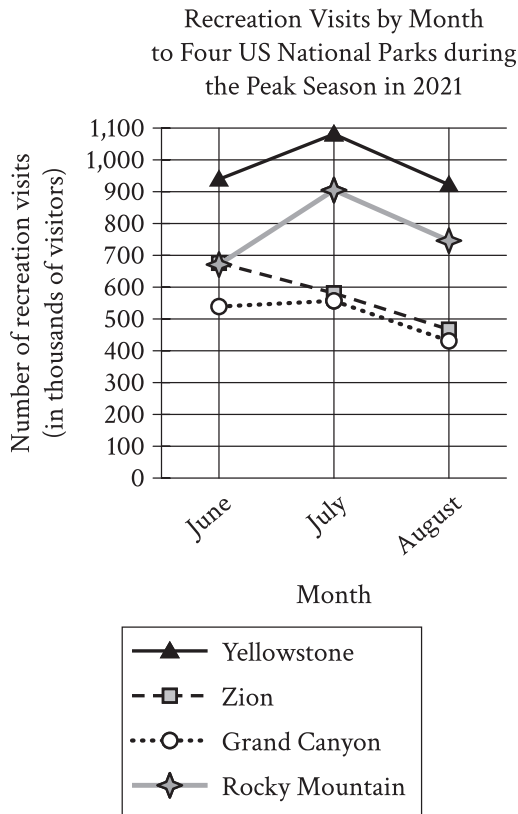
12

Some astronomers searching for extraterrestrial life have proposed that atmospheric NH_3 (ammonia) can serve as a biosignature gas—an indication that a planet harbors life. Jingcheng Huang, Sara Seager, and colleagues evaluated this possibility, finding that on rocky planets, atmospheric NH_3 likely couldn't reach detectably high levels in the absence of biological activity. But the team also found that on so-called mini-Neptunes—gas planets smaller than Neptune but with atmospheres similar to Neptune's—atmospheric pressure and temperature can be high enough to produce atmospheric NH_3 .

Based on the text, Huang, Seager, and colleagues would most likely agree with which statement about atmospheric NH_3 ?

- A) Its presence is more likely to indicate that a planet is a mini-Neptune than that the planet is a rocky planet that could support life.
- B) Its absence from a planet that's not a mini-Neptune indicates that the planet probably doesn't have life.
- C) It should be treated as a biosignature gas if detected in the atmosphere of a rocky planet but not if detected in the atmosphere of a mini-Neptune.
- D) It doesn't reliably reach high enough concentrations in the atmospheres of rocky planets or mini-Neptunes to be treated as a biosignature gas.

13



In 2021, four of the United States national parks that were among the most visited were Grand Canyon National Park, Rocky Mountain National Park, Yellowstone National Park, and Zion National Park. The graph shows the number of visits for recreation to each of these parks during the three-month period with the highest number of visitors. A student notes that among the parks shown in the graph, the park with the highest monthly recreation visits in all three months was _____

Which choice most effectively uses data from the graph to complete the text?

- A) Zion National Park.
- B) Rocky Mountain National Park.
- C) Yellowstone National Park.
- D) Grand Canyon National Park.

14

“Lines Written in Early Spring” is a 1798 poem by William Wordsworth. In the poem, the speaker describes having contradictory feelings while experiencing the sights and sounds of a spring day: _____

Which quotation from “Lines Written in Early Spring” most effectively illustrates the claim?

- A) “Through primrose-tufts, in that sweet bower, / The periwinkle trail’d its wreathes; / And ’tis my faith that every flower / Enjoys the air it breathes.”
- B) “The budding twigs spread out their fan, / To catch the breezy air; / And I must think, do all I can, / That there was pleasure there.”
- C) “The birds around me hopp’d and play’d: / Their thoughts I cannot measure, / But the least motion which they made, / It seem’d a thrill of pleasure.”
- D) “I heard a thousand blended notes, / While in a grove I [sat] reclined, / In that sweet mood when pleasant thoughts / Bring sad thoughts to the mind.”

Mean Ratings for Patients after 21 Days

Measure	Mean rating for participants aware of taking a placebo	Mean rating for participants in the control group
Global improvement	5.0	3.9
Symptom severity reduction	92.00	46.00
Quality of life improvement	11.4	5.4

To test whether a medication is effective, scientists compare outcomes for patients taking it and patients taking a placebo (a medically inactive substance). Patients normally aren't told they're receiving a placebo, but a research team conducted a study to see if there might be a medical benefit to telling them. The team used various measures to evaluate participants, with higher ratings indicating greater well-being in each measure. Compared to the mean ratings after 21 days for participants in the control group, the mean ratings for participants who were aware of taking a placebo _____

Which choice most effectively uses data from the table to complete the statement?

- A) ranged from 5.0 to 92.00, indicating that well-being varied widely from participant to participant.
- B) were lower for two measures, with the rating for only one measure indicating greater well-being for these participants.
- C) ranged from 3.9 to 46.00, with no rating indicating greater well-being in any measure for these participants.
- D) were higher for all three measures, indicating greater overall well-being for these participants.

16

As media consumption has become increasingly multiplatform and socially mediated, active news acquisition has diminished in favor of an attitude known as “news finds me” (NFM), in which people passively rely on their social networks and ambient media environments for information about current events. Homero Gil de Zúñiga and Trevor Diehl examined data on a representative group of adults in the United States to determine participants’ strength of NFM attitude, political knowledge, and political interest. Although no major election took place sufficiently near the study for Gil de Zúñiga and Diehl to identify causality between NFM and voting behavior, they did posit that NFM may reduce voting probability through an indirect effect.

Which finding, if true, would most directly support the idea advanced by Gil de Zúñiga and Diehl?

- A) NFM attitude tends to increase in strength as major elections approach, and people are significantly more likely to vote in major elections than in minor elections.
- B) NFM attitude has a strong negative effect on political knowledge and interest, and there is known to be a strong positive correlation between political knowledge and interest and the likelihood of voting.
- C) Political interest is known to have a strong positive effect on likelihood of voting but shows only a weak positive effect on political knowledge, and NFM attitude shows little correlation with either political knowledge or political interest.
- D) The likelihood of voting increases as political knowledge increases, and the relationship between NFM attitude and political knowledge tends to strengthen as the size of people’s social networks increases.

17

The practice of logging (cutting down trees for commercial and other uses) is often thought to be at odds with forest conservation (the work of preserving forests). However, a massive study in forest management and preservation spanning 700,000 hectares in Oregon’s Malheur National Forest calls that view into question. So far, results of the study suggest that forest plots that have undergone limited logging (the careful removal of a controlled number of trees) may be more robust than plots that haven’t been logged at all. These results, in turn, suggest that _____

Which choice most logically completes the text?

- A) logging may be useful for maintaining healthy forests, provided it is limited.
- B) other forest management strategies are more effective than limited logging.
- C) as time passes, it will be difficult to know whether limited logging has any benefits.
- D) the best way to support forest health may be to leave large forests entirely untouched.

18

Even with the widespread adoption of personal computers, many authors still choose to write and revise their novels by hand and only then transcribe the final version on a computer. It may be tempting to speculate about how a novel written this way would be affected if it had been exclusively typed instead, but each novel is a unique entity resulting from a specific set of circumstances. Therefore,

Which choice most logically completes the text?

- A) in order to increase their efficiency, authors who currently write their novels largely by hand should instead work only on a computer.
- B) authors who do most of their drafting and revising by hand likely have more success than those who work entirely on a computer.
- C) novels written by hand take less time to produce, on average, than novels written on a computer do.
- D) there is no way to reasonably evaluate how a work would be different if it had been written by other means.

19

In forecasting weather events, meteorologists sometimes discuss the role of atmospheric rivers. What are atmospheric rivers, and how _____ Part of the water cycle, atmospheric rivers are narrow channels of moisture moving through the atmosphere. In certain conditions, these “rivers” can release some of their moisture as precipitation.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) do they affect our weather.
- B) they do affect our weather.
- C) do they affect our weather?
- D) they do affect our weather?

20

One of the few African American global explorers during the turn of the 20th century, _____

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Matthew Henson made several treks across Greenland between 1891 and 1909.
- B) 1891 and 1909 were the years between which Matthew Henson made several treks across Greenland.
- C) Greenland was where Matthew Henson made several treks between 1891 and 1909.
- D) several treks across Greenland were made by Matthew Henson between 1891 and 1909.

21

Woven from recycled yarn and hand tufted using a carpet weaving technique passed down by the artist’s Turkish grandmother, _____ so lush and tactilely inviting that you are tempted to reach out and touch them.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) the topological tapestries of Argentine textile artist Alexandra Kehayoglou are
- B) the Argentine textile artist Alexandra Kehayoglou creates topological tapestries that are
- C) when she creates her topological tapestries, Argentine textile artist Alexandra Kehayoglou makes them
- D) Alexandra Kehayoglou is an Argentine textile artist whose topological tapestries are

22

Physical materials can be classified by how much light passes through them. Clear glass, which is classified as transparent, allows all (or almost all) light to pass _____ wax paper, which is classified as translucent, allows only some light to pass through.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) through,
- B) through
- C) through;
- D) through and

23

Using natural debris, such as dried _____ such as plastic bags; and more traditional art supplies, such as tree glue, Ghanaian artist Ed Franklin Gavua creates his striking Yiiiikakaii African masks, which he hopes can help viewers rethink how waste is used in their communities.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) leaves, man-made trash:
- B) leaves; man-made trash,
- C) leaves, man-made trash,
- D) leaves; man-made trash;

24

Latin America is known to have dozens, if not hundreds, of popular dance forms. Only five of these dances are included in international ballroom dance _____ rumba, samba, cha-cha-cha, paso doble, and jive—the last of which is grouped with the other Latin dances despite not having Latin roots.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) competitions, however:
- B) competitions, however,
- C) competitions, however;
- D) competitions; however,

25

For thousands of years, humans have used domesticated goats (*Capra hircus*) to clear land of unwanted vegetation. When it comes to their diets, goats are notoriously _____ they will devour all kinds of shrubs and weeds, leaving virtually no part of any plant unconsumed.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) indiscriminate and
- B) indiscriminate,
- C) indiscriminate
- D) indiscriminate:

26

A species of *Byropsis* algae produces toxins to avoid being eaten by predators. However, in some cases, the toxins the organism uses to protect itself from predation actually _____ its attractiveness to predators. The Hawaiian sea slug, for example, not only tolerates *Byropsis* toxins but actually uses them for protection in the same way the algae does.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) is increasing
- B) increase
- C) increases
- D) has increased

27

After appropriate permissions are granted, a typical archaeological dig begins with a surveyor making a detailed grid of the excavation site. Then, the site is carefully dug, and any artifacts found are recorded and mapped onto the site grid. _____ the artifacts are removed, cataloged, and analyzed in a laboratory.

Which choice completes the text with the most logical transition?

- A) For instance,
- B) On the contrary,
- C) Earlier,
- D) Finally,

28

The liquid metals in Earth's core circulate constantly, and this circulation generates electrical currents that flow between Earth's North and South magnetic poles. These electrical currents, _____ create a barrier around Earth that protects us from radiation and charged particles coming from space.

Which choice completes the text with the most logical transition?

- A) in turn,
- B) likewise,
- C) nevertheless,
- D) in reality,

29

Biographer Michael Gorra notes that the novelist Henry James "lived in a world of second thoughts," frequently tinkering with his novels and stories after their initial publication. However, the differences between the 1881 first edition and the 1908 edition of his novel *A Portrait of a Lady* are extreme, even by James's standards; _____ some critics regard the two editions as two different novels altogether.

Which choice completes the text with the most logical transition?

- A) by contrast,
- B) in fact,
- C) nevertheless,
- D) in other words,

30

While researching a topic, a student has taken the following notes:

- In World War I, US soldiers who were members of the Choctaw Nation in Oklahoma participated in the Choctaw Code Talkers program.
- The Choctaw Code Talkers were trained to relay coded military information in their native language.
- In World War II, the US Army recruited Navajo (Diné) soldiers to transmit coded messages in their native language.
- These soldiers were known as the Navajo Code Talkers.

The student wants to emphasize a similarity between the Choctaw Code Talkers and the Navajo Code Talkers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) US soldiers who were members of the Choctaw Nation in Oklahoma used their native language to relay coded information.
- B) In World War II, one group of Navajo (Diné) soldiers was known as the Navajo Code Talkers.
- C) Both the Choctaw Code Talkers and the Navajo Code Talkers transmitted coded military messages in the soldiers' native languages.
- D) The Choctaw Code Talkers, not the Navajo Code Talkers, served in World War I.

31

While researching a topic, a student has taken the following notes:

- Meteorites found on Earth are divided into two categories.
- A meteorite that was observed falling to Earth before being recovered is known as a meteorite fall.
- All other meteorites found on Earth are known as meteorite finds.
- There have been about 1,200 recorded meteorite falls.
- There have been over 60,000 recorded meteorite finds.

The student wants to contrast the number of meteorite falls with the number of meteorite finds. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A meteorite that was observed falling to Earth before being recovered is known as a meteorite fall; all others are known as meteorite finds.
- B) Meteorites found on Earth are divided into two categories: meteorite falls and meteorite finds.
- C) There have been about 1,200 recorded meteorite falls, or meteorites observed falling to Earth.
- D) While there have been only about 1,200 recorded meteorite falls, there have been over 60,000 meteorite finds.

32

While researching a topic, a student has taken the following notes:

- Doña María do Carmo Bandeira was a Brazilian botanist.
- Between 1924 and 1941, she collected approximately 800 botanical samples.
- She collected a sample of *Polytrichum juniperinum* from Serra de Itatiaia in Mauá in February of 1925.
- She collected a sample of *Sphagnum gracilescen* from Ponte do Inferno in Corcovado in March of 1925.
- *Polytrichum juniperinum* and *Sphagnum gracilescen* are both species of moss.

The student wants to emphasize the sample collected from Serra de Itatiaia. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Doña María do Carmo Bandeira was a botanist notable for collecting approximately 800 botanical samples between 1924 and 1941.
- B) Among the many botanical samples Doña María do Carmo Bandeira collected was *Polytrichum juniperinum*, a species of moss she collected from Serra de Itatiaia in 1925.
- C) Between 1924 and 1941, Doña María do Carmo Bandeira collected many botanical samples, such as *Polytrichum juniperinum* from Serra de Itatiaia and *Sphagnum gracilescen* from Ponte do Inferno.
- D) Between 1924 and 1941, Doña María do Carmo Bandeira collected samples of *Polytrichum juniperinum* and *Sphagnum gracilescen*, both species of moss.

33

While researching a topic, a student has taken the following notes:

- The US government classifies sensitive information according to the degree to which disclosure could affect the nation's security.
- Information that could cause "damage" to national security is classified as Confidential.
- Information that could cause "serious damage" to national security is classified as Secret.
- Most routine diplomatic correspondence, if disclosed, could cause damage but not serious damage to national security.
- Diplomatic correspondence includes communication with both allies and adversaries.

The student wants to indicate which category most routine diplomatic correspondence belongs in, based on how sensitive information is classified. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) According to the US government, which classifies such sensitive information as routine diplomatic correspondence, Confidential information could damage national security if disclosed.
- B) Most routine diplomatic correspondence is classified according to the degree to which disclosure could affect the nation's security.
- C) Having the potential to damage national security if disclosed, most routine diplomatic correspondence is classified as Confidential.
- D) If disclosed, communication with both allies and adversaries could affect the nation's security.

STOP

**If you finish before time is called, you may check your work on this module only.
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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

On the basis of extensive calculations and models, astronomers in the 1990s predicted that the collision of two neutron stars or a neutron star and a black hole could release a massive burst of gamma rays in an event called a kilonova. This _____ was confirmed with observations in 2017.

Which choice completes the text with the most logical and precise word or phrase?

- A) theory
- B) evidence
- C) constant
- D) experiment

2

The following text is from John Muir's 1913 autobiography *The Story of My Boyhood and Youth*. Muir describes being on a boat.

The water was so clear that it was almost invisible, and when we floated slowly out over the plants and fishes, we seemed to be miraculously sustained in the air while exploring a veritable fairyland.

As used in the text, what does the word "clear" most nearly mean?

- A) Simple
- B) Understandable
- C) Obvious
- D) Transparent

3

At the turn of the twentieth century, Black residents of Richmond, Virginia, had few formal options for banking and other financial services. To _____ this situation, Maggie Lena Walker chartered the St. Luke Penny Savings Bank in 1903. The bank went on to provide home loans and savings opportunities to thousands of Black families over the following decades.

Which choice completes the text with the most logical and precise word or phrase?

- A) prolong
- B) rectify
- C) retain
- D) highlight

4

The results of randomized clinical trials testing the efficacy of common medical interventions sometimes fail to _____ conclusions that practitioners reach based on their real-world observations of patients. While there are several possible reasons for this, one is that practitioners may overlook confounding variables that account for the results they attribute to the interventions in question.

Which choice completes the text with the most logical and precise word or phrase?

- A) circumvent
- B) corroborate
- C) disseminate
- D) implement

5

Diadromous fish migrate between freshwater and marine biomes during their life cycle. The migration's obligate nature is why diadromous fish can be _____ those that are merely euryhaline (able to tolerate high salinity): the euryhaline blackchin tilapia can survive high salinity, but its life cycle does not involve relocation to a different biome, as does that of the diadromous wild salmon.

Which choice completes the text with the most logical and precise word or phrase?

- A) demarcated from
- B) reconstituted as
- C) conflated with
- D) derived from

6

The following text is from Joan Didion's memoir *The Year of Magical Thinking*. In the text, the author discusses her home life.

[I]n California we heated our houses by building fires. We built fires even on summer evenings, because the fog came in. Fires said we were home, we had drawn the circle, we were safe through the night.

©2005 by Joan Didion

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It illustrates that a fire provides comfort beyond physical warmth.
- B) It summarizes the information that came before it in the text.
- C) It explains that the house remains cold even in summer.
- D) It suggests that the author feels comfortable in her home with or without a fire.

7

The majority of plastics today wind up in landfills or are, at best, recycled into materials that have a very limited range of applications. To address this problem, chemist Guoliang Liu and colleagues designed a reactor that melts polyethylene and polypropylene—two widely used plastics—into a wax. The wax can then be transformed into a surfactant (a chemical compound usable as a detergent). With this promising new method, plastic waste could be turned into a range of useful cleaning products.

Which choice best states the function of the underlined portion of the text?

- A) It clarifies the meaning of a scientific term.
- B) It describes an environmental concern.
- C) It explains the significance of a scientific discovery.
- D) It identifies a result that confused the team.

8

The following text is from H.D.'s 1916 poem "Mid-Day." In the poem, the speaker is on a path in an outdoor setting.

A slight wind shakes the seed-pods—
my thoughts are spent
as the black seeds.
My thoughts tear me,
I dread their fever.
I am scattered in its whirl.
I am scattered like
the hot shrivelled seeds.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It illustrates a change in the natural environment that the speaker implies is responsible for the growing misgivings described in the text.
- B) It establishes an example of consistency in the natural landscape that the speaker then contrasts with the unpredictability of human emotions.
- C) It presents an observation of an occurrence in the natural world that the speaker then expands on to convey a sense of a turbulent interior state.
- D) It evokes the ordinariness of an event in nature to suggest that the critical self-evaluation the speaker engages in is a common pursuit.

9

In 2023 literary scholar Jeremy Douglass cautioned technology investors and enthusiasts who predict conventional books' ultimate displacement by newer forms of media. Douglass observed that the concept of an "interactive" text is much older than technologists assume, extending back to the first time readers scratched notes into a text's margins. In addition, newer media, such as video games, haven't replaced older forms of entertainment, such as comic books, but rather exist alongside them. Douglass believes that rather than supplanting books, technology is simply making new forms of expression possible.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It challenges the stance of the investors and enthusiasts who are mentioned earlier in the text.
- B) It explains the basis for the claim made by the technologists mentioned in the text.
- C) It suggests that academics are better suited than investors to see the potential uses of contemporary interactive texts.
- D) It provides a historical anecdote about the technological challenges involved in reading the earliest interactive texts.

10

In 2018, scientists discovered an immense aggregation of *Muusoctopus robustus* (pearl octopuses) along a hydrothermal vent 3,200 meters beneath the ocean's surface. Water temperatures at this site—named the Octopus Garden—climb as high as 11°C, much warmer than the ambient 1.6°C typical at this depth. Based on observations made over three years, scientists concluded that temperatures at the site likely confer reproductive benefits and that the site is used exclusively for reproduction—6,000 *M. robustus* adults, hatchlings, and eggs were observed at the garden, but no juveniles were present.

Which statement about *M. robustus* and the Octopus Garden is best supported by the text?

- A) *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.
- B) The *M. robustus* population at the Octopus Garden remains stable despite variations in water temperature.
- C) *M. robustus* nests in the Octopus Garden contain on average fewer but larger eggs than nests at similar ocean depths.
- D) The Octopus Garden provides an ideal feeding ground for *M. robustus* hatchlings.

11

The following text is from Thomas Mann's 1924 novel *The Magic Mountain*, translated by John E. Woods in 1995.

The story of Hans Castorp that we intend to tell here—not for his sake (for the reader will come to know him as a perfectly ordinary, if engaging young man), but for the sake of the story itself, which seems to us to be very much worth telling (although in Hans Castorp's favor it should be noted that it is *his* story, and that not every story happens to everybody)—is a story that took place long ago, and is, so to speak, covered with the patina of history and must necessarily be told with verbs whose tense is that of the deepest past.

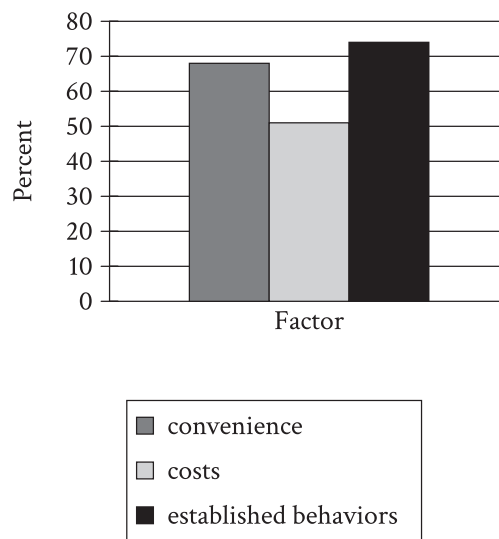
©1995 by Alfred A. Knopf, Inc.

What does the text most strongly suggest about the story of Hans Castorp?

- A) Though it is true that stories of even the most uninteresting people are themselves interesting because all people are unique, the reason this story is interesting is nonetheless difficult to understand because of the passage of time.
- B) Even though it is a story of a person of no particular importance, its age and the manner in which it therefore must be told are both indicators that the story itself is important.
- C) Like all stories about the lives of inconsequential people, this story must necessarily be related in a particular way if the reason the story is consequential is to be made evident to the audience.
- D) It is a remarkable story that happened to an unremarkable person, though one could plausibly argue that because the story is valuable, some of its value accrues to the person at its center.

12

Percentage of Participants
Who Mentioned Factors



Researcher Judith Hilton and her team interviewed 55 people about which factors would make them switch from using single-use plastic containers to reusable containers. The graph shows three of the factors mentioned in the interviews and the percentage of participants who mentioned them.

According to the graph, about what percentage of participants mentioned costs in the interviews?

- A) 10%
- B) 95%
- C) 25%
- D) 50%

13

A student in a political science course is writing a paper on Aristotle's *The Politics*, in which Aristotle offers his opinion on political instability and gives advice on how constitutions can be preserved. Aristotle observes that different forms of government can fall in different ways—for example, oligarchies might grant power to military leaders during wartime who refuse to relinquish that power during peacetime—but some methods of preserving order apply across all forms of government. The student claims that in particular Aristotle asserts that in a healthy state obedience to law must be as close to absolute as possible and that even minor infractions should not be ignored.

Which quotation from a philosopher's analysis of *The Politics* would best support the student's claim?

- A) "When constructing his argument regarding the characteristics of a well-functioning government, Aristotle asserts that 'Transgression creeps in unperceived and at last ruins the state,' illustrating this idea with a comparison to frequent small expenditures slowly and almost imperceptibly chipping away at a fortune until it is ultimately depleted."
- B) "When Aristotle writes on the necessity of avoiding corruption in government, he proposes that 'every state should be so administered and so regulated by law that its magistrates cannot possibly make money.' In particular, he thinks oligarchies are particularly susceptible to corruption through bribery."
- C) "When Aristotle considers the health of constitutions, he states that 'Constitutions are preserved when their destroyers are at a distance, and sometimes also because they are near, for the fear of them makes the government keep in hand the constitution.' He holds that rulers who wish to see constitutions preserved must continually remind the populace of the dangers that would result from a constitutional collapse."
- D) "When contrasting different forms of government, Aristotle holds that 'oligarchies may last, not from any inherent stability in such forms of government, but because the rulers are on good terms both with the unenfranchised and with the governing classes.' That is, oligarchic leaders who wish to hold on to power will introduce members of disenfranchised classes into government in a participatory role."

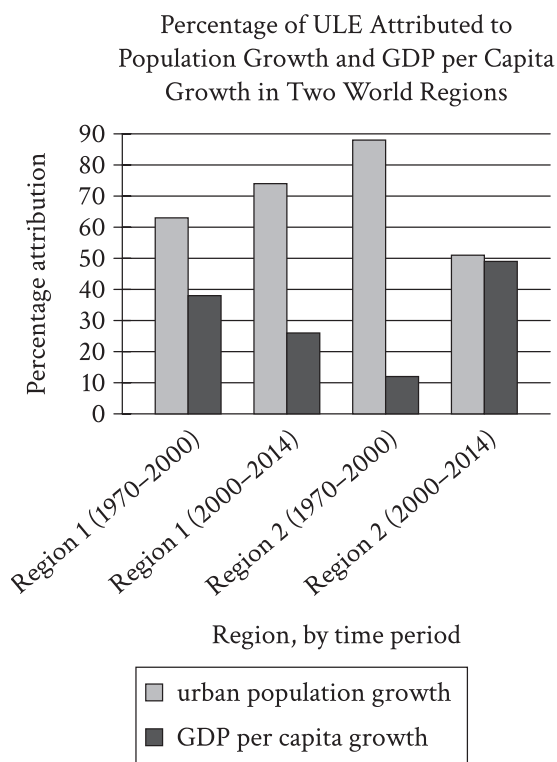
14

Almost all works of fiction contain references to the progression of time, including the time of day when events in a story take place. In a 2020 study, Allen Kim, Charuta Pethe, and Steven Skiena claim that an observable pattern in such references reflects a shift in human behavior prompted by the spread of electric lighting in the late nineteenth century. The researchers drew this conclusion from an analysis of more than 50,000 novels spanning many centuries and cultures, using software to recognize and tally both specific time references—that is, clock phrases, such as 7 a.m. or 2:30 p.m.—and implied ones, such as mentions of meals typically associated with a particular time of day.

Which finding from the study, if true, would most directly support the researchers' conclusion?

- A) Novels published after the year 1800 include the clock phrase 10 a.m. less often than novels published before the year 1800 do.
- B) Novels published after 1880 contain significantly more references to activities occurring after 10 p.m. than do novels from earlier periods.
- C) Among novels published in the nineteenth century, implied time references become steadily more common than clock phrases as publication dates approach 1900.
- D) The time references of noon (12 p.m.) and midnight (12 a.m.) are used with roughly the same frequency in the novels.

15

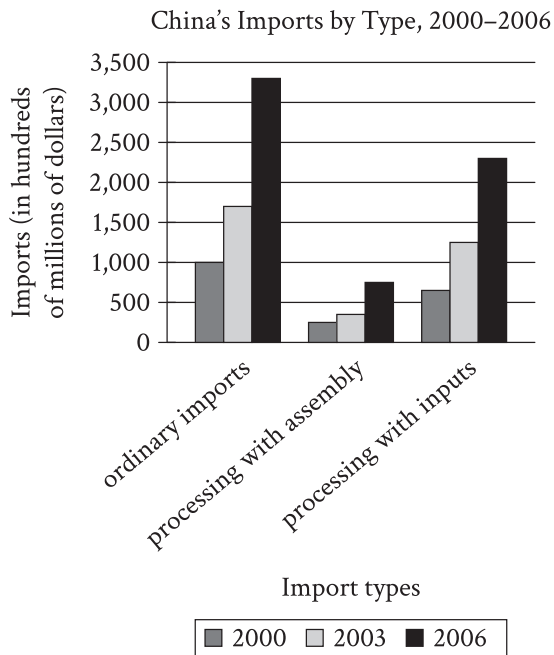


In a study of urban physical expansion, Richa Mahtta et al. conducted a meta-analysis of more than 300 cities worldwide to determine whether urban land expansion (ULE) was more strongly influenced by urban population growth or by growth in gross domestic product (GDP) per capita, a measure of economic activity. Because efficient national government is necessary to provide urban services and infrastructure that attract economic investment, Mahtta et al. propose that absent other factors, the importance of GDP per capita growth to ULE would likely increase relative to the importance of population growth as governments become more efficient. If true, this suggests the possibility that _____

Which choice most effectively uses data from the graph to complete the statement?

- A) national governments of countries in Region 1 experienced declines in efficiency in the period from 2000 to 2014, relative to the period from 1970 to 2000.
- B) countries in Region 1 experienced a slower rate of economic growth in the period from 2000 to 2014 than countries in Region 2 did, despite increasing national government efficiency in Region 1.
- C) national governments of most countries in Region 2 became more efficient in the period from 2000 to 2014 than they had been in the period from 1970 to 2000, but those of several countries in this region did not.
- D) national governments of countries in Region 1 and in Region 2 generally became more efficient in the period from 2000 to 2014 than they had been in the period from 1970 to 2000, but at different rates.

16



A student is researching the Chinese government's 1992 shift to a market economy that emphasizes trade liberalization. One means of trade liberalization involves expanding from ordinary imports into an emphasis on processing imports, which have two types: processing with assembly (in which a firm obtains raw materials from a foreign trading partner without payment and sells the final goods to that partner, charging for assembly) and processing with inputs (in which a firm expends capital to buy raw materials from a trading partner, processes them into final goods, and sells those goods to whichever trading partner it chooses). The student asserts that while initial efforts at trade liberalization were shaped by Chinese firms' limited capital, this situation resolved during the 2000s.

Which choice best describes data from the graph that support the student's assertion?

- A) Processing imports with inputs were greater than both ordinary imports and processing imports with assembly in 2006.
- B) From 2000 to 2006, processing imports with inputs rose much more sharply than processing imports with assembly did.
- C) From 2000 to 2006, neither processing imports with inputs nor processing imports with assembly were greater than ordinary imports.
- D) Processing imports with assembly were greater in 2006 than processing imports with inputs in 2000.

17

Narwhals are shy whales that live in the remote Arctic Ocean. Some of them have a long tusk, like a unicorn horn, with sensitive nerves. Narwhals are known for this tusk, but many actually don't have one and its purpose is unknown. One group of scientists came up with a possible purpose in 2014. The scientists suggested that the tusk may help narwhals determine when water around them is likely to start freezing and become dangerous for them. Marine biologist Kristin Laidre disagrees with that idea, though. She reasons that if the narwhal's tusk serves such an important purpose, then it's most likely that _____

Which choice most logically completes the text?

- A) some narwhals would seek a new habitat.
- B) fewer marine animals would also have tusks.
- C) more narwhals would have a tusk.
- D) narwhals would become less shy over time.

18

To address the susceptibility of materials used in components of high-performance machinery, such as aircraft engines, to creep (deformation that is induced by persistent mechanical stress and that often occurs at elevated temperatures), materials researchers have developed silicon carbide (SiC) fibers for producing aerospace composites. Testing the thermomechanical properties of several commercially available SiC fibers, Ramakrishna T. Bhatt et al. found that in comparison with two polymer-derived SiC fibers, a nitrogen-treated SiC fiber exhibited a lower minimum creep rate, a measure of the rate at which a stress-exposed material deforms at a constant temperature and uniaxial load. The finding suggests that _____

Which choice most logically completes the text?

- A) unlike the two polymer-derived SiC fibers, the nitrogen-treated SiC fiber can substantially inhibit creep, provided that temperatures and loads are consistent.
- B) the two polymer-derived SiC fibers likely hold similar potential for reducing the creep resistance of materials exposed to stress and elevated temperatures, thus prolonging the life span of aerospace machinery.
- C) composites based on the two polymer-derived SiC fibers have chemical properties that may improve the mechanical and thermal stability of aerospace equipment to a greater extent than do composites based on the nitrogen-treated SiC fiber.
- D) aerospace composites containing the nitrogen-treated SiC fiber may have the ability to withstand mechanical stress for a longer period of time than can aerospace composites containing either of the two polymer-derived SiC fibers.

19

One of the earliest known maps is a Babylonian clay tablet thought to be almost 4,500 years old. The map _____ the area of a plot of land, shows a river valley, and includes the cardinal directions.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) describes
- B) describe
- C) have described
- D) are describing

20

Eighteen letters written by Louisa May Alcott, author of the popular novel *Little Women* (1868), can be found at the New York Historical Society. _____ letters demonstrate Alcott's keen business sense in her interactions with publishers.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) One
- B) That
- C) This
- D) These

21

The Dust Bowl was a period of severe drought that plagued the Great Plains of the US during the 1930s. During this time, dust storms _____ over 100 million acres of land. They even reached as far east as New York City.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are affecting
- B) will have affected
- C) will affect
- D) affected

22

Many mechanical calculators were powered by a notched cylinder mechanism called the Leibniz wheel. Leibniz wheel calculators were popular in the first half of the twentieth _____ these ingenious devices were eventually replaced by electronic calculators.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) century
- B) century,
- C) century, but
- D) century that

23

Featuring jagged peaks of black ink surrounded by hazy swirls of blue and green paint, Zhang Daqian's 1983 painting *Panorama of Mount Lu* is inspired by the tradition of *qingliu shanshui*, a type of Chinese landscape painting _____ by the use of blue and green hues to depict ethereal, otherworldly landscapes.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) has been characterized
- B) will be characterized
- C) characterized
- D) is characterized

24

Increasing the heat on an uncovered boiling pot of water does not increase the temperature of the water. What increases is the rate at which the water turns to _____ a pressure cooker pot, though, an airtight seal traps the vapor in the pot, creating pressure that allows the temperature of the water to increase past its boiling point.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) vapor. With
- B) vapor with
- C) vapor, with
- D) vapor and with

25

Wanting to celebrate the 100th anniversary of the Alaska Purchase, _____ up with a motto that best captured the state's unique character. The commission selected "North to the Future," submitted by Juneau journalist Richard Peter, as its winning entry.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) a contest sponsored by the Alaska Centennial Commission would award \$300 to an individual who came
- B) an award of \$300 would go to an individual in a contest sponsored by the Alaska Centennial Commission for coming
- C) \$300 would be awarded to an individual by the Alaska Centennial Commission in a contest for coming
- D) the Alaska Centennial Commission sponsored a contest that would award \$300 to an individual who came

26

Recently unearthed Neronian tools in France dating to 54,000 years ago and attributed to *Homo sapiens* may provide evidence that interactions between Neanderthals and modern humans occurred 10,000 years earlier than was previously _____ finding that, if true, would overturn current theories about *H. sapiens* migration during the Upper Paleolithic.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) supposed; a
- B) supposed. A
- C) supposed a
- D) supposed, a

27

Guard cells are specialized cells that are part of a plant's pores. These cells help regulate the amount of carbon dioxide a plant takes in. _____ they help regulate a plant's water loss.

Which choice completes the text with the most logical transition?

- A) Additionally,
- B) Previously,
- C) In conclusion,
- D) Instead,

28

While researching a topic, a student has taken the following notes:

- Angana Chaudhuri is a scientist.
- Chaudhuri studies sedimentary rocks.
- A scientist who studies sedimentary rocks is called a sedimentologist.
- Shale, chalk, and sandstone are examples of sedimentary rocks.

The student wants to identify what type of scientist Chaudhuri is. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Chalk is a type of sedimentary rock.
- B) Some scientists study shale, chalk, and sandstone.
- C) There are scientists who study sedimentary rocks.
- D) Chaudhuri is a sedimentologist.

29

While researching a topic, a student has taken the following notes:

- “Raymond’s Run” is a short story.
- It was written by African American author Toni Cade Bambara.
- It was first published in her book *Gorilla, My Love* in 1972.
- It is told from a first person perspective.
- It takes place in Harlem.

The student wants to indicate where the short story takes place. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) “Raymond’s Run” takes place in Harlem.
- B) “Raymond’s Run” was published in *Gorilla, My Love*.
- C) “Raymond’s Run” is told from a first person perspective.
- D) “Raymond’s Run” was written by Toni Cade Bambara.

30

While researching a topic, a student has taken the following notes:

- The Royal Alcázar of Seville is a historic royal palace in Andalucía, Spain.
- The palace is famous for its intricate tilework.
- The palace features majolica and arista tiles.
- In the majolica style, designs are painted directly on the ceramic tiles.
- In the arista style, designs are stamped into the ceramic tiles.

The student wants to contrast the two styles of tiles. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Tiles in the majolica and arista styles can be found in the Royal Alcázar of Seville in Andalucía, Spain.
- B) Featuring tiles in the majolica and arista styles, the Royal Alcázar of Seville in Spain is famous for its intricate tilework.
- C) In the arista style, designs are stamped into the ceramic tiles, whereas in the majolica style, the designs are painted directly on them.
- D) Among the famous tilework of the Royal Alcázar of Seville are majolica style tiles, made by painting designs directly on the ceramic tiles.

31

While researching a topic, a student has taken the following notes:

- Musicians around the world have used protest songs to raise awareness about human rights violations.
- US folk singer Aunt Molly Jackson released the protest song “Poor Miner’s Farewell” in 1932.
- It exposed the unlivable wages and dangerous working conditions coal miners faced in Kentucky during the 1920s and 1930s.
- South African singer-songwriter Hugh Masekela released the protest song “Bring Him Back Home” in 1987.
- It called on the South African government to free Nelson Mandela, an anti-apartheid leader who’d been unjustly imprisoned.

The student wants to contrast the song “Poor Miner’s Farewell” with the song “Bring Him Back Home.” Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The songs “Poor Miner’s Farewell” and “Bring Him Back Home” both raised awareness about human rights violations.
- B) While both are protest songs, “Poor Miner’s Farewell” is about coal miners in Kentucky, whereas “Bring Him Back Home” is about the anti-apartheid leader Nelson Mandela.
- C) Hugh Masekela’s song “Bring Him Back Home,” released in 1987, called on the South African government to free Nelson Mandela.
- D) Released in 1932 by Aunt Molly Jackson, the song “Poor Miner’s Farewell” was a protest against the unlivable wages and dangerous working conditions faced by Kentucky coal miners.

32

While researching a topic, a student has taken the following notes:

- Political scientist Graham Allison is known for his Thucydides trap theory.
- Allison's theory states that whenever "a rising power is threatening to displace a ruling power," conflict is likely.
- The theory is based on Thucydides's explanation of the conflict between Athens and Sparta.
- Thucydides wrote that "the rise of Athens and the fear this instilled in Sparta" made conflict "inevitable."
- History professor Edmund Stewart recently challenged the historical basis of the theory.
- Stewart claimed that Athens was not a rising power and that the rivals experienced a "clash of cultures" instead.

The student wants to use a quotation to challenge Thucydides's explanation of the conflict between Athens and Sparta. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) According to Allison's Thucydides trap theory, whenever "a rising power is threatening to displace a ruling power," conflict is likely.
- B) Thucydides wrote that conflict between the two powers was "inevitable," although Stewart later challenged the historical basis of this claim.
- C) According to Stewart, a "clash of cultures" between Athens and Sparta caused the conflict, not Athens's rise.
- D) Thucydides explained that conflict was caused by "the rise of Athens and the fear this instilled in Sparta," but Allison disagreed, seeing the conflict as an example of the Thucydides trap.

33

While researching a topic, a student has taken the following notes:

- Researchers in a 2021 study wanted to determine the rate at which 17 languages conveyed both information and syllables.
- They calculated the bits of information conveyed per second (the IR, or information rate).
- The IR was found to be approximately consistent across the 17 languages (an average of 39 bits per second).
- They calculated the number of syllables spoken per second (the SR, or syllable rate).
- Spanish had the second-fastest SR (7.7 syllables per second).
- Vietnamese had the sixteenth-fastest SR (5.3 syllables per second).

The student wants to present an overview of the study's findings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The 2021 study determined the information rate (IR) of 17 languages in bits of information conveyed per second.
- B) Researchers found that information was conveyed more quickly in Spanish, at 7.7 syllables per second, than in Vietnamese, at 5.3 syllables per second.
- C) Vietnamese had the sixteenth-fastest syllable rate, lower than that of Spanish, which had the second-fastest; however, Spanish had the lower information rate of the two.
- D) Though some of the languages differed in number of syllables spoken per second, all 17 conveyed information at roughly the same rate.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Taking photographs in the mid-1800s was complicated and expensive, but this changed with the 1854 invention of the *carte de visite*, a small photo that cost little to make. *Carte de visite* photos helped to _____ photography: they made it easy and enjoyable for everyday people to have their pictures taken, and people at the time loved exchanging these small photos with friends and family.

Which choice completes the text with the most logical and precise word or phrase?

- A) weaken
- B) praise
- C) popularize
- D) isolate

2

Painter Alma W. Thomas was fascinated by the colors and shapes found in nature. The flowers and trees in the garden at her home in Washington, DC, _____ her work. For example, Thomas's use of broken brushstrokes was inspired by the way that light would shine through the leaves of a tree in front of her house.

Which choice completes the text with the most logical and precise word or phrase?

- A) restricted
- B) announced
- C) distracted
- D) influenced

3

In the 1990s, conservationists began planting more than 500,000 native trees in the habitat of the Azores bullfinch to boost the bird's numbers. This approach was apparently _____: the Azores bullfinch's population size increased from as few as 100 birds at the end of the 1980s to around 1,300 in 2023.

Which choice completes the text with the most logical and precise word or phrase?

- A) amusing
- B) costly
- C) successful
- D) disastrous

4

The recently observed gamma ray burst GRB 230307A lasted for 200 seconds, _____ for a burst generated by the merger of neutron stars. Bursts caused by neutron mergers typically last fewer than 2 seconds.

Which choice completes the text with the most logical and precise word or phrase?

- A) a coincidence
- B) a reprieve
- C) an incident
- D) an oddity

5

In 1776, the United States sent Benjamin Franklin to France to try to win the country's support in the United States' fight for independence from Great Britain. Franklin was very popular in France. This _____ surely helped him to convince France to assist the United States.

Which choice completes the text with the most logical and precise word or phrase?

- A) thoughtfulness
- B) esteem
- C) controversy
- D) sincerity

6

In the 1950s, scientists didn't know much about the ocean floor. Many scientists at the time believed that the ocean floor was mostly flat. But geologist Marie Tharp and her research partner, Bruce Heezen, proved that this idea was wrong. Using sonar data collected from the Atlantic Ocean, Tharp and Heezen showed that the floor was filled with canyons, mountains, and valleys.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It identifies a scientific belief that Tharp and Heezen showed to be wrong.
- B) It describes the design of Tharp and Heezen's experiment.
- C) It emphasizes a disagreement between Tharp and Heezen.
- D) It presents data to support a claim that Tharp and Heezen made.

7

In the early days of television in the 1940s, many people thought that US television programs would rely on the financial support of ad agencies and commercial sponsors, much like radio did. But advertisers hesitated to jump into a new space, particularly at a time when the manufacturing of new television sets was stalled due to the US's involvement in World War II. Broadcasters, like the National Broadcasting Company (NBC), needed to persuade advertisers to support their programming despite not knowing whether there would be a robust television audience to begin with.

Which choice best describes the function of the underlined phrase in the text as a whole?

- A) It compares the beginnings of radio programming with the beginnings of television programming in the United States.
- B) It identifies a specific reason behind some advertisers' hesitance to support television.
- C) It describes how broadcasters attempted to convince advertisers to support television.
- D) It explains why a type of television programming was popular at the time.

8

The Bayeux Tapestry, from eleventh-century France, depicts 75 scenes over 250 feet of fabric. It was likely produced by workers embroidering in sections and then joining the resulting panels together. It's plausible that the workshop that produced the tapestry had never produced one so large, and some researchers claim that a close examination of the joins—the places where the panels are stitched together—suggests that the workers developed and refined their joining process over the course of production. For example, the first join the workers completed exhibits a clear misalignment of the borders of the two panels, whereas the later joins are virtually invisible.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It identifies the people and events depicted in the Bayeux Tapestry.
- B) It supports an argument about the workers who produced the Bayeux Tapestry.
- C) It compares the Bayeux Tapestry with other tapestries from eleventh-century France.
- D) It describes how researchers determined where the Bayeux Tapestry was produced.

9

Text 1

Little is known about how plate tectonics—wherein slabs of Earth’s crust move over, under, away from, and against one another—began. Some researchers contend that tectonic movements began around 3 billion years ago, often noting that computer models of Earth’s mantle temperature at the time indicate that the mantle would have been sufficiently molten to enable the plates to move.

Text 2

Ultimately, any plausible claim about the inception of tectonic movement must rest on empirical evidence from the geological record. Researcher Wriju Chowdhury and his team analyzed the geochemistry of zircon crystals to gain insight into the chemical composition of the magma from which the crystals formed and, based on the data, compellingly argue that plate tectonics may have been occurring as early as 4.2 billion years ago.

Based on the texts, how would the author of Text 2 most likely respond to what “some researchers contend” as described in Text 1?

- A) By suggesting that the temperature of Earth’s mantle 3 billion years ago was likely insufficient to allow for the level of tectonic movement predicted by computer models
- B) By distinguishing between computer models of Earth’s mantle temperature that reliably predict the onset of plate tectonics and those that do not
- C) By indicating that computer models of Earth’s mantle temperature are still being improved such that new models tend to be much more reliable than their predecessors
- D) By asserting that a more definitive form of evidence than the computer models suggests a different timeline for the onset of plate tectonics on Earth

10

Hevea brasiliensis, a tree in the Amazon rainforest, is the world’s main source of natural rubber. The tree produces a milky substance called latex that is used to make rubber. The bark of *Hevea brasiliensis* is helpful for the process of making rubber because it has a unique structure that makes it easy to collect latex. A network of tubes in the tree’s inner bark helps the latex to flow out easily when people make small cuts into the bark.

What feature of *Hevea brasiliensis* does the text say is helpful for the process of making rubber?

- A) Its latex produces rubber of an especially high quality.
- B) Its bark has a unique structure that makes it easy to collect latex.
- C) It is able to grow in a wide variety of climates around the world.
- D) It is one of only two trees in the Amazon that produce latex.

11

Conservationists worldwide are working to protect ecosystems from habitat destruction and biodiversity loss, and in many cases, initiatives that rely on natural features or processes can help address such challenges. In response to a rapidly dwindling population of blueback salmon, the Quinault Indian Nation (a tribe in Washington State) partnered with the conservation organization Wild Salmon Center to restore naturally occurring logjams in the Quinault River. The logjams create shady pools where the blueback salmon can rest and spawn, thus promoting blueback population recovery.

Which choice best states the main idea of the text?

- A) A partnership between the Quinault Indian Nation and Wild Salmon Center shows the importance of collaborative approaches to preserving biodiversity.
- B) Nature-based approaches can be effective ways to achieve conservation goals.
- C) As indicated by a recent project, logjams help the blueback salmon thrive and reproduce.
- D) Scientists now realize that nature-based conservation methods offer better long-term solutions to environmental issues than methods that are not nature-based do.

12

Percentage of Bus Shelters with Shade in a County by Areas' Highest Average Summer Surface Temperature

Highest average surface temperature (Fahrenheit)	Percentage of bus stops with shaded shelter
90.2°	15%
97.7°	22%
102.7°	24%
111.2°	28%
125.6°	29%

A student is researching a bus system in a large county where surface temperatures vary by area and are hot in the summer. The student claims that all areas of the county should have more bus stops with shaded shelter, noting that the highest percentage of bus stops with shaded shelter for any area is only _____

Which choice most effectively uses data from the table to complete the student's claim?

- A) 50%.
- B) 15%.
- C) 90%.
- D) 29%.

13

Total Areas and 2022 Populations of Smallest
Arabian Peninsula Countries

Country	Total area (square miles)	Population
Kuwait	6,880	4,268,873
Bahrain	304	1,472,233
Qatar	4,471	2,695,122

In terms of area and population, the three smallest Arabian Peninsula countries are Bahrain, Qatar, and Kuwait.

According to the table, what is the total area of Bahrain?

- A) 4,268,873 square miles
- B) 4,471 square miles
- C) 304 square miles
- D) 6,880 square miles

14

Janet Echelman is a sculptor and fiber artist. She has installed giant sculptures all over the world. Echelman uses bright and flowing materials, which mimic the wind. However, while her sculptures appear as delicate as a breeze, they are actually very durable.

Which quotation from an article about Echelman's sculptures, if true, would most effectively illustrate the underlined claim?

- A) "Echelman uses a special program that makes a 3D model of the sculpture."
- B) "The first part of planning a new sculpture is done using paper and pencil, and then a digital program is used to finalize the design."
- C) "The materials that Echelman uses to build her sculptures are both flexible and strong."
- D) "Each sculpture is designed to reflect local landmarks from the area in which it is eventually installed."

15

Early Earth is thought to have been characterized by a stagnant lid tectonic regime, in which the upper lithosphere (the outer rocky layer) was essentially immobile and there was no interaction between the lithosphere and the underlying mantle. Researchers investigated the timing of the transition from a stagnant lid regime to a tectonic plate regime, in which the lithosphere is fractured into dynamic plates that in turn allow lithospheric and mantle material to mix. Examining chemical data from lithospheric and mantle-derived rocks ranging from 285 million to 3.8 billion years old, the researchers dated the transition to 3.2 billion years ago.

Which finding, if true, would most directly support the researchers' conclusion?

- A) Among rocks known to be older than 3.2 billion years, significantly more are mantle derived than lithospheric, but the opposite is true for the rocks younger than 3.2 billion years.
- B) Mantle-derived rocks older than 3.2 billion years show significantly more compositional diversity than lithospheric rocks older than 3.2 billion years do.
- C) There is a positive correlation between the age of lithospheric rocks and their chemical similarity to mantle-derived rocks, and that correlation increases significantly in strength at around 3.2 billion years old.
- D) Mantle-derived rocks younger than 3.2 billion years contain some material that is not found in older mantle-derived rocks but is found in older and contemporaneous lithospheric rocks.

16

The Uto-Aztecan language family is divided into a northern branch, which includes the Shoshone language of present-day Idaho and Utah, and a southern one, whose best-known representative is Nahuatl, the language of the Aztec Empire in Mexico. Lexical similarities across the family, including of botanical terms, confirm descent from a single language spoken millennia ago, and the family's geographical distribution suggests an origin in what is now the US Southwest. However, vocabulary pertaining to maize isn't shared between northern and southern branches, despite the crop's universal cultivation among Uto-Aztecan tribes. Given archaeological evidence that maize originated in Mexico and diffused northward into what became the US Southwest, some linguists reason that _____

Which choice most logically completes the text?

- A) northern Uto-Aztecan tribes likely obtained the crop directly from a southern Uto-Aztecan tribe rather than from a non-Uto-Aztecan tribe.
- B) variation in maize-related vocabulary within each branch of the Uto-Aztecan family likely reflects regionally specific methods for cultivating the crop.
- C) southern Uto-Aztecan tribes likely acquired maize at roughly the same time as northern Uto-Aztecan tribes did, though from different sources.
- D) the family's division into northern and southern branches likely preceded the acquisition of the crop by the Uto-Aztecan tribes.

17

Data collected by the Mars rover *Curiosity* at the Gale Crater's Murray Formation are suggestive of hydrological deposition of sediment in the distant past. To characterize the nature of the depositional environment, Frances Rivera-Hernández et al. analyzed the grain size of Murray Formation sediment, finding that although there are intervals of coarse grains, most of the sediment consists of fine grains that show signs of cracking due to episodic desiccation. Rivera-Hernández et al. concluded that the coarse grains are sandstone, which tends to be deposited by flowing water, whereas the fine grains are mudstone, which is slowly deposited by settling out of suspension in low-flow water, leading the researchers to posit that _____

Which choice most logically completes the text?

- A) although the area of the Murray Formation experienced a prolonged period of dryness that prevented a lake from forming, water flowing from a distant source was present.
- B) a lake existed at the Murray Formation for a prolonged period, though the lake occasionally experienced drying and there were periods in which one or more streams were present.
- C) one or more streams existed at the Murray Formation for an extended period until being replaced by a lake that persisted for only a brief period before permanently drying.
- D) a stream-fed lake was present at the Murray Formation for an extended period, and although the streams experienced occasional drying, the lake did not.

18

An analysis by Alain Elayi and colleagues of coins minted in Sidon in the fifth and fourth centuries BCE reveals a change in their composition over time: while a coin from circa 450 BCE contains about 98% silver and 1% copper, a coin from 367 BCE (the end of Ba'alšilem II's reign) contains 74.2% silver and 24.7% copper, giving it a relatively yellowish appearance that traders would have noticed. Because coins with a silver content below 80% were widely considered unsuitable for trade, Elayi et al. speculate that a crisis in confidence in the currency occurred in Sidon around 367 BCE, which was likely relieved—despite Sidon's persistent oppressive financial obligations—as a result of Ba'alšilem II's successor Abd'aštart I's decision to _____

Which choice most logically completes the text?

- A) proclaim that the percentage of silver in coins suitable for trade would be raised to a threshold higher than 80%.
- B) keep the amount of silver in Sidonian coins consistent with that in coins minted in 367 BCE but decrease their weight.
- C) begin minting heavier coins with a proportion of silver to copper similar to that in coins minted in 367 BCE.
- D) fund the mining of some copper deposits that were not available to Ba'alšilem II.

19

Nowadays, tug-of-war is usually seen as an informal game one might play at a picnic or in gym class. Surprisingly, the Olympic committee once decided _____ tug-of-war as an official Olympic event! Nations competed in the event at the Olympic Games from 1900 to 1920.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) included
- B) including
- C) include
- D) to include

20

The Globe Theatre in London is a reconstruction of the famed venue where many of Shakespeare's plays were first performed. In 1613, a prop cannon _____ during a performance and ignited the Globe's thatched roof. No one was hurt, but in two hours the original Globe was gone.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) malfunctions
- B) will malfunction
- C) has malfunctioned
- D) malfunctioned

21

Nigerian American artist Toyin Ojih Odutola uses black-ink pens to create highly detailed drawings of human figures. Her portrait of novelist Zadie _____ is displayed in the National Portrait Gallery in London.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Smith:
- B) Smith—
- C) Smith
- D) Smith,

22

When a given industry—water and electricity are two well-known examples—carries high infrastructural start-up costs and other barriers that discourage competition, _____ of just one or two suppliers per municipality. Such industries are known as natural monopolies.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) these often consist
- B) they often consist
- C) it often consists
- D) this often consists

23

As the fourteenth US librarian of Congress, Carla Hayden has many responsibilities. These include overseeing the Library of Congress's collections, which boast more than 162 million _____ the US Copyright Office, which registers copyright claims and advises Congress on copyright law; and appointing the US poet laureate.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) items managing
- B) items, managing
- C) items; managing
- D) items. Managing

24

Digital artist Jung (Lulu) Chen primarily uses a suite of software tools to create illustrations for children's books. To manifest the warm and welcoming atmospheres that are a signature of her _____ she occasionally relies on more traditional art techniques, such as painting with watercolors.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) work, though,
- B) work, though
- C) work; though,
- D) work, though;

25

Chondrites are stony meteorites that are undifferentiated—that is, their contents have not melted and separated into distinct layers. They are hardly _____ many chondrites experience aqueous alteration as a result of exposure to fluids, as well as fracturing, veining, and localized melting due to collisions with other objects.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) pristine, though
- B) pristine, though;
- C) pristine; though
- D) pristine, though,

26

That the geographic center of North America lay in the state of North Dakota was conceded by all _____ establishing its precise coordinates proved more divisive.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) involved:
- B) involved,
- C) involved
- D) involved;

27

Famous for its four-degree tilt, the leaning Garisenda Tower is a popular attraction in Bologna's city center. However, measurements taken in 2023 showed that the tower was rotating in a concerning way. _____ city officials closed the area around the tower so experts could explore solutions to stabilize the historical twelfth-century structure.

Which choice completes the text with the most logical transition?

- A) Similarly,
- B) As a result,
- C) For example,
- D) In comparison,

28

In 2021, a model developed by astrophysicist Catherine Zucker and her research team revealed that the same supernovas responsible for the creation and ongoing expansion of the Local Bubble—a 14-million-year-old cavity in the Milky Way—are likely responsible for the formation of new stars. _____ this model detailed how the bubble's expansion trapped interstellar clouds of gas and dust that became stars upon their eventual collapse.

Which choice completes the text with the most logical transition?

- A) Hence,
- B) However,
- C) Admittedly,
- D) Specifically,

29

Following the American Revolutionary War, North American foodways underwent a radical transformation, fueled in large part by spiking consumer demand for certain grains. The cultivation, trade, and transportation of maize and wheat, _____ reconfigured the continent's existing regional foodways into a globally oriented food system.

Which choice completes the text with the most logical transition?

- A) in particular,
- B) alternatively,
- C) by comparison,
- D) second of all,

30

When, in 2017, Cambridge University students Lucy Moss and Toby Marlow decided they wanted to develop a musical together, one of their goals was for their female actor friends to have good parts to play. _____ they created the show *Six*, a retelling of the history of King Henry VIII's wives in which each of the six queens has a starring role.

Which choice completes the text with the most logical transition?

- A) In other words,
- B) In summary,
- C) For example,
- D) To that end,

31

Mountain climbing routes that incorporate metal rungs and cables are known as via ferratas, from the Italian phrase for "iron path." As climbing these routes has shifted from a mode of travel to a sporting activity, modern via ferratas are rarely designed to simply reach a summit. _____ new routes favor recreation over utility, aiming to provide a challenging climb or showcase dramatic scenery.

Which choice completes the text with the most logical transition?

- A) Additionally,
- B) On the other hand,
- C) More often,
- D) Nonetheless,

32

While researching a topic, a student has taken the following notes:

- Samuel Delany is a US writer known for his science fiction.
- Delany's science fiction novel *Babel-17* was published in 1966.
- The novel won a Nebula Award in 1967.
- The Nebula Awards are given each year to the best works of science fiction published in the US.

The student wants to indicate the title of a novel that won a Nebula Award. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) *Babel-17*, by Samuel Delany, won a Nebula Award in 1967.
- B) Samuel Delany published a science fiction novel in 1966.
- C) Samuel Delany is an award-winning US writer known for his science fiction.
- D) One of Samuel Delany's novels was among the best works of science fiction published in the US.

33

While researching a topic, a student has taken the following notes:

- Chiura Obata was a Japanese American artist who lived in California.
- *Yosemite Falls* is a notable painting by Obata.
- It uses a Japanese method of black ink painting called sumi-e.
- This painting was completed in 1930.

The student wants to indicate the year *Yosemite Falls* was completed. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) While living in California, Obata created black ink paintings.
- B) Obata, a Japanese American artist, created a notable painting.
- C) *Yosemite Falls* was completed in 1930.
- D) Obata used a Japanese painting method called sumi-e.

STOP

**If you finish before time is called, you may check your work on this module only.
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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

As Mexico's first president from an Indigenous community, Benito Juarez became one of the most _____ figures in his country's history: among the many significant accomplishments of his long tenure in office (1858–1872), Juarez consolidated the authority of the national government and advanced the rights of Indigenous peoples.

Which choice completes the text with the most logical and precise word or phrase?

- A) unpredictable
- B) important
- C) secretive
- D) ordinary

2

Mônica Lopes-Ferreira and others at Brazil's Butantan Institute are studying the freshwater stingray species *Potamotrygon rex* to determine whether biological characteristics such as the rays' age and sex have _____ effect on the toxicity of their venom—that is, to see if differences in these traits are associated with considerable variations in venom potency.

Which choice completes the text with the most logical and precise word or phrase?

- A) a disconcerting
- B) an acceptable
- C) an imperceptible
- D) a substantial

3

Kelp forests grow underwater along the eastern Pacific Coast. These underwater forests are important to fish and other marine animals. Ocean currents can be powerful and rough, making it difficult for animals to find safe places to hide from predators. The underwater forests slow down the currents. This creates a more _____ environment with calmer waters where animals can take shelter.

Which choice completes the text with the most logical and precise word or phrase?

- A) tranquil
- B) dangerous
- C) imaginative
- D) surprising

4

The Mule Bone, a 1930 play written by Zora Neale Hurston and Langston Hughes, is perhaps the best-known of the few examples of _____ in literature. Most writers prefer working alone, and given that working together cost Hurston and Hughes their friendship, it is not hard to see why.

Which choice completes the text with the most logical and precise word or phrase?

- A) characterization
- B) interpretation
- C) collaboration
- D) commercialization

5

Ofelia Zepeda's contributions to the field of linguistics are _____: her many accomplishments include working as a linguistics professor and bilingual poet, authoring the first Tohono O'odham grammar book, and co-founding the American Indian Language Development Institute.

Which choice completes the text with the most logical and precise word or phrase?

- A) pragmatic
- B) controversial
- C) extensive
- D) universal

6

Archaeologists studying the ancient city of Pompeii in Italy recently discovered a well-preserved food shop known as a *thermopolium*. The site contains food remains, artworks, and decorations. These items give researchers a better understanding of what daily life in Pompeii may have been like. For example, the archaeologists found a ceramic jar that they believe likely contained a meat and seafood stew.

Which choice best states the main purpose of the text?

- A) To compare ancient artworks with modern ones
- B) To discuss the political system of Italy
- C) To present a recent archaeological discovery
- D) To describe a region's climate

7

The following text is from Sarah Orne Jewett's 1899 short story "Martha's Lady." Martha is employed by Miss Pyne as a maid.

Miss Pyne sat by the window watching, in her best dress, looking stately and calm; she seldom went out now, and it was almost time for the carriage. Martha was just coming in from the garden with the strawberries, and with more flowers in her apron. It was a bright cool evening in June, the golden robins sang in the elms, and the sun was going down behind the apple-trees at the foot of the garden. The beautiful old house stood wide open to the long-expected guest.

Which choice best states the main purpose of the text?

- A) To convey the worries brought about by a new guest
- B) To describe how the characters have changed over time
- C) To contrast the activity indoors with the stillness outside
- D) To depict the setting as the characters await a visitor's arrival

8

The following text is adapted from Aphra Behn's 1689 novel *The Lucky Mistake*. Atlante and Rinaldo are neighbors who have been secretly exchanging letters through Charlot, Atlante's sister.

[Atlante] gave this letter to Charlot; who immediately ran into the balcony with it, where she still found Rinaldo in a melancholy posture, leaning his head on his hand: She showed him the letter, but was afraid to toss it to him, for fear it might fall to the ground; so he ran and fetched a long cane, which he cleft at one end, and held it while she put the letter into the cleft, and stayed not to hear what he said to it. But never was man so transported with joy, as he was at the reading of this letter; it gives him new wounds; for to the generous, nothing obliges love so much as love.

Which choice best describes the overall structure of the text?

- A) It describes the delivery of a letter, and then portrays a character's happiness at reading that letter.
- B) It establishes that a character is desperate to receive a letter, and then explains why another character has not yet written that letter.
- C) It presents a character's concerns about delivering a letter, and then details the contents of that letter.
- D) It reveals the inspiration behind a character's letter, and then emphasizes the excitement that another character feels upon receiving that letter.

9

According to historian Vicki L. Ruiz, Mexican American women made crucial contributions to the labor movement during World War II. At the time, food processing companies entered into contracts to supply United States armed forces with canned goods. Increased production quotas conferred greater bargaining power on the companies' employees, many of whom were Mexican American women: employees insisted on more favorable benefits, and employers, who were anxious to fulfill the contracts, complied. Thus, labor activism became a platform for Mexican American women to assert their agency.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It elaborates on a claim about labor relations in a particular industry made earlier in the text.
- B) It offers an example of a trend in the World War II-era economy discussed earlier in the text.
- C) It notes a possible exception to the historical narrative of labor activism sketched earlier in the text.
- D) It provides further details about the identities of the workers discussed earlier in the text.

10

For many years, the only existing fossil evidence of mixopterid eurypterids—an extinct family of large aquatic arthropods known as sea scorpions and related to modern arachnids and horseshoe crabs—came from four species living on the paleocontinent of Laurussia. In a discovery that expands our understanding of the geographical distribution of mixopterids, paleontologist Bo Wang and others have identified fossilized remains of a new mixopterid species, *Terropterus xiushanensis*, that lived over 400 million years ago on the paleocontinent of Gondwana.

According to the text, why was Wang and his team's discovery of the *Terropterus xiushanensis* fossil significant?

- A) The fossil constitutes the first evidence found by scientists that mixopterids lived more than 400 million years ago.
- B) The fossil helps establish that mixopterids are more closely related to modern arachnids and horseshoe crabs than previously thought.
- C) The fossil helps establish a more accurate timeline of the evolution of mixopterids on the paleocontinents of Laurussia and Gondwana.
- D) The fossil constitutes the first evidence found by scientists that mixopterids existed outside the paleocontinent of Laurussia.

11

Video Game Availability by Initial Release Years

Initial release years	Percentage of games still available
1975–1979	0.89
1980–1984	3.65
1985–1989	15.38
1990–1994	19.33
1995–1999	14.22

In a recent study, researchers found that relatively few video games released over the decades remain available today. For example, only 14.22 percent of games are still available that were initially released in _____

Which choice most effectively uses data from the table to complete the statement?

- A) 2000–2004.
- B) 1995–1999.
- C) 1970–1974.
- D) 1985–1989.

12

Depths at Which Four Deep-Sea Fish Species Live

Species	Depth below the ocean surface
Footballfish	200–1,000 meters
Southern stoplight loosejaw	500–2,000 meters
Black seadevil	250–2,000 meters
Bollons' rattail	300–800 meters

Some oceanic fish species live very deep underwater. Researchers collected data about the depths at which various species live.

Based on the information in the table, at what depth does the southern stoplight loosejaw live?

- A) More than 2,000 meters below the surface
- B) 150 to 400 meters below the surface
- C) 500 to 2,000 meters below the surface
- D) 250 to 500 meters below the surface

13

Housing Starts in the US, January–April 2022
(in thousands)

Month	Housing starts
January	1,669
February	1,771
March	1,713
April	1,803

When construction of a single-family house begins, it is called a housing start. In the first four months of 2022, the highest number of housing starts in the United States was in _____

Which choice most effectively uses data from the table to complete the statement?

- A) April.
- B) March.
- C) January.
- D) February.

14

Born in 1891 to a Quechua-speaking family in the Andes Mountains of Peru, Martín Chambi is today considered to be one of the most renowned figures of Latin American photography. In a paper for an art history class, a student claims that Chambi's photographs have considerable ethnographic value—in his work, Chambi was able to capture diverse elements of Peruvian society, representing his subjects with both dignity and authenticity.

Which finding, if true, would most directly support the student's claim?

- A) Chambi took many commissioned portraits of wealthy Peruvians, but he also produced hundreds of images carefully documenting the peoples, sites, and customs of Indigenous communities of the Andes.
- B) Chambi's photographs demonstrate a high level of technical skill, as seen in his strategic use of illumination to create dramatic light and shadow contrasts.
- C) During his lifetime, Chambi was known and celebrated both within and outside his native Peru, as his work was published in places like Argentina, Spain, and Mexico.
- D) Some of the peoples and places Chambi photographed had long been popular subjects for Peruvian photographers.

15

Researchers hypothesized that a decline in the population of dusky sharks near the mid-Atlantic coast of North America led to a decline in the population of eastern oysters in the region. Dusky sharks do not typically consume eastern oysters but do consume cownose rays, which are the main predators of the oysters.

Which finding, if true, would most directly support the researchers' hypothesis?

- A) Declines in the regional abundance of dusky sharks' prey other than cownose rays are associated with regional declines in dusky shark abundance.
- B) Eastern oyster abundance tends to be greater in areas with both dusky sharks and cownose rays than in areas with only dusky sharks.
- C) Consumption of eastern oysters by cownose rays in the region substantially increased before the regional decline in dusky shark abundance began.
- D) Cownose rays have increased in regional abundance as dusky sharks have decreased in regional abundance.

16

In the mountains of Brazil, *Barbacenia tomentosa* and *Barbacenia macrantha*—two plants in the Velloziaceae family—establish themselves on soilless, nutrient-poor patches of quartzite rock. Plant ecologists Anna Abrahão and Patricia de Britto Costa used microscopic analysis to determine that the roots of *B. tomentosa* and *B. macrantha*, which grow directly into the quartzite, have clusters of fine hairs near the root tip; further analysis indicated that these hairs secrete both malic and citric acids. The researchers hypothesize that the plants depend on dissolving underlying rock with these acids, as the process not only creates channels for continued growth but also releases phosphates that provide the vital nutrient phosphorus.

Which finding, if true, would most directly support the researchers' hypothesis?

- A) Other species in the Velloziaceae family are found in terrains with more soil but have root structures similar to those of *B. tomentosa* and *B. macrantha*.
- B) Though *B. tomentosa* and *B. macrantha* both secrete citric and malic acids, each species produces the acids in different proportions.
- C) The roots of *B. tomentosa* and *B. macrantha* carve new entry points into rocks even when cracks in the surface are readily available.
- D) *B. tomentosa* and *B. macrantha* thrive even when transferred to the surfaces of rocks that do not contain phosphates.

17

Many mosquito repellents contain natural components that work by activating multiple odor receptors on mosquitoes' antennae. As the insects develop resistance, new repellents are needed. Ke Dong and her team found that EBF, a molecular component of a chrysanthemum-flower extract, can repel mosquitoes by activating just one odor receptor—and this receptor, Or31, is present in all mosquito species known to carry diseases. Therefore, the researchers suggest that in developing new repellents, it would be most useful to _____

Which choice most logically completes the text?

- A) identify molecular components similar to EBF that target the activation of Or31 receptors.
- B) investigate alternative methods for extracting EBF molecules from chrysanthemums.
- C) verify the precise locations of Or31 and other odor receptors on mosquitoes' antennae.
- D) determine the maximum number of different odor receptors that can be activated by a single molecule.

18

Birds of many species ingest foods containing carotenoids, pigmented molecules that are converted into feather coloration. Coloration tends to be especially saturated in male birds' feathers, and because carotenoids also confer health benefits, the deeply saturated colors generally serve to communicate what is known as an honest signal of a bird's overall fitness to potential mates. However, ornithologist Allison J. Shultz and others have found that males in several species of the tanager genus *Ramphocelus* use microstructures in their feathers to manipulate light, creating the appearance of deeper saturation without the birds necessarily having to maintain a carotenoid-rich diet. These findings suggest that _____

Which choice most logically completes the text?

- A) individual male tanagers can engage in honest signaling without relying on carotenoid consumption.
- B) feather microstructures may be less effective than deeply saturated feathers for signaling overall fitness.
- C) scientists have yet to determine why tanagers have a preference for mates with colorful appearances.
- D) a male tanager's appearance may function as a dishonest signal of the individual's overall fitness.

19

When writing *The Other Black Girl* (2021), novelist Zakiya Dalila Harris drew on her own experiences working at a publishing office. The award-winning book is Harris's first novel, but her writing _____ honored before. At the age of twelve, she entered a contest to have a story published in *American Girl* magazine—and won.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) were
- B) have been
- C) has been
- D) are

20

In order to prevent nonnative fish species from moving freely between the Mediterranean and Red Seas, marine biologist Bella Galil has proposed that a saline lock system be installed along the Suez Canal in Egypt's Great Bitter Lakes. The lock would increase the salinity of the lakes and _____ a natural barrier of water most marine creatures would be unable to cross.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) creates
- B) create
- C) creating
- D) created

21

Paintings by the renowned twentieth-century US _____ were featured in *Artist to Artist*, an exhibition at the Smithsonian Art Museum that paired the works of artists whose career trajectories intersected in meaningful ways.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) artists: Thomas Hart Benton and Jackson Pollock,
- B) artists Thomas Hart Benton and Jackson Pollock
- C) artists Thomas Hart Benton, and Jackson Pollock,
- D) artists, Thomas Hart Benton and Jackson Pollock

22

In 1943, in the midst of World War II, mathematics professor Grace Hopper was recruited by the US military to help the war effort by solving complex equations. Hopper's subsequent career would involve more than just _____ as a pioneering computer programmer, Hopper would help usher in the digital age.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) equations, though:
- B) equations, though,
- C) equations. Though,
- D) equations though

23

Increased gender diversity is revitalizing the field of economics, according to Harvard's Claudia Goldin. The trailblazing accomplishments of Goldin, winner of the 2023 Nobel Prize in Economics for her work on women in the labor force, _____ to the value of scholars of diverse backgrounds in spurring research into previously unexplored, but vitally important, topics.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) attests
- B) has attested
- C) is attesting
- D) attest

24

During the English neoclassical period (1660–1789), many writers imitated the epic poetry and satires of ancient Greece and Rome. They were not the first in England to adopt the literary modes of classical _____ some of the most prominent figures of the earlier Renaissance period were also influenced by ancient Greek and Roman literature.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) antiquity, however
- B) antiquity, however,
- C) antiquity, however;
- D) antiquity; however,

25

English poet and Shakespeare contemporary John Donne's _____ much admired during his lifetime (1572–1631) and in the decades that followed, had, at the time of their enthusiastic rediscovery by the early twentieth-century modernists, been essentially gathering dust for the intervening 250 years.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) works were
- B) works, were
- C) works,
- D) works had been

26

Compared to that of alumina glass, _____ silica glass atoms are so far apart that they are unable to re-form bonds after being separated.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) silica glass is at a significant disadvantage due to its more dispersed atomic arrangement:
- B) silica glass has a more dispersed atomic arrangement, resulting in a significant disadvantage:
- C) a significant disadvantage of silica glass is that its atomic arrangement is more dispersed:
- D) silica glass's atomic arrangement is more dispersed, resulting in a significant disadvantage:

27

In November 1934, Amrita Sher-Gil was living in what must have seemed like the ideal city for a young artist: Paris. She was studying firsthand the color-saturated style of France's modernist masters and beginning to make a name for herself as a painter. _____ Sher-Gil longed to return to her childhood home of India; only there, she believed, could her art truly flourish.

Which choice completes the text with the most logical transition?

- A) Still,
- B) Therefore,
- C) Indeed,
- D) Furthermore,

28

In 1974, Mexican chemist Mario Molina and US chemist F. Sherwood Rowland discovered that chemicals called CFCs were harmful to the ozone layer. Their research was extremely influential in the fight against CFCs. _____ it laid the foundation for a 1987 treaty that phased out the use of CFCs across the globe.

Which choice completes the text with the most logical transition?

- A) Regardless,
- B) Specifically,
- C) However,
- D) Earlier,

29

With his room-sized installation *Unicorn/My Private Sky*, Norwegian artist Børre Sæthre succeeds in creating a whimsical yet perplexing experience. _____ when visitors set foot inside the fantastically blue room and encounter the life-sized stuffed unicorn preening at the far end of it, they are both dazzled and confused—as if stepping into a strange and enchanting new world.

Which choice completes the text with the most logical transition?

- A) Second,
- B) Instead,
- C) Indeed,
- D) Nevertheless,

30

While researching a topic, a student has taken the following notes:

- The Philadelphia and Lancaster Turnpike was a road built between 1792 and 1794.
- It was the first private turnpike in the United States.
- It connected the cities of Philadelphia and Lancaster in the state of Pennsylvania.
- It was sixty-two miles long.

The student wants to emphasize the distance covered by the Philadelphia and Lancaster Turnpike. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The sixty-two-mile-long Philadelphia and Lancaster Turnpike connected the Pennsylvania cities of Philadelphia and Lancaster.
- B) The Philadelphia and Lancaster Turnpike was the first private turnpike in the United States.
- C) The Philadelphia and Lancaster Turnpike, which connected two Pennsylvania cities, was built between 1792 and 1794.
- D) A historic Pennsylvania road, the Philadelphia and Lancaster Turnpike was completed in 1794.

31

While researching a topic, a student has taken the following notes:

- Most, but not all, of the Moon's oxygen comes from the Sun, via solar wind.
- Cosmochemist Kentaro Terada from Osaka University wondered if some of the unaccounted-for oxygen could be coming from Earth.
- In 2008, he analyzed data from the Japanese satellite Kaguya.
- Kaguya gathered data about gases and particles it encountered while orbiting the Moon.
- Based on the Kaguya data, Terada confirmed his suspicion that Earth is sending oxygen to the Moon.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) As it orbited the Moon, the Kaguya satellite collected data that was later analyzed by cosmochemist Kentaro Terada.
- B) Before 2008, Kentaro Terada wondered if the Moon was receiving some of its oxygen from Earth.
- C) Cosmochemist Kentaro Terada set out to determine whether some of the Moon's oxygen was coming from Earth.
- D) Kentaro Terada's study determined that Earth is sending a small amount of oxygen to the Moon.

32

While researching a topic, a student has taken the following notes:

- The factors that affect clutch size (the number of eggs laid at one time) have been well studied in birds but not in lizards.
- A team led by Shai Meiri of Tel Aviv University investigated which factors influence lizard clutch size.
- Meiri's team obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.
- Larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- Lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.

The student wants to emphasize the aim of the research study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Researchers wanted to know which factors influence lizard egg clutch size because such factors have been well studied in birds but not in lizards.
- B) After they obtained data for over 3,900 lizard species, researchers determined that larger clutch size was associated with environments in higher latitudes that have more seasonal change.
- C) We now know that lizards in higher-latitude environments may lay larger clutches to take advantage of shorter windows of favorable conditions.
- D) Researchers obtained clutch-size and habitat data for over 3,900 lizard species and analyzed the data with statistical models.

33

While researching a topic, a student has taken the following notes:

- British musicians John Lennon and Paul McCartney shared writing credit for numerous Beatles songs.
- Many Lennon-McCartney songs were actually written by either Lennon or McCartney, not by both.
- The exact authorship of specific parts of many Beatles songs, such as the verse for “In My Life,” is disputed.
- Mark Glickman, Jason Brown, and Ryan Song used statistical methods to analyze the musical content of Beatles songs.
- They concluded that there is 18.9% probability that McCartney wrote the verse for “In My Life,” stating that the verse is “consistent with Lennon’s songwriting style.”

The student wants to make a generalization about the kind of study conducted by Glickman, Brown, and Song. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Based on statistical analysis, Glickman, Brown, and Song claim that John Lennon wrote the verse of “In My Life.”
- B) There is only an 18.9% probability that Paul McCartney wrote the verse for “In My Life”; John Lennon is the more likely author.
- C) It is likely that John Lennon, not Paul McCartney, wrote the verse for “In My Life.”
- D) Researchers have used statistical methods to address questions of authorship within the field of music.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

Art scholars have noted that some colors seem to be more _____ viewers than others. For example, people tend to find paintings featuring blues and greens more appealing than paintings featuring yellows and oranges.

Which choice completes the text with the most logical and precise word or phrase?

- A) confusing for
- B) attractive to
- C) corrected by
- D) similar to

2

Researchers and conservationists stress that biodiversity loss due to invasive species is _____. For example, people can take simple steps such as washing their footwear after travel to avoid introducing potentially invasive organisms into new environments.

Which choice completes the text with the most logical and precise word or phrase?

- A) preventable
- B) undeniable
- C) common
- D) concerning

3

The process of mechanically recycling plastics is often considered _____ because of the environmental impact and the loss of material quality that often occurs. But chemist Takunda Chazovachii has helped develop a cleaner process of chemical recycling that converts superabsorbent polymers from diapers into a desirable reusable adhesive.

Which choice completes the text with the most logical and precise word or phrase?

- A) resilient
- B) inadequate
- C) dynamic
- D) satisfactory

4

In the Indigenous intercropping system known as the Three Sisters, maize, squash, and beans form an _____ web of relations: maize provides the structure on which the bean vines grow; the squash vines cover the soil, discouraging competition from weeds; and the beans aid their two “sisters” by enriching the soil with essential nitrogen.

Which choice completes the text with the most logical and precise word or phrase?

- A) indecipherable
- B) ornamental
- C) obscure
- D) intricate

5

Within baleen whale species, some individuals develop an accessory spleen—a seemingly functionless formation of splenic tissue outside the normal spleen. Given the formation’s greater prevalence among whales known to make deeper dives, some researchers hypothesize that its role isn’t _____; rather, the accessory spleen may actively support diving mechanisms.

Which choice completes the text with the most logical and precise word or phrase?

- A) replicable
- B) predetermined
- C) operative
- D) latent

6

In 2007, computer scientist Luis von Ahn was working on converting printed books into a digital format. He found that some words were distorted enough that digital scanners couldn’t recognize them, but most humans could easily read them. Based on that finding, von Ahn invented a simple security test to keep automated “bots” out of websites. The first version of the reCAPTCHA test asked users to type one known word and one of the many words scanners couldn’t recognize. Correct answers proved the users were humans and added data to the book-digitizing project.

Which choice best states the main purpose of the text?

- A) To discuss von Ahn’s invention of reCAPTCHA
- B) To explain how digital scanners work
- C) To call attention to von Ahn’s book-digitizing project
- D) To indicate how popular reCAPTCHA is

7

The following text is adapted from Gwendolyn Bennett’s 1926 poem “Street Lamps in Early Spring.”

Night wears a garment
All velvet soft, all violet blue...
And over her face she draws a veil
As shimmering fine as floating dew...
And here and there
In the black of her hair
The subtle hands of Night
Move slowly with their gem-starred light.

Which choice best describes the overall structure of the text?

- A) It presents alternating descriptions of night in a rural area and in a city.
- B) It sketches an image of nightfall, then an image of sunrise.
- C) It makes an extended comparison of night to a human being.
- D) It portrays how night changes from one season of the year to the next.

8

The following text is adapted from Jane Austen’s 1814 novel *Mansfield Park*. The speaker, Tom, is considering staging a play at home with a group of his friends and family.

We mean nothing but a little amusement among ourselves, just to vary the scene, and exercise our powers in something new. We want no audience, no publicity. We may be trusted, I think, in choosing some play most perfectly unexceptionable; and I can conceive no greater harm or danger to any of us in conversing in the elegant written language of some respectable author than in chattering in words of our own.

Which choice best states the main purpose of the text?

- A) To offer Tom’s assurance that the play will be inoffensive and involve only a small number of people
- B) To clarify that the play will not be performed in the manner Tom had originally intended
- C) To elaborate on the idea that the people around Tom lack the skills to successfully stage a play
- D) To assert that Tom believes the group performing the play will be able to successfully promote it

9

Text 1

Ecologists have long wondered how thousands of microscopic phytoplankton species can live together near ocean surfaces competing for the same resources. According to conventional wisdom, one species should emerge after outcompeting the rest. So why do so many species remain? Ecologists’ many efforts to explain this phenomenon still haven’t uncovered a satisfactory explanation.

Text 2

Ecologist Michael Behrenfeld and colleagues have connected phytoplankton’s diversity to their microscopic size. Because these organisms are so tiny, they are spaced relatively far apart from each other in ocean water and, moreover, experience that water as a relatively dense substance. This in turn makes it hard for them to move around and interact with one another. Therefore, says Behrenfeld’s team, direct competition among phytoplankton probably happens much less than previously thought.

Based on the texts, how would Behrenfeld and colleagues (Text 2) most likely respond to the “conventional wisdom” discussed in Text 1?

- A) By arguing that it is based on a misconception about phytoplankton species competing with one another
- B) By asserting that it fails to recognize that routine replenishment of ocean nutrients prevents competition between phytoplankton species
- C) By suggesting that their own findings help clarify how phytoplankton species are able to compete with larger organisms
- D) By recommending that more ecologists focus their research on how competition among phytoplankton species is increased with water density

10

The following text is adapted from Sylvia Acevedo's 2018 memoir *Path to the Stars: My Journey from Girl Scout to Rocket Scientist*. The narrator is traveling by car with her family to Mexico City. Mario and Laura are her brother and sister.

Mario and I played games to see how many different license plates we could spot, and Laura liked to look for children in the back seats of the cars we passed. We were used to the forty-five-minute drive to El Paso and familiar with the six-hour ride to Chihuahua, but I wondered what the long journey to Mexico City would be like.

©2018 by Sylvia Acevedo

According to the text, what did the narrator and Mario do while riding in the car?

- A) They read books.
- B) They sang songs.
- C) They went to sleep.
- D) They played games.

11

In the 1700s and 1800s, European composers experimented with volume in their musical works. They did so by increasing the number of musicians playing in the orchestra. For example, in some of his operas, German composer Richard Wagner added more horns, trombones, and tubas to the orchestra. With more instruments playing at the same time, the orchestra could play extremely loudly at key moments in his operas.

According to the text, how did Richard Wagner achieve moments of extremely high volume in his operas?

- A) By moving the performances of his operas from outdoor stages to indoor ones
- B) By increasing the number of musicians playing horns, trombones, and tubas in the orchestra
- C) By building a concert hall whose shape would cause sounds to echo
- D) By insisting that the singers undergo special training to sing for extended periods of time

12

Poetry in Classical Nahuatl, the language of the Aztec Empire, relies on *difrasismo*, or a parallel noun construction that conventionally operates as a single metaphor. For example, the common *difrasismo* in *cuauhtli in ocelotl* (literally, “the eagle, the jaguar”) signifies “warrior.” The device’s function is both formal—providing structure to lines of verse—and ritual: semantic relations among the two nouns and the concept they signify can be tenuous, as in the previous example, such that *difrasismos* are often only intelligible according to the conceptual associations observed in Aztec ceremonial culture.

Which statement about the *difrasismo* in *cuauhtli in ocelotl* is most strongly supported by the text?

- A) Its metaphorical significance derives from the semantic equivalence of the two nouns constituting the *difrasismo*.
- B) Its unintelligibility may cause its formal function within a line of verse to go unnoticed by present-day readers.
- C) Its apparent obscurity can be resolved when considered in the proper cultural context.
- D) Its frequency in Classical Nahuatl poetry confirms its intelligibility to the Aztec audience.

13

Eighteenth-century economist Adam Smith is famed for his metaphor of the invisible hand, which he putatively used to illustrate a robust model of how individuals produce aggregate benefits by pursuing their own economic interests. Note “putatively”: as Gavin Kennedy has shown, Smith deploys this metaphor only once in his economic writings—to make a narrow point about the then-dominant economic theory of mercantilism—and it was largely ignored until some twentieth-century economists eager to secure an intellectual pedigree for their views elevated it to a fully-fledged paradigm.

Which choice best states the main idea of the text?

- A) Although Smith is famed for his metaphor of the invisible hand, the metaphor was largely ignored until economists in the twentieth century came to realize that the metaphor was a robust model that anticipated their own views.
- B) Some twentieth-century economists gave Smith’s metaphor of the invisible hand a significance it does not have in Smith’s work, but it is nevertheless a useful model of how individuals produce aggregate benefits by pursuing their own economic interests.
- C) Smith’s metaphor of the invisible hand has been interpreted as a model of how individuals acting in their own interest produce aggregate benefits, but it was intended as a subtle critique of the economic theory of mercantilism.
- D) The reputation of Smith’s metaphor of the invisible hand is not due to the importance of the metaphor in Smith’s work but rather to the promotion of the metaphor by some later economists for their own ends.

Approximate Rates of Speech and
Information Conveyed for Five Languages

Language	Rate of speech (syllables per second)	Rate of information conveyed (bits per second)
Serbian	7.2	39.1
Spanish	7.7	42.0
Vietnamese	5.3	42.5
Thai	4.7	33.8
Hungarian	5.9	34.6

A group of researchers working in Europe, Asia, and Oceania conducted a study to determine how quickly different Eurasian languages are typically spoken (in syllables per second) and how much information they can effectively convey (in bits per second). They found that, although languages vary widely in the speed at which they are spoken, the amount of information languages can effectively convey tends to vary much less. Thus, they claim that two languages with very different spoken rates can nonetheless convey the same amount of information in a given amount of time.

Which choice best describes data from the table that support the researchers' claim?

- A) Among the five languages in the table, Thai and Hungarian have the lowest rates of speech and the lowest rates of information conveyed.
- B) Vietnamese conveys information at approximately the same rate as Spanish despite being spoken at a slower rate.
- C) Among the five languages in the table, the language that is spoken the fastest is also the language that conveys information the fastest.
- D) Serbian and Spanish are spoken at approximately the same rate, but Serbian conveys information faster than Spanish does.

Estimates of Tyrannosaurid Bite Force

Study	Year	Estimation method	Approximate bite force (newtons)
Cost et al.	2019	muscular and skeletal modeling	35,000–63,000
Gignac and Erickson	2017	tooth-bone interaction analysis	8,000–34,000
Meers	2002	body-mass scaling	183,000–235,000
Bates and Falkingham	2012	muscular and skeletal modeling	35,000–57,000

The largest tyrannosaurids—the family of carnivorous dinosaurs that includes *Tarbosaurus*, *Albertosaurus*, and, most famously, *Tyrannosaurus rex*—are thought to have had the strongest bites of any land animals in Earth’s history. Determining the bite force of extinct animals can be difficult, however, and paleontologists Paul Barrett and Emily Rayfield have suggested that an estimate of dinosaur bite force may be significantly influenced by the methodology used in generating that estimate.

Which choice best describes data from the table that support Barrett and Rayfield’s suggestion?

- A) The study by Meers used body-mass scaling and produced the lowest estimated maximum bite force, while the study by Cost et al. used muscular and skeletal modeling and produced the highest estimated maximum.
- B) In their study, Gignac and Erickson used tooth-bone interaction analysis to produce an estimated bite force range with a minimum of 8,000 newtons and a maximum of 34,000 newtons.
- C) The bite force estimates produced by Bates and Falkingham and by Cost et al. were similar to each other, while the estimates produced by Meers and by Gignac and Erickson each differed substantially from any other estimate.
- D) The estimated maximum bite force produced by Cost et al. exceeded the estimated maximum produced by Bates and Falkingham, even though both groups of researchers used the same method to generate their estimates.

16

Psychologists Dacher Keltner and Jonathan Haidt have argued that experiencing awe—a sensation of reverence and wonder typically brought on by perceiving something grand or powerful—can enable us to feel more connected to others and thereby inspire us to act more altruistically. Keltner, along with Paul K. Piff, Pia Dietze, and colleagues, claims to have found evidence for this effect in a recent study where participants were asked to either gaze up at exceptionally tall trees in a nearby grove (reported to be a universally awe-inspiring experience) or stare at the exterior of a nearby, nondescript building. After one minute, an experimenter deliberately spilled a box of pens nearby.

Which finding from the researchers' study, if true, would most strongly support their claim?

- A) Participants who had been looking at the trees helped the experimenter pick up significantly more pens than did participants who had been looking at the building.
- B) Participants who helped the experimenter pick up the pens used a greater number of positive words to describe the trees and the building in a postexperiment survey than did participants who did not help the experimenter.
- C) Participants who did not help the experimenter pick up the pens were significantly more likely to report having experienced a feeling of awe, regardless of whether they looked at the building or the trees.
- D) Participants who had been looking at the building were significantly more likely to notice that the experimenter had dropped the pens than were participants who had been looking at the trees.

17

The domestic sweet potato (*Ipomoea batatas*) descends from a wild plant native to South America. It also populates the Polynesian Islands, where evidence confirms that Native Hawaiians and other Indigenous peoples were cultivating the plant centuries before seafaring first occurred over the thousands of miles of ocean separating them from South America. To explain how the sweet potato was first introduced in Polynesia, botanist Pablo Muñoz-Rodríguez and colleagues analyzed the DNA of numerous varieties of the plant, concluding that Polynesian varieties diverged from South American ones over 100,000 years ago. Given that Polynesia was peopled only in the last three thousand years, the team concluded that _____

Which choice most logically completes the text?

- A) the cultivation of the sweet potato in Polynesia likely predates its cultivation in South America.
- B) Polynesian peoples likely acquired the sweet potato from South American peoples only within the last three thousand years.
- C) human activity likely played no role in the introduction of the sweet potato in Polynesia.
- D) Polynesian sweet potato varieties likely descend from a single South American variety that was domesticated, not wild.

18

The morphological novelty of echinoderms—marine invertebrates with radial symmetry, usually starlike, around a central point—impedes comparisons with most other animals, in which bilateral symmetry on an anterior-posterior (head to tail) axis through a trunk is typical. Particularly puzzling are sea stars, thought to have evolved a headless layout from a known bilateral origin. Applying genomic knowledge of *Saccoglossus kowalevskii* acorn worms (close relatives of sea stars, and thus expected to have similar markers for corresponding anatomical regions) to the body patterning genes of *Patiria miniata* sea stars, Laurent Formery et al. observed activity only in anterior genes across *P. miniata*'s entire body and some posterior genes limited to the edges, suggesting that _____

Which choice most logically completes the text?

- A) despite the greater prevalence of anterior genes in sea stars' genetic makeup, posterior genes active at the body's perimeter are primarily responsible for the starlike layout that distinguishes sea stars' radial symmetry from that of other echinoderms.
- B) contrary to the belief that they evolved from early ancestors with the bilateral form typical of many other animals, sea stars instead originated with an atypical body layout that was neither bilaterally nor radially symmetrical.
- C) although the two species are closely related, there is only minimal correspondence in the genetic markers for head, tail, and trunk region development in *P. miniata* sea stars and *S. kowalevskii* acorn worms.
- D) rather than undergoing changes resulting in the eventual elimination of a head region in their radial body plan, as previously assumed, sea stars' morphology evolved to completely lack a trunk and consist primarily of a head region.

19

To survive when water is scarce, embryos inside African turquoise killifish eggs _____ a dormant state known as diapause. In this state, embryonic development is paused for as long as two years—longer than the life span of an adult killifish.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) enter
- B) to enter
- C) having entered
- D) entering

20

Formed in 1967 to foster political and economic stability within the Asia-Pacific region, the Association of Southeast Asian Nations was originally made up of five members: Thailand, the Philippines, Singapore, Malaysia, and Indonesia. By the end of the 1990s, the organization _____ its initial membership.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) has doubled
- B) had doubled
- C) doubles
- D) will double

21

After the United Kingdom began rolling out taxes equivalent to a few cents on single-use plastic grocery bags in 2011, plastic-bag consumption decreased by up to ninety _____ taxes are subject to what economists call the “rebound effect”: as the change became normalized, plastic-bag use started to creep back up.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) percent, such
- B) percent and such
- C) percent. Such
- D) percent such

22

In 1966, Emmett Ashford became the first African American to umpire a Major League Baseball game. His energetic gestures announcing when a player had struck out and his habit of barreling after a hit ball to see if it would land out of _____ transform the traditionally solemn umpire role into a dynamic one.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) bounds helped
- B) bounds, helping
- C) bounds that helped
- D) bounds to help

23

The forty-seven geothermal springs of Arkansas’ Hot Springs National Park are sourced via a process known as natural groundwater recharge, in which rainwater percolates downward through the earth—in this case, the porous rocks of the hills around Hot _____ collect in a subterranean basin.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Springs to
- B) Springs: to
- C) Springs—to
- D) Springs, to

24

Over twenty years ago, in a landmark experiment in the psychology of choice, professor Sheena Iyengar set up a jam-tasting booth at a grocery store. The number of jams available for tasting _____ some shoppers had twenty-four different options, others only six. Interestingly, the shoppers with fewer jams to choose from purchased more jam.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) varied:
- B) varied,
- C) varied, while
- D) varied while

25

Nigerian author Buchi Emecheta's celebrated literary oeuvre includes *The Joys of Motherhood*, a novel about the changing roles of women in 1950s _____ a television play about the private struggles of a newlywed couple in Nigeria; and *Head Above Water*, her autobiography.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) Lagos, *A Kind of Marriage*,
- B) Lagos; *A Kind of Marriage*,
- C) Lagos, *A Kind of Marriage*:
- D) Lagos; *A Kind of Marriage*

26

Jetties—long, narrow structures that extend from a landmass into the water—are often constructed to protect coastlines from erosion. Jetties can sometimes have the opposite _____ obstructing the natural flow of sand along the shore can lead to increased erosion in some areas.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) effect, though;
- B) effect, though
- C) effect; though
- D) effect, though,

27

The Alaska Native Language Archive (ANLA) is known for its impressive audio collection. _____ the ANLA has more than 5,000 audio recordings of Native Alaskan languages dating as far back as 1943.

Which choice completes the text with the most logical transition?

- A) In fact,
- B) After,
- C) Regardless,
- D) Instead,

28

Etched into Peru's Nazca Desert are line drawings so large that they can only be fully seen from high above. Archaeologists have known of the lines since the 1920s, when a researcher spotted some from a nearby foothill, and they have been studying the markings ever since. _____ archaeologists' efforts are aided by drones that capture high-resolution aerial photographs of the lines.

Which choice completes the text with the most logical transition?

- A) Currently,
- B) In comparison,
- C) Still,
- D) However,

29

At two weeks old, the time their critical socialization period begins, wolves can smell but cannot yet see or hear. Domesticated dogs, _____ can see, hear, and smell by the end of two weeks. This relative lack of sensory input may help explain why wolves behave so differently around humans than dogs do: from a very young age, wolves are more wary and less exploratory.

Which choice completes the text with the most logical transition?

- A) in other words,
- B) for instance,
- C) by contrast,
- D) accordingly,

30

Upon first approaching artist Kurt Wenner's *Dies Irae*, a colorful scene painted on the surface of a cobblestone street in Mantua, Italy, one might assume a deep hole filled with life-sized, classically styled sculptures had opened up in the street. _____ by expertly applying the principles of perspective, Wenner created merely the illusion of depth.

Which choice completes the text with the most logical transition?

- A) Additionally,
- B) On the contrary,
- C) As a result,
- D) Next,

31

Economist Elinor Ostrom's studies of communities around the world have empirically demonstrated that common pool resources, such as grazing lands, can be sustainably managed by the people who use them (rather than through private entities or centralized governments). _____ Ostrom's work is a repudiation of the "tragedy of the commons," the view that individuals will inevitably overexploit a finite shared resource if given unfettered access to it.

Which choice completes the text with the most logical transition?

- A) By contrast,
- B) For example,
- C) That said,
- D) As such,

32

While researching a topic, a student has taken the following notes:

- Some sandstone arches in Utah’s Arches National Park have been defaced by tourists’ carvings.
- Park rangers can smooth away some carvings using power grinders.
- For deep carvings, power grinding is not always feasible because it can greatly alter or damage the rock.
- Park rangers can use an infilling technique, which involves filling in carvings with ground sandstone and a bonding agent.
- This technique is minimally invasive.

The student wants to explain an advantage of the infilling technique. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) To remove carvings from sandstone arches in Utah’s Arches National Park, power grinding is not always feasible.
- B) Filling in carvings with ground sandstone and a bonding agent is less invasive than smoothing them away with a power grinder, which can greatly alter or damage the sandstone arches.
- C) Park rangers can use a power grinding technique to smooth away carvings or fill them in with ground sandstone and a bonding agent.
- D) As methods for removing carvings from sandstone, power grinding and infilling differ in their level of invasiveness.

33

While researching a topic, a student has taken the following notes:

- *Las sergas de Esplandián* was a novel popular in sixteenth-century Spain.
- The novel featured a fictional island inhabited solely by Black women and known as California.
- That same century, Spanish explorers learned of an “island” off the west coast of Mexico.
- They called it California after the island in the novel.
- The “island” was actually the peninsula now known as Baja California (“Lower California”), which lies to the south of the US state of California.

The student wants to emphasize the role a misconception played in the naming of a place. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The novel *Las sergas de Esplandián* featured a fictional island known as California.
- B) To the south of the US state of California lies Baja California (“Lower California”), originally called California after a fictional place.
- C) In the sixteenth century, Spanish explorers learned of a peninsula off the west coast of Mexico and called it California.
- D) Thinking it was an island, Spanish explorers called a peninsula California after an island in a popular novel.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**



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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The following text is adapted from Ida B. Wells's 1970 autobiography *A Crusade for Justice*. Mr. Watts is a reference to George Frederic Watts, an English painter.

[Manchester's] art galleries are so arranged that the name of every picture is plainly seen and one has no need of a catalogue to pick out the name and the artist. This is a convenience to the general public, which other art galleries, which shall be nameless, might copy to advantage. To her treasure of art Manchester has added Mr. Watts' latest picture, the Good Samaritan.

©1970 by the University of Chicago Press

As used in the text, what does the word "arranged" most nearly mean?

- A) Organized
- B) Ranked
- C) Scheduled
- D) Discussed

2

One challenge of generating electricity from ocean waves is that wave power isn't _____: it varies in unpredictable ways that pose technological and planning problems for electricity generation.

Which choice completes the text with the most logical and precise word or phrase?

- A) accidental
- B) confident
- C) expensive
- D) consistent

3

Due to their often strange images, highly experimental syntax, and opaque subject matter, many of John Ashbery's poems can be quite difficult to _____ and thus are the object of heated debate among scholars.

Which choice completes the text with the most logical and precise word or phrase?

- A) delegate
- B) compose
- C) interpret
- D) renounce

4

Diego Velázquez was the leading artist in the court of King Philip IV of Spain during the seventeenth century, but his influence was hardly _____ Spain: realist and impressionist painters around the world employed his techniques and echoed elements of his style.

Which choice completes the text with the most logical and precise word or phrase?

- A) derived from
- B) recognized in
- C) confined to
- D) repressed by

5

Although science fiction was dominated mostly by white male authors when Octavia Butler, a Black woman, began writing, she did not view the genre as _____. Butler broke into the field with the publication of several short stories and her 1976 novel *Patternmaster*, and she later became the first science fiction writer to win a prestigious MacArthur Fellowship.

Which choice completes the text with the most logical and precise word or phrase?

- A) legitimate
- B) impenetrable
- C) compelling
- D) indecipherable

6

The following text is adapted from Cynthia Kadohata's 2004 novel *Kira-Kira*.

[Uncle Katsuhisa] was as loud as my father was quiet. Even when he wasn't talking, he made a lot of noise, clearing his throat and sniffing and tapping his fingers.

©2004 by Cynthia Kadohata

Which choice best describes the function of the underlined sentence?

- A) It lists the kinds of topics Uncle Katsuhisa enjoys discussing.
- B) It suggests that Uncle Katsuhisa dislikes meeting new people.
- C) It contrasts Uncle Katsuhisa with the narrator's father.
- D) It describes a conversation between the narrator and the narrator's father.

7

Wakako Yamauchi is best known for *And the Soul Shall Dance*, her 1977 play about a Japanese American family in Southern California. The play is based on a short story Yamauchi had published three years earlier. Adapting the story wasn't easy. Theater relies on dialogue between characters, but the original story features little dialogue and instead describes its characters' silent thoughts. To transform the story into a play, Yamauchi created situations where characters reveal their thoughts by speaking them aloud during conversations with each other.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It offers information about how Yamauchi adapted her short story into a play.
- B) It argues that Yamauchi's play influenced later playwrights.
- C) It explains why Yamauchi's short story is better known than the play adaptation is.
- D) It describes how Yamauchi chose the actors who performed in the play.

8

In response to concerns that some recent financial crises were exacerbated by consumers misunderstanding risks associated with credit cards, loans, and other financial products, policymakers in many countries have instituted risk-disclosure requirements on sellers of those products. Enrique Seira et al. investigated a variety of risk-disclosure messages sent to thousands of credit card customers and found that the messages had only small and short-lived effects on behavior. Seira et al. asserted that such effects may nevertheless be worth pursuing, given the negligible cost of messaging.

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It notes a factor that led Seira et al. to not dismiss risk-disclosure messaging altogether despite their evidence of its limited utility.
- B) It acknowledges a type of risk-disclosure messaging that Seira et al. may not have fully accounted for in their study.
- C) It describes a consideration that explains why Seira et al. recommended risk-disclosure messaging even though its effects may be small relative to its costs.
- D) It points out a circumstance that Seira et al. conceded may make risk-disclosure messaging more effective than their study suggests.

9

Text 1

When companies in the same industry propose merging with one another, they often claim that the merger will benefit consumers by increasing efficiency and therefore lowering prices. Economist Ying Fan investigated this notion in the context of the United States newspaper market. She modeled a hypothetical merger of Minneapolis-area newspapers and found that subscription prices would rise following a merger.

Text 2

Economists Dario Focarelli and Fabio Panetta have argued that research on the effect of mergers on prices has focused excessively on short-term effects, which tend to be adverse for consumers. Using the case of consumer banking in Italy, they show that over the long term (several years, in their study), the efficiency gains realized by merged companies do result in economic benefits for consumers.

Based on the texts, how would Focarelli and Panetta (Text 2) most likely respond to Fan's findings (Text 1)?

- A) They would recommend that Fan compare the near-term effect of a merger on subscription prices in the Minneapolis area with the effect of a merger in another newspaper market.
- B) They would argue that over the long term the expenses incurred by the merged newspaper company will also increase.
- C) They would encourage Fan to investigate whether the projected effect on subscription prices persists over an extended period.
- D) They would claim that mergers have a different effect on consumer prices in the newspaper industry than in most other industries.

10

Utah is home to Pando, a colony of about 47,000 quaking aspen trees that all share a single root system. Pando is one of the largest single organisms by mass on Earth, but ecologists are worried that its growth is declining in part because of grazing by animals. The ecologists say that strong fences could prevent deer from eating young trees and help Pando start thriving again.

According to the text, why are ecologists worried about Pando?

- A) It isn't growing at the same rate it used to.
- B) It isn't producing young trees anymore.
- C) It can't grow into new areas because it is blocked by fences.
- D) Its root system can't support many more new trees.

11

Recordings of Female Bottlenose
Dolphins with Their Calves

Dolphin ID	Recording year
FB07	2012
FB25	1989
FB43	1992
FB79	2018

In a study of bottlenose dolphins, biologist Laela S. Sayigh and a team of researchers analyzed recordings of female bottlenose dolphins interacting with their calves.

According to the table, in which year was the dolphin with the ID FB43 recorded with her calf?

- A) 1999
- B) 2012
- C) 2020
- D) 1992

12

Maximum Height of Maple Trees
When Fully Grown

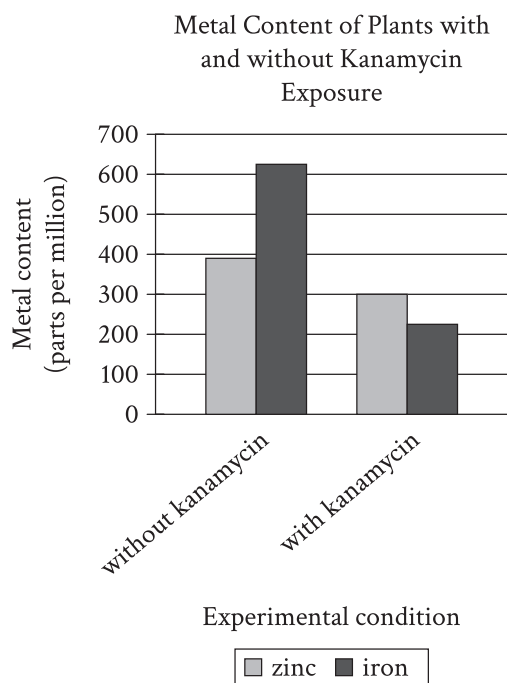
Tree type	Maximum height (feet)	Native to North America
Sugar maple	75	yes
Silver maple	70	yes
Red maple	60	yes
Japanese maple	25	no
Norway maple	50	no

For a school project, a forestry student needs to recommend a maple tree that is native to North America and won't grow more than 60 feet in height. Based on the characteristics of five common maple trees, she has decided to select a _____

Which choice most effectively uses data from the table to complete the text?

- A) silver maple.
- B) sugar maple.
- C) red maple.
- D) Norway maple.

13



Many plants lose their leaf color when exposed to kanamycin, an antibiotic produced by some soil microorganisms. Spelman College biologist Mentewab Ayalew and her colleagues hypothesized that plants' response to kanamycin exposure involves altering their uptake of metals, such as iron and zinc. The researchers grew two groups of seedlings of the plant *Arabidopsis thaliana*, half of which were exposed to kanamycin and half of which were a control group without exposure to kanamycin, and measured the plants' metal content five days after germination.

Which choice best describes data in the graph that support Ayalew and her colleagues' hypothesis?

- A) The control plants contained higher levels of zinc than iron, but plants exposed to kanamycin contained higher levels of iron than zinc.
- B) Both groups of plants contained more than 200 parts per million of both iron and zinc.
- C) Zinc levels were around 300 parts per million in the control plants but nearly 400 parts per million in the plants exposed to kanamycin.
- D) The plants exposed to kanamycin showed lower levels of iron and zinc than the control plants did.

14

Average Number and Duration of Torpor Bouts and Arousal Episodes for Alaska Marmots and Arctic Ground Squirrels, 2008–2011

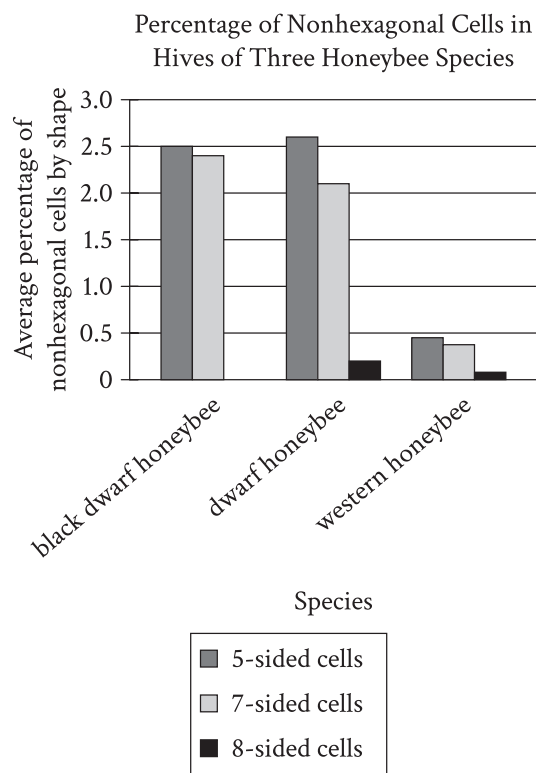
Feature	Alaska marmots	Arctic ground squirrels
torpor bouts	12	10.5
duration per bout	13.81 days	16.77 days
arousal episodes	11	9.5
duration per episode	21.2 hours	14.2 hours

When hibernating, Alaska marmots and Arctic ground squirrels enter a state called torpor, which minimizes the energy their bodies need to function. Often a hibernating animal will temporarily come out of torpor (called an arousal episode) and its metabolic rate will rise, burning more of the precious energy the animal needs to survive the winter. Alaska marmots hibernate in groups and therefore burn less energy keeping warm during these episodes than they would if they were alone. A researcher hypothesized that because Arctic ground squirrels hibernate alone, they would likely exhibit longer bouts of torpor and shorter arousal episodes than Alaska marmots.

Which choice best describes data from the table that support the researcher's hypothesis?

- A) The Alaska marmots' arousal episodes lasted for days, while the Arctic ground squirrels' arousal episodes lasted less than a day.
- B) The Alaska marmots and the Arctic ground squirrels both maintained torpor for several consecutive days per bout, on average.
- C) The Alaska marmots had shorter torpor bouts and longer arousal episodes than the Arctic ground squirrels did.
- D) The Alaska marmots had more torpor bouts than arousal episodes, but their arousal episodes were much shorter than their torpor bouts.

15



Honeybee hives consist mainly of hexagonal (six-sided) units called cells, in which queens lay eggs. Hexagonal cells for eggs that develop into nonreproductive workers are smaller than those for eggs that develop into reproductive drones, though the size difference varies by species. Difference in cell size results in a construction problem—it's hard to neatly connect sections of small cells to sections of large cells—that worsens as the difference increases. To fill in gaps between the sections when building a hive, bees rely on cells that have more or fewer than six sides. A student studying beehive structure consults data on three species, concluding that _____

Which choice most effectively uses data from the graph to complete the student's conclusion?

- A) cells for worker eggs are probably closer in size to cells for drone eggs in the hives of the western honeybee than in the hives of the dwarf honeybee and the black dwarf honeybee.
- B) both the western honeybee and the black dwarf honeybee probably reserve eight-sided cells for drone eggs, while the dwarf honeybee likely deposits drone eggs in seven-sided cells.
- C) the western honeybee probably relies on many more geometrical shapes when constructing cells than either the dwarf honeybee or the black dwarf honeybee does.
- D) the percentage of hexagonal cells is probably slightly lower in the hives of the western honeybee than in the hives of the dwarf honeybee and the black dwarf honeybee.

16

ALSOL is a microcredit program in Mexico that makes small loans to female entrepreneurs who lack the collateral and credit history to secure financing from conventional banks. Borrowers use their business proceeds to repay loans in equal weekly installments and incur no penalty for missed payments other than lack of access to larger loans. Economists Gustavo Barboza and Sandra Trejos analyzed ALSOL data and found that rural borrowers, who mostly make and sell handicrafts, miss payments more often than urban borrowers do, partly because they sell their goods less frequently than they could. Barboza and Trejos claim that this behavior reflects strategic decisions that enable rural women to increase their profits per unit sold.

Which finding, if true, would most directly support Barboza and Trejos's claim?

- A) Many marketplaces require entrepreneurs to pay marketplace operators a fixed percentage of each day's proceeds in exchange for permission to sell goods there.
- B) Rural entrepreneurs can typically sell their goods for higher prices in cities than in their home areas, but the number of people selling competing goods tends to be higher in cities.
- C) Due to the lower costs they incur, rural entrepreneurs tend to require smaller initial loans than urban entrepreneurs do.
- D) The cost to rural entrepreneurs to bring their goods to towns with marketplaces is high but largely independent of the number of goods they bring.

17

In documents called judicial opinions, judges explain the reasoning behind their legal rulings, and in those explanations they sometimes cite and discuss historical and contemporary philosophers. Legal scholar and philosopher Anita L. Allen argues that while judges are naturally inclined to mention philosophers whose views align with their own positions, the strongest judicial opinions consider and rebut potential objections; discussing philosophers whose views conflict with judges' views could therefore _____

Which choice most logically completes the text?

- A) allow judges to craft judicial opinions without needing to consult philosophical works.
- B) help judges improve the arguments they put forward in their judicial opinions.
- C) make judicial opinions more comprehensible to readers without legal or philosophical training.
- D) bring judicial opinions in line with views that are broadly held among philosophers.

18

Although military veterans make up a small proportion of the total population of the United States, they occupy a significantly higher proportion of the jobs in the civilian government. One possible explanation for this disproportionate representation is that military service familiarizes people with certain organizational structures that are also reflected in the civilian government bureaucracy, and this familiarity thus _____

Which choice most logically completes the text?

- A) makes civilian government jobs especially appealing to military veterans.
- B) alters the typical relationship between military service and subsequent career preferences.
- C) encourages nonveterans applying for civilian government jobs to consider military service instead.
- D) increases the number of civilian government jobs that require some amount of military experience to perform.

19

A member of the Cherokee Nation, Mary Golda Ross is renowned for her contributions to NASA's Planetary Flight Handbook, which _____ detailed mathematical guidance for missions to Mars and Venus.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) provided
- B) having provided
- C) to provide
- D) providing

20

The classic children's board game Chutes and Ladders is a version of an ancient Nepalese game, Paramapada Sopanapata. In both games, players encounter "good" or "bad" spaces while traveling along a path; landing on one of the good spaces _____ a player to skip ahead and arrive closer to the end goal.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) allows
- B) are allowing
- C) have allowed
- D) allow

21

In 1930, Japanese American artist Chiura Obata depicted the natural beauty of Yosemite National Park in two memorable woodcuts: *Evening at Carl Inn* and *Lake Basin in the High Sierra*. In 2019, _____ exhibited alongside 150 of Obata's other works in a single-artist show at the Smithsonian American Art Museum.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) it was
- B) they were
- C) this was
- D) some were

22

Journalists have dubbed Gil Scott-Heron the “godfather of rap,” a title that has appeared in hundreds of articles about him since the 1990s. Scott-Heron himself resisted the godfather _____ feeling that it didn’t encapsulate his devotion to the broader African American blues music tradition as well as “bluesologist,” the moniker he preferred.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) nickname, however
- B) nickname, however;
- C) nickname, however,
- D) nickname; however,

23

Stomata, tiny pore structures in a leaf that absorb gases needed for plant growth, open when guard cells surrounding each pore swell with water. In a pivotal 2007 article, plant cell _____ showed that lipid molecules called phosphatidylinositol phosphates are responsible for signaling guard cells to open stomata.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) biologist, Yuree Lee
- B) biologist Yuree Lee,
- C) biologist Yuree Lee
- D) biologist, Yuree Lee,

24

As cheesemaking practices spread throughout Europe and Asia during and after the Neolithic, divergent strategies for preserving milk _____ whereas rennet-coagulated cheesemaking became key to milk preservation in Europe and Southwest Asia, acid-heat coagulation methods became common among nomadic herding populations of the northeastern Eurasian steppe.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) emerged
- B) emerged and
- C) emerged:
- D) emerged,

25

Recent pollen analyses of the Aran Islands have led some researchers to propose that the now treeless islands were once wooded. This hypothesis _____ that certain trees, such as *P. sylvestris*, survived without interruption or human intervention throughout the Holocene cannot stand, researchers Michael O’Connell and Karen Molloy counter, unless other explanations can first be ruled out.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) suggesting
- B) suggested
- C) suggests
- D) has suggested

26

Researchers studying the “terra-cotta army,” the thousands of life-size statues of warriors found interred near the tomb of Emperor Qin Shi Huang of China, were shocked to realize that the shape of each statue’s ears, like the shape of each person’s ears, _____ unique.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are
- B) is
- C) were
- D) have been

27

Chimamanda Ngozi Adichie’s 2013 novel *Americanah* chronicles the divergent experiences of Ifemelu and Obinze, a young Nigerian couple, after high school. Ifemelu moves to the United States to attend a prestigious university. _____ Obinze travels to London, hoping to start a career there. However, frustrated with the lack of opportunities, he soon returns to Nigeria.

Which choice completes the text with the most logical transition?

- A) Meanwhile,
- B) Nevertheless,
- C) Secondly,
- D) In fact,

28

Some members of the US Supreme Court have resisted calls to televise the court’s oral arguments, concerned that the participants would be tempted to perform for the cameras (and thus lower the quality of the discourse). _____ the justices worry that most viewers would not even watch the full deliberations, only short clips that could be misinterpreted and mischaracterized.

Which choice completes the text with the most logical transition?

- A) However,
- B) Additionally,
- C) In comparison,
- D) For example,

29

The more diverse and wide ranging an animal’s behaviors, the larger and more energy demanding the animal’s brain tends to be. _____ from an evolutionary perspective, animals that perform only basic actions should allocate fewer resources to growing and maintaining brain tissue. The specialized subtypes of ants within colonies provide an opportunity to explore this hypothesis.

Which choice completes the text with the most logical transition?

- A) Subsequently,
- B) Besides,
- C) Nevertheless,
- D) Thus,

30

A firefly uses specialized muscles to draw oxygen into its lower abdomen through narrow tubes, triggering a chemical reaction whereby the oxygen combines with chemicals in the firefly's abdomen to produce a glow. _____ when the firefly stops drawing in oxygen, the reaction—and the glow—cease.

Which choice completes the text with the most logical transition?

- A) For instance,
- B) By contrast,
- C) Specifically,
- D) In conclusion,

31

While researching a topic, a student has taken the following notes:

- NASA uses rovers, large remote vehicles with wheels, to explore the surface of Mars.
- NASA's rovers can't explore regions inaccessible to wheeled vehicles.
- Rovers are also heavy, making them difficult to land on the planet's surface.
- Microprobes, robotic probes that weigh as little as 50 milligrams, could be deployed virtually anywhere on the surface of Mars.
- Microprobes have been proposed as an alternative to rovers.

The student wants to explain an advantage of microprobes. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Despite being heavy, NASA's rovers can land successfully on the surface of Mars.
- B) Microprobes, which weigh as little as 50 milligrams, could explore areas of Mars that are inaccessible to NASA's heavy, wheeled rovers.
- C) NASA currently uses its rovers on Mars, but microprobes have been proposed as an alternative.
- D) Though they are different sizes, both microprobes and rovers can be used to explore the surface of Mars.

32

While researching a topic, a student has taken the following notes:

- Severo Ochoa discovered the enzyme PNPase in 1955.
- PNPase is involved in both the creation and degradation of mRNA.
- Ochoa incorrectly hypothesized that PNPase provides the genetic blueprints for mRNA.
- The discovery of PNPase proved critical to deciphering the human genetic code.
- Deciphering the genetic code has led to a better understanding of how genetic variations affect human health.

The student wants to emphasize the significance of Ochoa's discovery. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Ochoa's 1955 discovery of PNPase proved critical to deciphering the human genetic code, leading to a better understanding of how genetic variations affect human health.
- B) Ochoa first discovered PNPase, an enzyme that he hypothesized contained the genetic blueprints for mRNA, in 1955.
- C) In 1955, Ochoa discovered the PNPase enzyme, which is involved in both the creation and degradation of mRNA.
- D) Though his discovery of PNPase was critical to deciphering the human genetic code, Ochoa incorrectly hypothesized that the enzyme was the source of mRNA's genetic blueprints.

33

While researching a topic, a student has taken the following notes:

- Cecilia Vicuña is a multidisciplinary artist.
- In 1971, her first solo art exhibition, *Pinturas, poemas y explicaciones*, was shown at the Museo Nacional de Bellas Artes in Santiago, Chile.
- Her poetry collection *Precario/Precarious* was published in 1983 by Tanam Press.
- Her poetry collection *Instan* was published in 2002 by Kelsey St. Press.
- She lives part time in Chile, where she was born, and part time in New York.

The student wants to introduce the artist's 1983 poetry collection. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Before she published the books *Precario/Precarious* (1983) and *Instan* (2002), Cecilia Vicuña exhibited visual art at the Museo Nacional de Bellas Artes in Santiago, Chile.
- B) Cecilia Vicuña is a true multidisciplinary artist whose works include numerous poetry collections and visual art exhibitions.
- C) Published in 1983 by Tanam Press, *Precario/Precarious* is a collection of poetry by the multidisciplinary artist Cecilia Vicuña.
- D) In 1971, Cecilia Vicuña exhibited her first solo art exhibition, *Pinturas, poemas y explicaciones*, in Chile, her country of birth.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The following text is from Anita Desai's 2011 novella *Translator Translated*. While working on her translation of a novel written in Odia (a language of India) into English, the narrator looks out her window at night to clear her mind.

I tried to distract myself with these sights of the ordinary world, but in my mind it was the lines I had been translating and the lines that I had been writing that remained in the forefront. I longed for sleep to obliterate them but it eluded me. Perhaps everything would be normal again once I had sent off the manuscript, I thought, and looked forward to completing the work.

©2011 by Anita Desai

As used in the text, what does the word “completing” most nearly mean?

- A) Destroying
- B) Finishing
- C) Advertising
- D) Rejecting

2

Predatory animals differ widely in how they _____ food for their young. Some leave dead prey nearby for their young to consume, some bring live prey to their young, and some feed their young directly from their own mouths.

Which choice completes the text with the most logical and precise word or phrase?

- A) avoid
- B) guess
- C) provide
- D) describe

3

Nigerian American author Teju Cole's _____ his two passions—photography and the written word—culminates in his 2017 book, *Blind Spot*, which evocatively combines his original photographs from his travels with his poetic prose.

Which choice completes the text with the most logical and precise word or phrase?

- A) indifference to
- B) enthusiasm for
- C) concern about
- D) surprise at

4

Artist Marilyn Dingle’s intricate, coiled baskets are _____ sweetgrass and palmetto palm. Following a Gullah technique that originated in West Africa, Dingle skillfully winds a thin palm frond around a bunch of sweetgrass with the help of a “sewing bone” to create the basket’s signature look that no factory can reproduce.

Which choice completes the text with the most logical and precise word or phrase?

- A) indicated by
- B) handmade from
- C) represented by
- D) collected with

5

Beginning in the 1950s, Navajo Nation legislator Annie Dodge Wauneka continuously worked to promote public health; this _____ effort involved traveling throughout the vast Navajo homeland and writing a medical dictionary for speakers of *Diné bizaad*, the Navajo language.

Which choice completes the text with the most logical and precise word or phrase?

- A) impartial
- B) offhand
- C) persistent
- D) mandatory

6

Oral histories—whether they consist of interviews or recordings of songs and stories—can offer researchers a rich view of people’s everyday experiences. For her book about coal mining communities in Kentucky during the twentieth century, Karida Brown therefore relied in part on interviews with coal miners and their families. By doing so, she gained valuable insights into her subjects’ day-to-day lives.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It provides a little-known geographical fact about Kentucky.
- B) It argues that Karida Brown is an expert on United States politics.
- C) It presents a major historical event that took place in the twentieth century.
- D) It describes how Karida Brown benefited from incorporating oral history in her book.

7

The following text is from Georgia Douglas Johnson's 1922 poem "Benediction."

Go forth, my son,
Winged by my heart's desire!
Great reaches, yet unknown,
Await
For your possession.
I may not, if I would,
Retrace the way with you,
My pilgrimage is through,
But life is calling you!

Which choice best states the main purpose of the text?

- A) To express hope that a child will have the same accomplishments as his parent did
- B) To suggest that raising a child involves many struggles
- C) To warn a child that he will face many challenges throughout his life
- D) To encourage a child to embrace the experiences life will offer

8

The following text is from Edith Wharton's 1905 novel *The House of Mirth*. Lily Bart and a companion are walking through a park.

Lily had no real intimacy with nature, but she had a passion for the appropriate and could be keenly sensitive to a scene which was the fitting background of her own sensations. The landscape outspread below her seemed an enlargement of her present mood, and she found something of herself in its calmness, its breadth, its long free reaches. On the nearer slopes the sugar-maples wavered like pyres of light; lower down was a massing of grey orchards, and here and there the lingering green of an oak-grove.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It creates a detailed image of the physical setting of the scene.
- B) It establishes that a character is experiencing an internal conflict.
- C) It makes an assertion that the next sentence then expands on.
- D) It illustrates an idea that is introduced in the previous sentence.

9

Individual elephants and Arctic herbivores such as caribou tend to have fixed geographic ranges throughout their lifetimes, which had prompted some researchers to speculate that the Arctic woolly mammoth, an extinct elephantid, might have exhibited similar behavior. Mammoth tusks grew in sequential layers, incorporating ingested minerals and organics, and so each ivory stratum reflects the ratio of strontium isotopes ($^{87}\text{Sr}/^{86}\text{Sr}$) in the local environment; thus, the sequence of strata shows where the animal roamed during life. Recent analysis of the strontium ratios in the strata of one Arctic woolly mammoth tusk in relation to the geographic distribution of strontium ratios in the environment shows the animal's range begin to expand as it reached sexual maturity, only to contract again in its final 1.5 years.

Which choice best describes the function of the underlined statement in the text as a whole?

- A) It discusses a characteristic shared by certain animals in order to explain why researchers raised a possibility that turned out not to be supported by data described later in the text.
- B) It illustrates a pattern of behavior among certain animals in order to present a theory about exceptions to that pattern that is weakened by a finding described later in the text.
- C) It describes a similarity in the behavior of certain animals in order to show why a method described later in the text did not reveal whether another animal also showed that behavior.
- D) It introduces a trait shared by certain animals in order to contextualize a hypothesis about the origin of that trait that is advanced later in the text.

10

In the 1960s, Gloria Richardson led a movement to promote racial equality. Her involvement in this effort was inspired by her daughter, Donna Richardson. In 1961, Donna joined protests organized by the Student Nonviolent Coordinating Committee in Cambridge, Maryland. Following her daughter, Gloria joined these protests too. Gloria soon became the cochair of the Cambridge Nonviolent Action Committee. She was also the leader of what became known as the Cambridge movement.

According to the text, what did Gloria Richardson lead?

- A) The Cambridge movement
- B) Her daughter Donna's high school
- C) Protests to support environmental protections
- D) A new business in Cambridge, Maryland

11

The following text is from Jane Austen’s 1811 novel *Sense and Sensibility*. Elinor lives with her younger sisters and her mother, Mrs. Dashwood.

Elinor, this eldest daughter, whose advice was so effectual, possessed a strength of understanding, and coolness of judgment, which qualified her, though only nineteen, to be the counsellor of her mother, and enabled her frequently to counteract, to the advantage of them all, that eagerness of mind in Mrs. Dashwood which must generally have led to imprudence. She had an excellent heart;—her disposition was affectionate, and her feelings were strong; but she knew how to govern them: it was a knowledge which her mother had yet to learn; and which one of her sisters had resolved never to be taught.

According to the text, what is true about Elinor?

- A) Elinor often argues with her mother but fails to change her mind.
- B) Elinor can be overly sensitive with regard to family matters.
- C) Elinor thinks her mother is a bad role model.
- D) Elinor is remarkably mature for her age.

12

Culinary anthropologist Vertamae Smart-Grosvenor may be known for her decades of work in national public television and radio, but her book *Vibration Cooking: or, the Travel Notes of a Geechee Girl* is likely her most influential project. The 1970 book, whose title refers to Smart-Grosvenor’s roots in the Low Country of South Carolina, was unusual for its time. It combined memoir, recipes, travel writing, and social commentary and challenged notions about conventions of food and cooking. Long admired by many, the book and its author have shaped contemporary approaches to writing about cuisine.

Which choice best describes the main idea of the text?

- A) Smart-Grosvenor’s unconventional book *Vibration Cooking: or, the Travel Notes of a Geechee Girl* is an important contribution to food writing.
- B) Smart-Grosvenor held many different positions over her life, including reporter and food writer.
- C) Smart-Grosvenor’s groundbreaking book *Vibration Cooking: or, the Travel Notes of a Geechee Girl* didn’t receive the praise it deserved when it was first published in 1970.
- D) Smart-Grosvenor was a talented chef whose work inspired many people to start cooking for themselves.

13

The following text is adapted from Charles W. Chesnutt's 1901 novel *The Marrow of Tradition*.

Mrs. Ochiltree was a woman of strong individuality, whose comments upon her acquaintance[s], present or absent, were marked by a frankness at times no less than startling. This characteristic caused her to be more or less avoided. Mrs. Ochiltree was aware of this sentiment on the part of her acquaintance[s], and rather exulted in it.

Based on the text, what is true about Mrs. Ochiltree's acquaintances?

- A) They try to refrain from discussing topics that would upset Mrs. Ochiltree.
- B) They are unable to spend as much time with Mrs. Ochiltree as she would like.
- C) They are too preoccupied with their own concerns to speak with Mrs. Ochiltree.
- D) They are likely offended by what Mrs. Ochiltree has said about them.

14

"Mrs. Spring Fragrance" is a 1912 short story by Sui Sin Far. In the story, Mrs. Spring Fragrance, a Chinese immigrant living in Seattle, is traveling in California. In letters to her husband and friend, she demonstrates her concern for what's happening at her home in Seattle while she is away: _____

Which quotation from Mrs. Spring Fragrance's letters most effectively illustrates the claim?

- A) "My honorable cousin is preparing for the Fifth Moon Festival, and wishes me to compound for the occasion some American 'fudge,' for which delectable sweet, made by my clumsy hands, you have sometimes shown a slight prejudice."
- B) "Next week I accompany Ah Oi to the beauteous town of San José. There will we be met by the son of the Illustrious Teacher."
- C) "Forget not to care for the cat, the birds, and the flowers. Do not eat too quickly nor fan too vigorously now that the weather is warming."
- D) "I am enjoying a most agreeable visit, and American friends, as also our own, strive benevolently for the accomplishment of my pleasure."

15

When digging for clams, their primary food, sea otters damage the roots of eelgrass plants growing on the seafloor. Near Vancouver Island in Canada, the otter population is large and well established, yet the eelgrass meadows are healthier than those found elsewhere off Canada's coast. To explain this, conservation scientist Erin Foster and colleagues compared the Vancouver Island meadows to meadows where otters are absent or were reintroduced only recently. Finding that the Vancouver Island meadows have a more diverse gene pool than the others do, Foster hypothesized that damage to eelgrass roots increases the plant's rate of sexual reproduction; this, in turn, boosts genetic diversity, which benefits the meadows' health overall.

Which finding, if true, would most directly undermine Foster's hypothesis?

- A) At some sites in the study, eelgrass meadows are found near otter populations that are small and have only recently been reintroduced.
- B) At several sites not included in the study, there are large, well-established sea otter populations but no eelgrass meadows.
- C) At several sites not included in the study, eelgrass meadows' health correlates negatively with the length of residence and size of otter populations.
- D) At some sites in the study, the health of plants unrelated to eelgrass correlates negatively with the length of residence and size of otter populations.

16

In the twentieth century, ethnographers made a concerted effort to collect Mexican American folklore, but they did not always agree about that folklore's origins. Scholars such as Aurelio Espinosa claimed that Mexican American folklore derived largely from the folklore of Spain, which ruled Mexico and what is now the southwestern United States from the sixteenth to early nineteenth centuries. Scholars such as Américo Paredes, by contrast, argued that while some Spanish influence is undeniable, Mexican American folklore is mainly the product of the ongoing interactions of various cultures in Mexico and the United States.

Which finding, if true, would most directly support Paredes's argument?

- A) The folklore that the ethnographers collected included several songs written in the form of a *décima*, a type of poem originating in late sixteenth-century Spain.
- B) Much of the folklore that the ethnographers collected had similar elements from region to region.
- C) Most of the folklore that the ethnographers collected was previously unknown to scholars.
- D) Most of the folklore that the ethnographers collected consisted of *corridos*—ballads about history and social life—of a clearly recent origin.

17

Among social animals that care for their young, such as chickens, macaque monkeys, and humans, newborns appear to show an innate attraction to faces and face-like stimuli. Elisabetta Versace and her colleagues used an image of three black dots arranged in the shape of eyes and a nose or mouth to test whether this trait also occurs in *Testudo* tortoises, which live alone and do not engage in parental care. They found that tortoise hatchlings showed a significant preference for the image, suggesting that _____

Which choice most logically completes the text?

- A) face-like stimuli are likely perceived as harmless by newborns of social species that practice parental care but as threatening by newborns of solitary species without parental care.
- B) researchers should not assume that an innate attraction to face-like stimuli is necessarily an adaptation related to social interaction or parental care.
- C) researchers can assume that the attraction to face-like stimuli that is seen in social species that practice parental care is learned rather than innate.
- D) newly hatched *Testudo* tortoises show a stronger preference for face-like stimuli than adult *Testudo* tortoises do.

18

Aptamers—synthetic DNA or RNA molecules that bind to target molecules—can be used to test for foodborne bacterial pathogens, though their specificity (the probability of returning a negative result in the absence of the focal pathogen) in real-world foods has been unclear. Sandeep Somvanshi et al. fabricated test paper incorporating aptamers targeting strain O157:H7 of the bacteria *Escherichia coli*; the paper shifts from pink to purple as the aptamers bind to target molecules. Somvanshi et al. tested the paper in store-bought pear juice they treated with *E. coli* O157:H7, other strains of *E. coli*, or other bacteria species. Following exposure, the paper from the O157:H7 test was purple while papers from the other tests were pink, suggesting that _____

Which choice most logically completes the text?

- A) aptamer-based tests in real-world foods are more likely to show a high degree of specificity if the focal pathogen is *E. coli* O157:H7 than if the focal pathogen is another strain of *E. coli* or another species.
- B) uncertainty about the specificity of aptamer-based tests for pathogens in real-world foods may be due to the similarity between *E. coli* O157:H7 and other *E. coli* strains.
- C) the specificity of the tests in a real-world food was unaffected by the aptamers' tendency to bind to different strains of *E. coli*.
- D) the aptamers successfully bound to *E. coli* O157:H7 and the tests displayed a high degree of specificity in a real-world food.

19

British scientists James Watson and Francis Crick won the Nobel Prize in part for their 1953 paper announcing the double helix structure of DNA, but it is misleading to say that Watson and Crick discovered the double helix. _____ findings were based on a famous X-ray image of DNA fibers, “Photo 51,” developed by X-ray crystallographer Rosalind Franklin and her graduate student Raymond Gosling.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) They’re
- B) It’s
- C) Their
- D) Its

20

In the historical novel *The Surrender Tree*, Cuban American author Margarita Engle uses poetry rather than prose _____ the true story of Cuban folk hero Rosa La Bayamesa.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) tells
- B) told
- C) is telling
- D) to tell

21

Interest in mechanotransduction, the mechanism by which cells sense and convert mechanical stimuli into biochemical signals, is expanding because of innovative work by biomedical scientists—many of whom, like neuroscience and biophysics expert Elba Serrano, _____ this mechanism to better understand how the body’s neurological and biomechanical systems interact.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) is studying
- B) has studied
- C) study
- D) studies

22

In crafting her fantasy fiction, Nigerian-born British author Helen Oyeyemi has drawn inspiration from the classic nineteenth-century fairy tales of the Brothers Grimm. Her 2014 novel *Boy, Snow, Bird*, for instance, is a complex retelling of the story of Snow White, while her 2019 novel _____ offers a delicious twist on the classic tale of Hansel and Gretel.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) *Gingerbread*—
- B) *Gingerbread*,
- C) *Gingerbread*
- D) *Gingerbread*:

23

Sociologist Todd Gitlin co-opted the term “recombinant,” normally used in reference to genetic engineering, to describe serialized television shows of the 1980s. Gitlin’s use of the term referenced TV studios’ practice of repackaging successful narrative formulas as new _____ even shows that varied only slightly from other shows still attracted sizeable audiences.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) content, in that era
- B) content; in that era,
- C) content in that era,
- D) content, in that era,

24

When external forces are applied to common glass made from silicates, energy builds up around minuscule defects in the material, resulting in fractures. Recently, engineer Erkkka Frankberg of Tampere University in Finland used the chemical _____ to make a glassy solid that can withstand higher strain than silicate glass can before fracturing.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) compound, aluminum oxide
- B) compound aluminum oxide,
- C) compound, aluminum oxide,
- D) compound aluminum oxide

25

In the late nineteenth and early twentieth centuries, automobiles were commonly referred to as horseless carriages after the older technology they still resembled. Known as the Brass Era, this period in automotive design is remembered for its grandeur and artistry, its vehicles _____ by collectors for their ornate detailing and gleaming brass fittings.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are highly prized
- B) had been highly prized
- C) highly prized
- D) were highly prized

26

With the development of new technologies that use natural resources more efficiently, the overall consumption of those resources might be expected to decrease. Economists have observed that improvements in efficiency often correlate negatively with resource _____ efficiency gains, lowering the cost of use, may increase demand to the extent that resource consumption ultimately rises.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) conservation, though,
- B) conservation; though
- C) conservation, though;
- D) conservation, though

27

Before California's 1911 election to approve a proposition granting women the right to vote, activists across the state sold tea to promote the cause of suffrage. In San Francisco, the Woman's Suffrage Party sold Equality Tea at local fairs. _____ in Los Angeles, activist Nancy Tuttle Craig, who ran one of California's largest grocery store firms, distributed Votes for Women Tea.

Which choice completes the text with the most logical transition?

- A) For example,
- B) To conclude,
- C) Similarly,
- D) In other words,

28

Earth's auroras—colorful displays of light seen above the northern and southern poles—result, broadly speaking, from the Sun's activity. _____ the Sun releases charged particles that are captured by Earth's magnetic field and channeled toward the poles. These particles then collide with atoms in the atmosphere, causing the atoms to emit auroral light.

Which choice completes the text with the most logical transition?

- A) Specifically,
- B) Similarly,
- C) Nevertheless,
- D) Hence,

29

While researching a topic, a student has taken the following notes:

- The Seikan Tunnel is a rail tunnel in Japan.
- It connects the island of Honshu to the island of Hokkaido.
- It is roughly 33 miles long.
- The Channel Tunnel is a rail tunnel in Europe.
- It connects Folkestone, England, to Coquelles, France.
- It is about 31 miles long.

The student wants to compare the lengths of the two rail tunnels. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Some of the world's rail tunnels, including one tunnel that extends from Folkestone, England, to Coquelles, France, are longer than 30 miles.
- B) The Seikan Tunnel is roughly 33 miles long, while the slightly shorter Channel Tunnel is about 31 miles long.
- C) The Seikan Tunnel, which is roughly 33 miles long, connects the Japanese islands of Honshu and Hokkaido.
- D) Both the Seikan Tunnel, which is located in Japan, and the Channel Tunnel, which is located in Europe, are examples of rail tunnels.

30

While researching a topic, a student has taken the following notes:

- Pinnipeds, which include seals, sea lions, and walruses, live in and around water.
- Pinnipeds are descended not from sea animals but from four-legged, land-dwelling carnivores.
- Canadian paleobiologist Natalia Rybczynski recently found a fossil with four legs, webbed toes, and the skull and teeth of a seal.
- Rybczynski refers to her rare find as a “transitional fossil.”
- The fossil illustrates an early stage in the evolution of pinnipeds from their land-dwelling ancestors.

The student wants to emphasize the fossil’s significance. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Canadian paleobiologist Natalia Rybczynski’s fossil has the skull and teeth of a seal, which, like sea lions and walruses, is a pinniped.
- B) Pinnipeds are descended from four-legged, land-dwelling carnivores; a fossil that resembles both was recently found.
- C) Having four legs but the skull and teeth of a seal, the rare fossil illustrates an early stage in the evolution of pinnipeds from their land-dwelling ancestors.
- D) A “transitional fossil” was recently found by paleobiologist Natalia Rybczynski.

31

While researching a topic, a student has taken the following notes:

- Physicist Muluneh Abebe was working on a garment suited for both warm and cold conditions.
- He analyzed the emissivity, or ability to emit heat, of the materials he planned to use.
- Abebe found that reflective metal fibers emitted almost no heat and had an emissivity of 0.02.
- He found that silicon carbide fibers absorbed large amounts of heat and had an emissivity of 0.74.
- The amount of heat a material absorbs is equal to the amount of heat it emits.

The student wants to contrast the emissivity of reflective metal fibers with that of silicon carbide fibers. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The ability of reflective metal fibers and silicon carbide fibers to emit heat was determined by an analysis of each material’s emissivity.
- B) The amount of heat a material absorbs is equal to the amount it emits, as evidenced in Abebe’s analyses.
- C) Though the reflective metal fibers and silicon carbide fibers had different rates of emissivity, Abebe planned to use both in a garment.
- D) Whereas the reflective metal fibers had an emissivity of just 0.02, the silicon carbide fibers absorbed large amounts of heat, resulting in an emissivity of 0.74.

32

While researching a topic, a student has taken the following notes:

- In 2020, theater students at Radford and Virginia Tech chose an interactive, online format to present a play about woman suffrage activists.
- Their “Women and the Vote” website featured an interactive digital drawing of a Victorian-style house.
- Audiences were asked to focus on a room of their choice and select from that room an artifact related to the suffrage movement.
- One click took them to video clips, songs, artwork, and texts associated with the artifact.
- The play was popular with audiences because the format allowed them to control the experience.

The student wants to explain an advantage of the “Women and the Vote” format. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) “Women and the Vote” featured a drawing of a Victorian-style house with several rooms, each containing suffrage artifacts.
- B) To access video clips, songs, artwork, and texts, audiences had to first click on an artifact.
- C) The “Women and the Vote” format appealed to audiences because it allowed them to control the experience.
- D) Using an interactive format, theater students at Radford and Virginia Tech created “Women and the Vote,” a play about woman suffrage activists.

33

While researching a topic, a student has taken the following notes:

- Ducklings expend up to 62.8% less energy when swimming in a line behind their mother than when swimming alone.
- The physics behind this energy savings hasn’t always been well understood.
- Naval architect Zhiming Yuan used computer simulations to study the effect of the mother duck’s wake.
- The study revealed that ducklings are pushed in a forward direction by the wake’s waves.
- Yuan determined this push reduces the effect of wave drag on the ducklings by 158%.

The student wants to present the study and its methodology. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) A study revealed that ducklings, which expend up to 62.8% less energy when swimming in a line behind their mother, also experience 158% less drag.
- B) Seeking to understand how ducklings swimming in a line behind their mother save energy, Zhiming Yuan used computer simulations to study the effect of the mother duck’s wake.
- C) Zhiming Yuan studied the physics behind the fact that by being pushed in a forward direction by waves, ducklings save energy.
- D) Naval architect Zhiming Yuan discovered that ducklings are pushed in a forward direction by the waves of their mother’s wake, reducing the effect of drag by 158%.

STOP

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Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

The general store was essential to daily life in the rural United States during the 1800s because it provided the supplies that the people living in nearby communities needed. Also, the store was a _____ of information. People socializing at the general store would share news and help spread it throughout their communities.

Which choice completes the text with the most logical and precise word or phrase?

- A) source
- B) rival
- C) condition
- D) waste

2

For painter Jacob Lawrence, being _____ was an important part of the artistic process. Because he paid close attention to all the details of his Harlem neighborhood, Lawrence's artwork captured nuances in the beauty and vitality of the Black experience during the Harlem Renaissance and the Great Migration.

Which choice completes the text with the most logical and precise word or phrase?

- A) skeptical
- B) observant
- C) critical
- D) confident

3

Former astronaut Ellen Ochoa says that although she doesn't have a definite idea of when it might happen, she _____ that humans will someday need to be able to live in other environments than those found on Earth. This conjecture informs her interest in future research missions to the moon.

Which choice completes the text with the most logical and precise word or phrase?

- A) demands
- B) speculates
- C) doubts
- D) establishes

4

The parasitic dodder plant increases its reproductive success by flowering at the same time as the host plant it has latched onto. In 2020, Jianqiang Wu and his colleagues determined that the tiny dodder achieves this _____ with its host by absorbing and utilizing a protein the host produces when it is about to flower.

Which choice completes the text with the most logical and precise word or phrase?

- A) synchronization
- B) hibernation
- C) prediction
- D) moderation

5

Barring major archaeological discoveries, we are unlikely to ever have _____ account of ancient Egypt under the female pharaoh Hatshepsut, as much of the evidence of her reign was deliberately destroyed by her successors.

Which choice completes the text with the most logical and precise word or phrase?

- A) an imaginative
- B) a superficial
- C) an exhaustive
- D) a questionable

6

Jazz tap is a dance form that was first developed in African American communities. Jazz tap was heavily influenced by jazz music, which became widely popular in the United States in the 1920s. Tap dancers were inspired by jazz music's quick rhythms and by the way jazz musicians would make up melodies as they played. As jazz music continued to develop in the 1930s and 1940s, jazz tap evolved with it. Because of jazz music's influence, jazz tap quickly developed into a dance form that was very different from earlier kinds of tap dance.

Which choice best states the main purpose of the text?

- A) It explains why audiences prefer some kinds of music over others.
- B) It discusses the development of a dance form.
- C) It describes how to play a musical instrument.
- D) It emphasizes the popularity of a famous dancer.

7

The north celestial pole (NCP)—the fixed point around which stars in the Northern Hemisphere (including the Sun) appear to rotate—is discernible only at night. Inspired by the navigational strategies of some insects and birds, researchers devised a method for locating the NCP in daytime using skylight polarization, which occurs as atmospheric particles scatter sunlight. A polarimetric camera captures images of polarization patterns, which rotate as the Sun’s position in the sky changes; temporal variances across images can then be used to determine an observer’s latitude and bearing relative to the NCP.

Which choice best describes the overall structure of the text?

- A) It illustrates how most navigational tools utilize the NCP, recounts how researchers discovered that certain animals are able to navigate without using the NCP, and then proposes that this discovery could be used to avoid problems in navigation associated with reliance on the NCP.
- B) It presents a celestial-based method of navigation, enumerates the comparative benefits of an alternative method used by certain animals that is based on an unrelated natural occurrence, and then indicates how researchers assessed the relative accuracy of the two methods.
- C) It explains how the NCP is typically located, emphasizes a key difference between how humans and certain animals use the NCP for navigation, and then suggests an alternative way of using the NCP to improve existing navigational instruments.
- D) It notes an obstacle to observing an astronomical phenomenon, mentions a navigational ability of certain animals that inspired a solution to that obstacle, and then explains how researchers used an optical device to mimic that ability.

8

The following text is adapted from Zora Neale Hurston’s 1921 short story “John Redding Goes to Sea.” John is a child who lives in a town in the woods.

Perhaps ten-year-old John was puzzling to the folk there in the Florida woods for he was an imaginative child and fond of day-dreams. The St. John River flowed a scarce three hundred feet from his back door. On its banks at this point grow numerous palms, luxuriant magnolias and bay trees. On the bosom of the stream float millions of delicately colored hyacinths. [John Redding] loved to wander down to the water’s edge, and, casting in dry twigs, watch them sail away down stream to Jacksonville, the sea, the wide world and [he] wanted to follow them.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A) It provides an extended description of a location that John likes to visit.
- B) It reveals that some residents of John’s town are confused by his behavior.
- C) It illustrates the uniqueness of John’s imagination compared to the imaginations of other children.
- D) It suggests that John longs to experience a larger life outside the Florida woods.

9

Astronomers are confident that the star Betelgeuse will eventually consume all the helium in its core and explode in a supernova. They are much less confident, however, about when this will happen, since that depends on internal characteristics of Betelgeuse that are largely unknown. Astrophysicist Sarafina El-Badry Nance and colleagues recently investigated whether acoustic waves in the star could be used to determine internal stellar states but concluded that this method could not sufficiently reveal Betelgeuse’s internal characteristics to allow its evolutionary state to be firmly fixed.

Which choice best describes the function of the second sentence in the overall structure of the text?

- A) It describes a serious limitation of the method used by Nance and colleagues.
- B) It presents the central finding reported by Nance and colleagues.
- C) It identifies the problem that Nance and colleagues attempted to solve but did not.
- D) It explains how the work of Nance and colleagues was received by others in the field.

10

Text 1

Astronomer Mark Holland and colleagues examined four white dwarfs—small, dense remnants of past stars—in order to determine the composition of exoplanets that used to orbit those stars. Studying wavelengths of light in the white dwarf atmospheres, the team reported that traces of elements such as lithium and sodium support the presence of exoplanets with continental crusts similar to Earth’s.

Text 2

Past studies of white dwarf atmospheres have concluded that certain exoplanets had continental crusts. Geologist Keith Putirka and astronomer Siyi Xu argue that those studies unduly emphasize atmospheric traces of lithium and other individual elements as signifiers of the types of rock found on Earth. The studies don’t adequately account for different minerals made up of various ratios of those elements, and the possibility of rock types not found on Earth that contain those minerals.

Based on the texts, how would Putirka and Xu (Text 2) most likely characterize the conclusion presented in Text 1?

- A) As unexpected, because it was widely believed at the time that white dwarf exoplanets lack continental crusts
- B) As premature, because researchers have only just begun trying to determine what kinds of crusts white dwarf exoplanets had
- C) As questionable, because it rests on an incomplete consideration of potential sources of the elements detected in white dwarf atmospheres
- D) As puzzling, because it’s unusual to successfully detect lithium and sodium when analyzing wavelengths of light in white dwarf atmospheres

11

The following text is from David Barclay Moore's 2022 novel *Holler of the Fireflies*. The narrator has just arrived at summer camp, which is far away from his home.

This place was different than I thought it would be. I'd never been somewhere like this before. I did feel scared, but also excited.

©2022 by David Barclay Moore

According to the text, how does the narrator feel about being at summer camp?

- A) He feels overjoyed.
- B) He feels peaceful.
- C) He feels both scared and excited.
- D) He feels both angry and jealous.

12

The following text is adapted from Oscar Wilde's 1891 novel *The Picture of Dorian Gray*. Dorian Gray is taking his first look at a portrait that Hallward has painted of him.

Dorian passed listlessly in front of his picture and turned towards it. When he saw it he drew back, and his cheeks flushed for a moment with pleasure. A look of joy came into his eyes, as if he had recognized himself for the first time. He stood there motionless and in wonder, dimly conscious that Hallward was speaking to him, but not catching the meaning of his words. The sense of his own beauty came on him like a revelation. He had never felt it before.

According to the text, what is true about Dorian?

- A) He wants to know Hallward's opinion of the portrait.
- B) He is delighted by what he sees in the portrait.
- C) He prefers portraits to other types of paintings.
- D) He is uncertain of Hallward's talent as an artist.

13

Choctaw/Cherokee artist Jeffrey Gibson turns punching bags used by boxers into art by decorating them with beadwork and elements of Native dressmaking. These elements include leather fringe and jingles, the metal cones that cover the dresses worn in the jingle dance, a women's dance of the Ojibwe people. Thus, Gibson combines an object commonly associated with masculinity (a punching bag) with art forms traditionally practiced by women in most Native communities (beadwork and dressmaking). In this way, he rejects the division of male and female gender roles.

Which choice best describes Gibson's approach to art, as presented in the text?

- A) He draws from traditional Native art forms to create his original works.
- B) He has been influenced by Native and non-Native artists equally.
- C) He finds inspiration from boxing in designing the dresses he makes.
- D) He rejects expectations about color and pattern when incorporating beadwork.

14

O Pioneers! is a 1913 novel by Willa Cather. In the novel, Cather portrays Alexandra Bergson as having a deep emotional connection to her natural surroundings: _____

Which quotation from *O Pioneers!* most effectively illustrates the claim?

- A) "She had never known before how much the country meant to her. The chirping of the insects down in the long grass had been like the sweetest music. She had felt as if her heart were hiding down there, somewhere, with the quail and the plover and all the little wild things that crooned or buzzed in the sun. Under the long shaggy ridges, she felt the future stirring."
- B) "Alexandra talked to the men about their crops and to the women about their poultry. She spent a whole day with one young farmer who had been away at school, and who was experimenting with a new kind of clover hay. She learned a great deal."
- C) "Alexandra drove off alone. The rattle of her wagon was lost in the howling of the wind, but her lantern, held firmly between her feet, made a moving point of light along the highway, going deeper and deeper into the dark country."
- D) "It was Alexandra who read the papers and followed the markets, and who learned by the mistakes of their neighbors. It was Alexandra who could always tell about what it had cost to fatten each steer, and who could guess the weight of a hog before it went on the scales closer than John Bergson [her father] himself."

15

The novelist Toni Morrison was the first Black woman to work as an editor at the publishing company Random House, from 1967 to 1983. A scholar asserts that one of Morrison's likely aims during her time as an editor was to strengthen the presence of Black writers on the list of Random House's published authors.

Which finding, if true, would most strongly support the scholar's claim?

- A) The percentage of authors published by Random House who were Black rose in the early 1970s and stabilized throughout the decade.
- B) Black authors who were interviewed in the 1980s and 1990s were highly likely to cite Toni Morrison's novels as a principal influence on their work.
- C) The novels written by Toni Morrison that were published after 1983 sold significantly more copies and received wider critical acclaim than the novels she wrote that were published before 1983.
- D) Works that were edited by Toni Morrison during her time at Random House displayed stylistic characteristics that distinguished them from works that were not edited by Morrison.

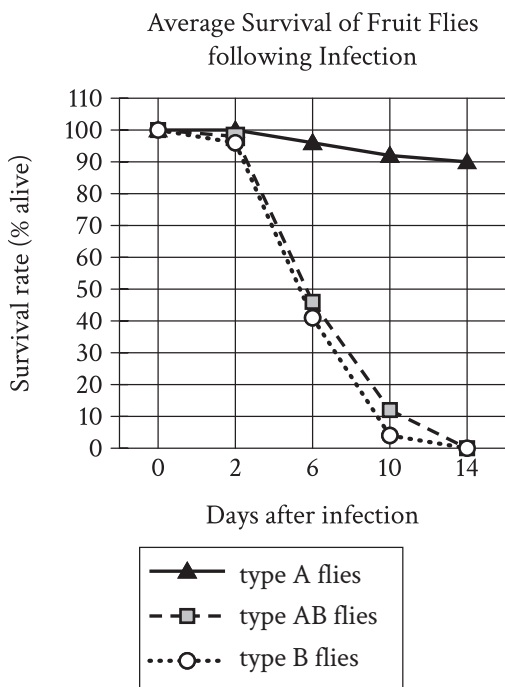
16

Archaeologist Petra Vaiglova, anthropologist Xinyi Liu, and their colleagues investigated the domestication of farm animals in China during the Bronze Age (approximately 2000 to 1000 BCE). By analyzing the chemical composition of the bones of sheep, goats, and cattle from this era, the team determined that wild plants made up the bulk of sheep's and goats' diets, while the cattle's diet consisted largely of millet, a crop cultivated by humans. The team concluded that cattle were likely raised closer to human settlements, whereas sheep and goats were allowed to roam farther away.

Which finding, if true, would most strongly support the team's conclusion?

- A) Analysis of the animal bones showed that the cattle's diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.
- B) Further investigation of sheep and goat bones revealed that their diets consisted of small portions of millet as well.
- C) Cattle's diets generally require larger amounts of food and a greater variety of nutrients than do sheep's and goats' diets.
- D) The diets of sheep, goats, and cattle were found to vary based on what the farmers in each Bronze Age settlement could grow.

17



In a study of the evolution of *DptA* and *DptB*—*Diptericin* genes encoding antimicrobial peptides that combat pathogens and foster beneficial microbes in fruit flies (*Drosophila*)—researchers assessed *Drosophila melanogaster* resistance to pathogenic infections by *Providencia rettgeri* and *Acetobacter sicerae*, bacteria common in the flies' environments. Subjects included flies identified by mutations silencing *DptA*, *DptB*, or both *DptA* and *DptB* (termed types A, B, and AB, respectively). In conjunction with the observation that resistance to *P. rettgeri* correlates with *DptA* activity but is not significantly affected by *DptB* activity, data in the graph of survival rates post-*A. sicerae* infection suggest that _____

Which completion of the text is best supported by data in the graph?

- A) *DptA* confers defense against *A. sicerae* regardless of the presence of *DptB*.
- B) *DptB* protects against only one bacteria species, whereas *DptA* protects against multiple species.
- C) *DptB* may have developed as a specific defense against *A. sicerae*.
- D) defense against *A. sicerae* is strongest when both *DptA* and *DptB* are present.

18

Euphorbia esula (leafy spurge) is a Eurasian plant that has become invasive in North America, where it displaces native vegetation and sickens cattle. *E. esula* can be controlled with chemical herbicides, but that approach can also kill harmless plants nearby. Recent research on introducing engineered DNA into plant species to inhibit their reproduction may offer a path toward exclusively targeting *E. esula*, consequently _____

Which choice most logically completes the text?

- A) making individual *E. esula* plants more susceptible to existing chemical herbicides.
- B) enhancing the ecological benefits of *E. esula* in North America.
- C) enabling cattle to consume *E. esula* without becoming sick.
- D) reducing invasive *E. esula* numbers without harming other organisms.

19

A team of biologists led by Jae-Hoon Jung, Antonio D. Barbosa, and Stephanie Hutin investigated the mechanism that allows *Arabidopsis thaliana* (thale cress) plants to accelerate flowering at high temperatures. They replaced the protein ELF3 in the plants with a similar protein found in another species (stiff brome) that, unlike *A. thaliana*, displays no acceleration in flowering with increased temperature. A comparison of unmodified *A. thaliana* plants with the altered plants showed no difference in flowering at 22° Celsius, but at 27° Celsius, the unmodified plants exhibited accelerated flowering while the altered ones did not, which suggests that _____

Which choice most logically completes the text?

- A) temperature-sensitive accelerated flowering is unique to *A. thaliana*.
- B) *A. thaliana* increases ELF3 production as temperatures rise.
- C) ELF3 enables *A. thaliana* to respond to increased temperatures.
- D) temperatures of at least 22° Celsius are required for *A. thaliana* to flower.

20

The Alvarez theory, developed in 1980 by physicist Luis Walter Alvarez and his geologist son Walter Alvarez, maintained that the secondary effects of an asteroid impact caused many dinosaurs and other animals to die _____ it left unexplored the question of whether unrelated volcanic activity might have also contributed to the mass extinctions.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) out but
- B) out, but
- C) out
- D) out,

21

Typically, underlines, scribbles, and notes left in the margins by a former owner lower a book's _____ when the former owner is a famous poet like Walt Whitman, such markings, known as marginalia, can be a gold mine to literary scholars.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) value, but
- B) value
- C) value,
- D) value but

22

In winter, the diets of Japanese macaques, also known as snow monkeys, are influenced more by food availability than by food preference. Although the monkeys prefer to eat vegetation and land-dwelling invertebrates, those food sources may become unavailable because of extensive snow and ice cover, _____ the monkeys to hunt for marine animals in any streams that have not frozen over.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) forces
- B) to force
- C) forcing
- D) forced

23

While many video game creators strive to make their graphics ever more _____ others look to the past, developing titles with visuals inspired by the “8-bit” games of the 1980s and 1990s. (The term “8-bit” refers to a console whose processor could only handle eight bits of data at once.)

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) lifelike but
- B) lifelike
- C) lifelike,
- D) lifelike, but

24

Food and the sensation of taste are central to Monique Truong's novels. In *The Book of Salt*, for example, the exiled character of Binh connects to his native Saigon through the food he prepares, while in *Bitter in the Mouth*, the character of Linda _____ a form of synesthesia whereby the words she hears evoke tastes.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) experienced
- B) had experienced
- C) experiences
- D) will be experiencing

25

Along with carbon dioxide concentration and temperature, light intensity affects the chemical reaction rate of _____ as light intensity increases, so does the rate at which the reactants (water and carbon dioxide) are converted into their products (glucose and oxygen).

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) photosynthesis and
- B) photosynthesis,
- C) photosynthesis:
- D) photosynthesis

26

In Marisol's 1968 sculpture *Mi Mama y Yo*, gone are the types of pop culture references that made the Parisian-born Venezuelan American artist a star at the height of the pop art movement. In _____ place is a far more personal subject: a sculptural depiction of the artist as a young girl with her mother.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) its
- B) they're
- C) their
- D) it's

27

The ghazal, a poetic form originating in seventh-century Arabic poetry, has an intricate structure. The twentieth-century Kashmiri American poet Agha Shahid Ali explains that each one of a ghazal's couplets, while adhering to the patterns of rhyme (*qafia*) and refrain (*radif*) established in the poem's opening lines (*matla*), _____ thematically and logically autonomous, resulting in a poem with "a stringently formal disunity."

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) is
- B) were
- C) have been
- D) are

28

Organisms have evolved a number of surprising adaptations to ensure their survival in adverse conditions. Tadpole shrimp (*Triops longicaudatus*) embryos, _____ can pause development for over ten years during extended periods of drought.

Which choice completes the text with the most logical transition?

- A) in contrast,
- B) for example,
- C) meanwhile,
- D) consequently,

29

When Chinese director Chloé Zhao accepted the Oscar in 2021 for her film *Nomadland*, she made Academy Award history. _____ only one other woman, Kathryn Bigelow of the United States, had been named best director at the Oscars, making Zhao the second woman and the first Asian woman to win the award.

Which choice completes the text with the most logical transition?

- A) As a result,
- B) Previously,
- C) However,
- D) Likewise,

30

If the formation of Earth's mantle had been purely a product of core differentiation—whereby heavier elements sink toward the core and lighter elements rise—the upper mantle would be depleted of heavy siderophile elements. Siderophiles are much more abundant in the mantle than predicted in that model, however. _____ extraterrestrial material containing siderophiles, likely from asteroid or comet impacts, almost certainly accreted to Earth following core differentiation.

Which choice completes the text with the most logical transition?

- A) That said,
- B) Hence,
- C) For example,
- D) Likewise,

31

While researching a topic, a student has taken the following notes:

- In 2013, archaeologists studied cat bone fragments they had found in the ruins of Quanhucun, a Chinese farming village.
- The fragments were estimated to be 5,300 years old.
- A chemical analysis of the fragments revealed that the cats had consumed large amounts of grain.
- The grain consumption is evidence that the Quanhucun cats may have been domesticated.

The student wants to present the Quanhucun study and its conclusions. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) As part of a 2013 study of cat domestication, a chemical analysis was conducted on cat bone fragments found in Quanhucun, China.
- B) A 2013 analysis of cat bone fragments found in Quanhucun, China, suggests that cats there may have been domesticated 5,300 years ago.
- C) In 2013, archaeologists studied what cats in Quanhucun, China, had eaten more than 5,000 years ago.
- D) Cat bone fragments estimated to be 5,300 years old were found in Quanhucun, China, in 2013.

32

While researching a topic, a student has taken the following notes:

- Gaspar Enriquez is an artist.
- He specializes in portraits of Mexican Americans.
- A portrait is an artistic representation of a person.
- Enriquez completed a painting of the sculptor Luis Jimenez in 2003.
- He completed a drawing of the writer Rudolfo Anaya in 2016.

The student wants to emphasize a difference between the two portraits. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) The portraits, or artistic representations, of Luis Jimenez and Rudolfo Anaya were both completed by Enriquez in the early 2000s.
- B) Enriquez has completed portraits of numerous Mexican Americans, including sculptor Luis Jimenez and writer Rudolfo Anaya.
- C) While both are by Enriquez, the 2003 portrait of Luis Jimenez is a painting, and the 2016 portrait of Rudolfo Anaya is a drawing.
- D) Luis Jimenez was a Mexican American sculptor, and Rudolfo Anaya was a Mexican American writer.

33

While researching a topic, a student has taken the following notes:

- The Gullah are a group of African Americans who have lived in parts of the southeastern United States since the 18th century.
- Gullah culture is influenced by West African and Central African traditions.
- Louise Miller Cohen is a Gullah historian, storyteller, and preservationist.
- She founded the Gullah Museum of Hilton Head Island, South Carolina, in 2003.
- Vermelle Rodrigues is a Gullah historian, artist, and preservationist.
- She founded the Gullah Museum of Georgetown, South Carolina, in 2003.

The student wants to emphasize the duration and purpose of Cohen's and Rodrigues's work. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) At the Gullah Museums in Hilton Head Island and Georgetown, South Carolina, visitors can learn more about the Gullah people who have lived in the region for centuries.
- B) Louise Miller Cohen and Vermelle Rodrigues have worked to preserve the culture of the Gullah people, who have lived in the United States since the 18th century.
- C) Since 2003, Louise Miller Cohen and Vermelle Rodrigues have worked to preserve Gullah culture through their museums.
- D) Influenced by the traditions of West and Central Africa, Gullah culture developed in parts of the southeastern United States in the 18th century.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**

Reading and Writing

33 QUESTIONS

DIRECTIONS

The questions in this section address a number of important reading and writing skills. Each question includes one or more passages, which may include a table or graph. Read each passage and question carefully, and then choose the best answer to the question based on the passage(s).

All questions in this section are multiple-choice with four answer choices. Each question has a single best answer.

1

In the early 1800s, the Cherokee scholar Sequoyah created the first script, or writing system, for an Indigenous language in the United States. Because it represented the sounds of spoken Cherokee so accurately, his script was easy to learn and thus quickly achieved _____ use: by 1830, over 90 percent of the Cherokee people could read and write it.

Which choice completes the text with the most logical and precise word or phrase?

- A) widespread
- B) careful
- C) unintended
- D) infrequent

2

Researchers have struggled to pinpoint specific causes for hiccups, which happen when a person's diaphragm contracts _____. However, neuroscientist Kimberley Whitehead has found that these uncontrollable contractions may play an important role in helping infants regulate their breathing.

Which choice completes the text with the most logical and precise word or phrase?

- A) involuntarily
- B) beneficially
- C) strenuously
- D) smoothly

3

The province of Xoconochco was situated on the Pacific coast, hundreds of kilometers southeast of Tenochtitlan, the capital of the Aztec Empire. Because Xoconochco's location within the empire was so _____, cacao and other trade goods produced there could reach the capital only after a long overland journey.

Which choice completes the text with the most logical and precise word or phrase?

- A) unobtrusive
- B) concealed
- C) approximate
- D) peripheral

4

Proposals to raise the age at which retirees begin receiving government transfers of funds are generally discussed in terms of the effects on transfer recipients, but Andria Smythe has argued that delaying such transfers could _____ wealth creation among working adults by lengthening the period in which they are providing financial support to their nonworking parents.

Which choice completes the text with the most logical and precise word or phrase?

- A) stymie
- B) compound
- C) disparage
- D) outstrip

5

Political blogs with conspicuous ideological alignments became an integral component of US media in the early 2000s. While some commentators lauded this development, asserting that such blogs had a welcome transparency missing from traditional news, less _____ observers countered that such blogs tended to ideological extremes that exacerbated political polarization to problematic levels.

Which choice completes the text with the most logical and precise word or phrase?

- A) sanguine
- B) recalcitrant
- C) misanthropic
- D) earnest

6

The following text is adapted from Pam Muñoz Ryan’s 2020 novel *Mañanaland*. In the village where Max lives, there is an old fortress called La Reina. Children in the village say that the fortress is haunted.

For as long as he could remember, Max had begged Papá [his father] to take him to see La Reina and the ruins up close. He’d be a hero among his friends if he was the first boy to cross the haunted gates! Just because Papá didn’t believe in ghosts didn’t mean they weren’t there. Maybe this summer Papá would finally take him. He *was* almost twelve.

©2020 by Pam Muñoz Ryan

Which choice best describes the overall purpose of the text?

- A) To portray how proud Max’s father is of Max
- B) To explain why Max doesn’t want to grow up yet
- C) To criticize Max for disliking summer
- D) To show how much Max wants to visit La Reina

7

The following text is adapted from George Bernard Shaw’s 1912 play *Pygmalion*. Henry Higgins has just arrived at the house of his mother (Mrs. Higgins). She is expecting her friends to visit soon.

MRS. HIGGINS: I’m serious, Henry. You offend all my friends: they stop coming whenever they meet you.

HIGGINS: Nonsense! I know I have no small talk; but people don’t mind.

MRS. HIGGINS: Oh! don’t they? Small talk indeed! What about your large talk? Really, dear, you mustn’t stay.

Which choice best states the main purpose of the text?

- A) To describe what Henry’s mother does when she goes out with her friends
- B) To show that Henry’s mother wants him to leave
- C) To present a detailed account of what Henry’s home looks like
- D) To explain why Henry often visits his mother

8

The following text is from Charlotte Forten Grimké's 1888 poem "At Newport."

Oh, deep delight to watch the gladsome waves
Exultant leap upon the rugged rocks;
Ever repulsed, yet ever rushing on—
Filled with a life that will not know defeat;
To see the glorious hues of sky and sea.
The distant snowy sails, glide spirit like,
Into an unknown world, to feel the sweet
Enchantment of the sea thrill all the soul,
Clearing the clouded brain, making the heart
Leap joyous as it own bright, singing waves!

Which choice best describes the function of the underlined portion in the text as a whole?

- A) It portrays the surroundings as an imposing and intimidating scene.
- B) It characterizes the sea's waves as a relentless and enduring force.
- C) It conveys the speaker's ambivalence about the natural world.
- D) It draws a contrast between the sea's waves and the speaker's thoughts.

9

Believing that living in an impractical space can heighten awareness and even improve health, conceptual artists Madeline Gins and Shusaku Arakawa designed an apartment building in Japan to be more fanciful than functional. A kitchen counter is chest-high on one side and knee-high on the other; a ceiling has a door to nowhere. The effect is disorienting but invigorating: after four years there, filmmaker Nobu Yamaoka reported significant health benefits.

Which choice best states the main idea of the text?

- A) Although inhabiting a home surrounded by fanciful features such as those designed by Gins and Arakawa can be rejuvenating, it is unsustainable.
- B) Designing disorienting spaces like those in the Gins and Arakawa building is the most effective way to create a physically stimulating environment.
- C) As a filmmaker, Yamaoka has long supported the designs of conceptual artists such as Gins and Arakawa.
- D) Although impractical, the design of the apartment building by Gins and Arakawa may improve the well-being of the building's residents.

10

The following text is adapted from Lewis Carroll's 1889 satirical novel *Sylvie and Bruno*. A crowd has gathered outside a room belonging to the Warden, an official who reports to the Lord Chancellor.

One man, who was more excited than the rest, flung his hat high into the air, and shouted (as well as I could make out) "Who roar for the Sub-Warden?" Everybody roared, but whether it was for the Sub-Warden, or not, did not clearly appear: some were shouting "Bread!" and some "Taxes!", but no one seemed to know what it was they really wanted.

All this I saw from the open window of the Warden's breakfast-saloon, looking across the shoulder of the Lord Chancellor.

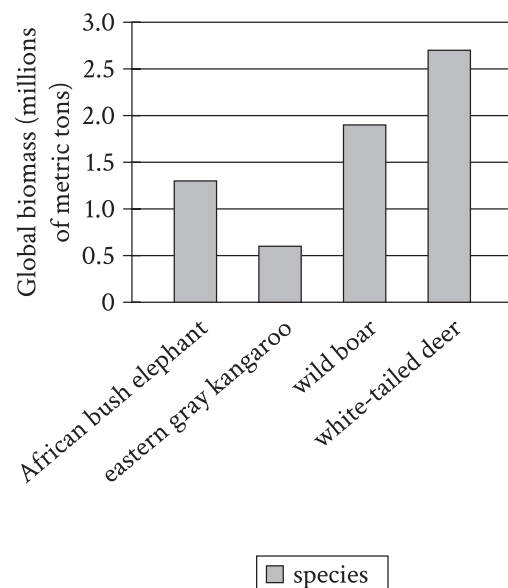
"What can it all mean?" he kept repeating to himself. "I never heard such shouting before—and at this time of the morning, too! And with such unanimity!"

Based on the text, how does the Lord Chancellor respond to the crowd?

- A) He asks about the meaning of the crowd's shouting, even though he claims to know what the crowd wants.
- B) He indicates a desire to speak to the crowd, even though the crowd has asked to speak to the Sub-Warden.
- C) He expresses sympathy for the crowd's demands, even though the crowd's shouting annoys him.
- D) He describes the crowd as being united, even though the crowd clearly appears otherwise.

11

Top Four Species of Wild Land Mammals by Global Biomass



Global biomass is the total mass of living material, such as animals and plants, on Earth. A team of scientists estimated the global biomass, by species, of various wild land mammals. The team found that the species with the highest global biomass is the _____

Which choice most effectively uses data from the graph to complete the sentence?

- A) wild boar.
- B) eastern gray kangaroo.
- C) African bush elephant.
- D) white-tailed deer.

12

Number and Origin of Clamshell Tools Found at Different Levels Below the Surface in Neanderthal Cave

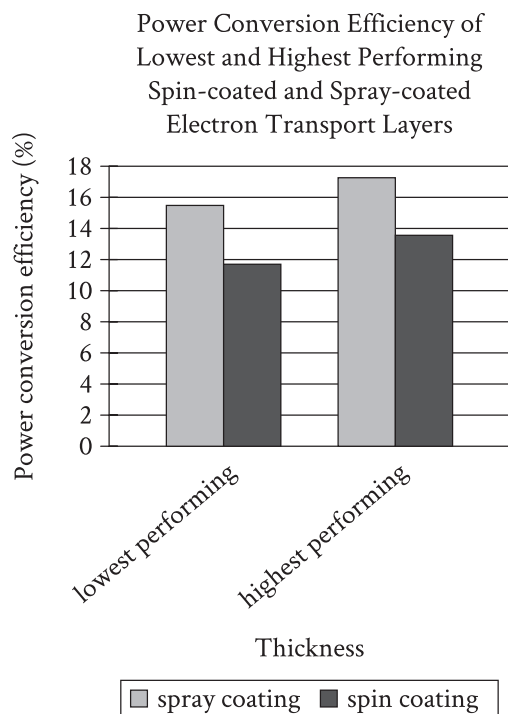
Depth of tools found below surface in cave (meters)	Clamshells that Neanderthals collected from the beach	Clamshells that Neanderthals harvested from the seafloor
3–4	99	33
6–7	1	0
4–5	2	0
2–3	7	0
5–6	18	7

Studying tools unearthed at a cave site on the western coast of Italy, archaeologist Paola Villa and colleagues have determined that prehistoric Neanderthal groups fashioned them from shells of clams that they harvested from the seafloor while wading or diving or that washed up on the beach. Clamshells become thin and eroded as they wash up on the beach, while those on the seafloor are smooth and sturdy, so the research team suspects that Neanderthals prized the tools made with seafloor shells. However, the team also concluded that those tools were likely more challenging to obtain, noting that _____

Which choice most effectively uses data from the table to support the research team's conclusion?

- A) at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that shells were easier to collect from the beach than to harvest from the seafloor.
- B) the highest number of tools were at a depth of 3–4 meters below the surface, which suggests that the Neanderthal population at the site was highest during the related period of time.
- C) at each depth below the surface in the cave, the difference in the numbers of tools of each type suggests that Neanderthals preferred to use clamshells from the beach because of their durability.
- D) the higher number of tools at depths of 5–6 meters below the surface in the cave than at depths of 4–5 meters below the surface suggests that the size of clam populations changed over time.

13



Perovskite solar cells convert light into electricity more efficiently than earlier kinds of solar cells, and manufacturing advances have recently made them commercially attractive. One limitation of the cells, however, has to do with their electron transport layer (ETL), through which absorbed electrons must pass. Often the ETL is applied through a process called spin coating, but such ETLs are fairly inefficient at converting input power to output power. André Taylor and colleagues tested a novel spray coating method for applying the ETL. The team produced ETLs of various thicknesses and concluded that spray coating holds promise for improving the power conversion efficiency of ETLs in perovskite solar cells.

Which choice best describes data from the graph that support Taylor and colleagues' conclusion?

- A) Both the ETL applied through spin coating and the ETL applied through spray coating showed a power conversion efficiency greater than 10% at their lowest performing thickness.
- B) The lowest performing ETL applied through spray coating had a higher power conversion efficiency than the highest performing ETL applied through spin coating.
- C) The highest performing ETL applied through spray coating showed a power conversion efficiency of approximately 13%, while the highest performing ETL applied through spin coating showed a power conversion efficiency of approximately 11%.
- D) There was a substantial difference in power conversion efficiency between the lowest and highest performing ETLs applied through spray coating.

14

Employment by Sector in France and the United States, 1800–2012
(% of total employment)

Year	Agriculture in France	Manufacturing in France	Services in France	Agriculture in US	Manufacturing in US	Services in US
1800	64	22	14	68	18	13
1900	43	29	28	41	28	31
1950	32	33	35	14	33	53
2012	3	21	76	2	18	80

Rows in table may not add up to 100 due to rounding.

Over the past two hundred years, the percentage of the population employed in the agricultural sector has declined in both France and the United States, while employment in the service sector (which includes jobs in retail, consulting, real estate, etc.) has risen. However, this transition happened at very different rates in the two countries. This can be seen most clearly by comparing the employment by sector in both countries in _____

Which choice most effectively uses data from the table to complete the statement?

- A) 1900 with the employment by sector in 1950.
- B) 1800 with the employment by sector in 2012.
- C) 1900 with the employment by sector in 2012.
- D) 1800 with the employment by sector in 1900.

15

The linguistic niche hypothesis (LNH) posits that the exotericity of languages (how prevalent non-native speakers are) and grammatical complexity are inversely related, which the LNH ascribes to attrition of complex grammatical rules as more non-native speakers adopt the language but fail to acquire those rules. Focusing on two characteristics that are positive indices of grammatical complexity, fusion (when new phonemes arise from the merger of previously distinct ones) and informativity (languages' capacity for meaningful variation), Olena Shcherbakova and colleagues conducted a quantitative analysis for more than 1,300 languages and claim the outcome is inconsistent with the LNH.

Which finding, if true, would most directly support Shcherbakova and colleagues' claim?

- A) Shcherbakova and colleagues' analysis showed a slightly negative correlation between grammatical complexity and fusion and between grammatical complexity and informativity.
- B) Shcherbakova and colleagues' analysis showed a slightly negative correlation between grammatical complexity and exotericity.
- C) Shcherbakova and colleagues' analysis showed a slightly positive correlation between grammatical complexity and fusion.
- D) Shcherbakova and colleagues' analysis showed a slightly positive correlation between fusion and exotericity and between informativity and exotericity.

16

Archaeologist Christiana Kohler and her team excavated the Egyptian tomb of Queen Merneith, the wife of a First Dynasty pharaoh. Some scholars claim that she also ruled Egypt on her own and was actually the first female pharaoh. The team found a tablet in Merneith's tomb with writing suggesting that she was in charge of the country's treasury and other central offices. Whether Merneith was a pharaoh or not, this discovery supports the idea that Merneith likely _____

Which choice most logically completes the text?

- A) had an important role in Egypt's government.
- B) lived after rather than before the First Dynasty of Egypt.
- C) traveled beyond Egypt's borders often.
- D) created a new form of writing in Egypt.

17

In a study of the cognitive abilities of white-faced capuchin monkeys (*Cebus imitator*), researchers neglected to control for the physical difficulty of the tasks they used to evaluate the monkeys. The cognitive abilities of monkeys given problems requiring little dexterity, such as sliding a panel to retrieve food, were judged by the same criteria as were those of monkeys given physically demanding problems, such as unscrewing a bottle and inserting a straw. The results of the study, therefore, _____

Which choice most logically completes the text?

- A) could suggest that there are differences in cognitive ability among the monkeys even though such differences may not actually exist.
- B) are useful for identifying tasks that the monkeys lack the cognitive capacity to perform but not for identifying tasks that the monkeys can perform.
- C) should not be taken as indicative of the cognitive abilities of any monkey species other than *C. imitator*.
- D) reveal more about the monkeys' cognitive abilities when solving artificial problems than when solving problems encountered in the wild.

18

Public-awareness campaigns about the need to reduce single-use plastics can be successful, says researcher Kim Borg of Monash University in Australia, when these campaigns give consumers a choice: for example, Japan achieved a 40 percent reduction in plastic-bag use after cashiers were instructed to ask customers whether _____ wanted a bag.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) they
- B) one
- C) you
- D) it

19

Lucía Michel of the University of Chile observed that alkaline soils contain an insoluble form of iron that blueberry plants cannot absorb, thus inhibiting blueberry growth. If these plants were grown in alkaline soil alongside grasses that aid in iron solubilization, _____ Michel was determined to find out.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) could the blueberries thrive.
- B) the blueberries could thrive.
- C) the blueberries could thrive?
- D) could the blueberries thrive?

20

Atoms in a synchrotron, a type of circular particle accelerator, travel faster and faster until they _____ a desired energy level, at which point they are diverted to collide with a target, smashing the atoms.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) will reach
- B) reach
- C) had reached
- D) are reaching

21

In his 1963 exhibition *Exposition of Music—Electronic Television*, Korean American artist Nam June Paik showed how television images could be manipulated to express an artist's perspective. Today, Paik _____ considered the first video artist.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) will be
- B) had been
- C) was
- D) is

22

Former First Lady of the United States Eleanor Roosevelt and Indian activist and educator Hansa Mehta were instrumental in drafting the United Nations' Universal Declaration of Human Rights, a document that _____ the basic freedoms to which all people are entitled.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) have outlined
- B) were outlining
- C) outlines
- D) outline

23

In February 1919, following the end of the First World War, women from ten countries around the world convened the Inter-Allied Women's Conference in Paris. The conference's goals were _____ ensure women's participation in the proceedings of the Paris Peace Conference, to secure the right of women to serve in the League of Nations, and to advocate for human rights.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) threefold: to
- B) threefold. To
- C) threefold to
- D) threefold; to

24

Mathematician Grigori Perelman, sometimes in conjunction with mathematicians Richard S. Hamilton and Shing-Tung Yau, _____ credited with proving the Poincaré conjecture. Having built on Hamilton's previous work to solve the proof, Perelman has insisted that Hamilton receive credit. Yau later found and closed gaps in Perelman's proof, persuading some mathematicians that he deserves credit as well.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) are
- B) have been
- C) are being
- D) is

25

Researchers studying magnetosensation have determined why some soil-dwelling roundworms in the Southern Hemisphere move in the opposite direction of Earth's magnetic field when searching for _____ in the Northern Hemisphere, the magnetic field points down, into the ground, but in the Southern Hemisphere, it points up, toward the surface and away from worms' food sources.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A) food:
- B) food,
- C) food while
- D) food

26

Most conifers (trees belonging to the phylum Coniferophyta) are evergreen. That is, they keep their green leaves or needles year-round. However, not all conifer species are evergreen. Larch trees, _____ lose their needles every fall.

Which choice completes the text with the most logical transition?

- A) for instance,
- B) nevertheless,
- C) meanwhile,
- D) in addition,

27

Neuroscientist Karen Konkoly wanted to determine whether individuals can understand and respond to questions during REM sleep. She first taught volunteers eye movements they would use to respond to basic math problems while asleep (a single left-right eye movement indicated the number one). _____ she attached electrodes to the volunteers' faces to record their eye movements during sleep.

Which choice completes the text with the most logical transition?

- A) Specifically,
- B) Next,
- C) For instance,
- D) In sum,

28

In his 1925 book *The Morphology of Landscape*, US geographer Carl Sauer challenged prevailing views about how natural landscapes influence human cultures. _____ Sauer argued that instead of being shaped entirely by their natural surroundings, cultures play an active role in their own development by virtue of their interactions with the environment.

Which choice completes the text with the most logical transition?

- A) Similarly,
- B) Finally,
- C) Therefore,
- D) Specifically,

29

In her 2012 analysis of tree rings from Japan's Yaku Island, cosmic ray physicist Fusa Miyake noted an anomalous carbon-14 spike dating to 774–775 CE, indicating that a massive burst of radiation reached Earth during that time. _____ this unprecedented radiocarbon surge was dubbed a "Miyake event" in honor of its discoverer.

Which choice completes the text with the most logical transition?

- A) Fittingly,
- B) Similarly,
- C) However,
- D) In other words,

30

Researchers Helena Mihaljević-Brandt, Lucía Santamaría, and Marco Tullney report that while mathematicians may have traditionally worked alone, evidence points to a shift in the opposite direction. _____ mathematicians are choosing to collaborate with their peers—a trend illustrated by a rise in the number of mathematics publications credited to multiple authors.

Which choice completes the text with the most logical transition?

- A) Similarly,
- B) For this reason,
- C) Furthermore,
- D) Increasingly,

31

While researching a topic, a student has taken the following notes:

- Shaun Tan is an Australian author.
- In 2008, he published *Tales from Outer Suburbia*, a book of fifteen short stories.
- The stories describe surreal events occurring in otherwise ordinary suburban neighborhoods.
- In 2018, he published *Tales from the Inner City*, a book of twenty-five short stories.
- The stories describe surreal events occurring in otherwise ordinary urban settings.

The student wants to emphasize a similarity between the two books by Shaun Tan. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- A) Shaun Tan’s book *Tales from Outer Suburbia*, which describes surreal events occurring in otherwise ordinary places, contains fewer short stories than *Tales from the Inner City* does.
- B) *Tales from Outer Suburbia* was published in 2008, and *Tales from the Inner City* was published in 2018.
- C) Unlike *Tales from the Inner City*, Shaun Tan’s book *Tales from Outer Suburbia* is set in suburban neighborhoods.
- D) Shaun Tan’s books *Tales from Outer Suburbia* and *Tales from the Inner City* both describe surreal events occurring in otherwise ordinary places.

32

While researching a topic, a student has taken the following notes:

- Started in 1925, the Scripps National Spelling Bee is a US-based spelling competition.
- The words used in the competition have diverse linguistic origins.
- In 2008, Sameer Mishra won by correctly spelling the word “guerdon.”
- “Guerdon” derives from the Anglo-French word “guerdun.”
- In 2009, Kavya Shivashankar won by correctly spelling the word “Laodicean.”
- “Laodicean” derives from the ancient Greek word “Laodíkeia.”

The student wants to emphasize a difference in the origins of the two words. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- “Guerdon,” the final word of the 2008 Scripps National Spelling Bee, is of Anglo-French origin, while the following year’s final word, “Laodicean,” derives from ancient Greek.
- In 2008, Sameer Mishra won the Scripps National Spelling Bee by correctly spelling the word “guerdon”; however, the following year, Kavya Shivashankar won based on spelling the word “Laodicean.”
- Kavya Shivashankar won the 2009 Scripps National Spelling Bee by correctly spelling “Laodicean,” which derives from the ancient Greek word “Laodíkeia.”
- The Scripps National Spelling Bee uses words from diverse linguistic origins, such as “guerdon” and “Laodicean.”

33

While researching a topic, a student has taken the following notes:

- In 1851, German American artist Emanuel Leutze painted *Washington Crossing the Delaware*.
- His huge painting (149 × 255 inches) depicts the first US president crossing a river with soldiers in the Revolutionary War.
- In 2019, Cree artist Kent Monkman painted *mistikôsiwak (Wooden Boat People): Resurgence of the People*.
- Monkman’s huge painting (132 × 264 inches) was inspired by Leutze’s.
- It portrays Indigenous people in a boat rescuing refugees.

The student wants to emphasize a similarity between the two paintings. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- Monkman, a Cree artist, finished his painting in 2019; Leutze, a German American artist, completed his in 1851.
- Although Monkman’s painting was inspired by Leutze’s, the people and actions the two paintings portray are very different.
- Leutze’s and Monkman’s paintings are both huge, measuring 149 × 255 inches and 132 × 264 inches, respectively.
- Leutze’s painting depicts Revolutionary War soldiers, while Monkman’s depicts Indigenous people and refugees.

STOP

**If you finish before time is called, you may check your work on this module only.
Do not turn to any other module in the test.**